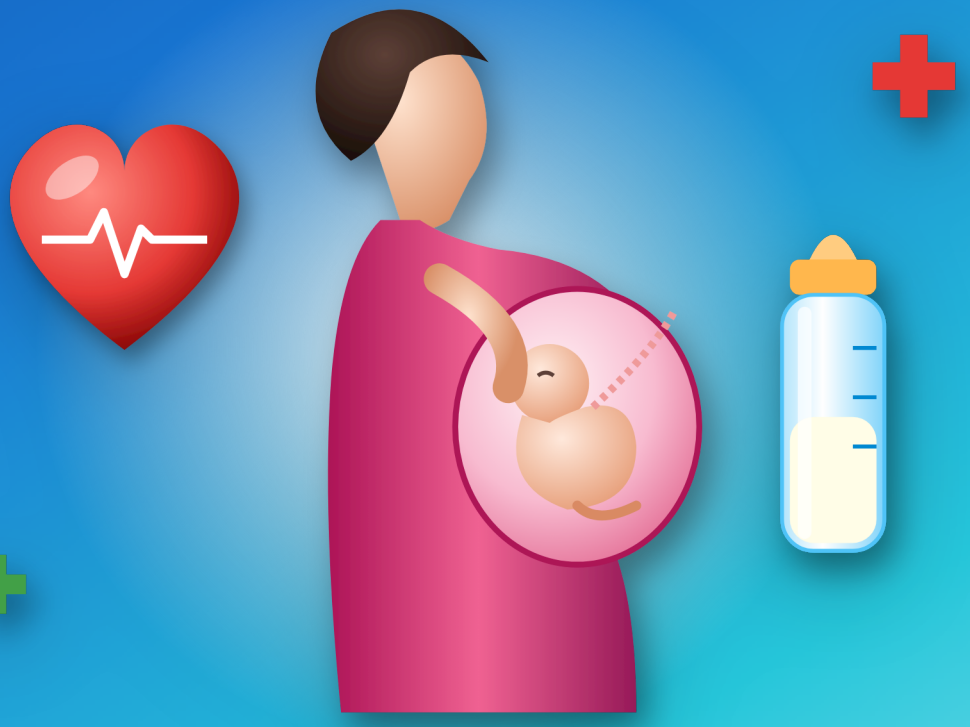
GSSSB · STAFF NURSE EXAM

Midwifery & Obstetrics (OBG)

Complete Colourful Notes · 3D Diagrams · Mnemonics · Exam Points



20 Chapters | 3D Coloured Diagrams | ANC · Labour · PNC · Newborn | National programmes | Obstetric drugs | Exam Points (PYQ style)

Hinglish explanation + English exam terms



Sister Asha
Aapki Nursing teacher

Index – Kya padhenge?

1 **Introduction to Midwifery, MCH Indicators & National Programmes**
Midwifery history, terms, maternal & child health indicators, MMR/IMR, JSY, JSSK, PMSMA, LaQshya, RMNCH+A, SUMAN, MCP card

2 **Female Reproductive Anatomy, Pelvis & Fetal Skull**
External & internal genitalia, uterus, tubes, ovaries, perineum, breast, pelvis bones, planes, diameters, types; fetal skull sutures, fontanelles, diameters, moulding

3 **Menstrual Cycle, Fertilisation, Implantation & Fetal Development**
Hormones, phases, ovulation, gametogenesis, fertilisation, implantation, decidua, embryo & fetal growth month-wise, teratogens

4 **Placenta, Membranes, Amniotic Fluid, Cord & Fetal Circulation**
Placental development, functions, hormones, abnormalities; amnion/ chorion; liquor volume & disorders; cord; fetal circulation & changes at birth

5 **Physiological Changes in Pregnancy**
Reproductive, breast, cardiovascular, blood, respiratory, renal, GI, skin, endocrine, metabolic & weight changes; normal values in pregnancy

6 **Diagnosis of Pregnancy & Antenatal Care**
Presumptive, probable, positive signs; pregnancy tests; Naegele's rule; ANC visits & schedule; investigations; Td, IFA, calcium, deworming; nutrition; minor ailments; danger signs; birth preparedness

7 **Obstetric Abdominal Examination & Fetal Wellbeing**
Lie, presentation, position, attitude, engagement; fundal height; Leopold grips; FHS; USG; NST, CST, BPP, Doppler, kick count, amniocentesis, CVS

8 **Normal Labour – Stages, Mechanism & Management**
Onset theories, true vs false labour, 3 Ps, stages & duration, mechanism (cardinal movements), partograph, management of each stage, AMTSL, episiotomy, perineal tears

9 **Normal Puerperium, Postnatal Care & Breastfeeding**
Involution, lochia, physiological changes, postnatal check-ups, care, exercises; lactation physiology, breastfeeding technique, problems, expressed milk

10 **Care of the Newborn**
Essential newborn care, APGAR, resuscitation (NRP golden minute), physical features, reflexes, vitamin K, birth doses, physiological changes, jaundice, LBW & preterm, KMC, thermal care

11 **High-Risk Pregnancy & Medical Disorders in Pregnancy**
High-risk factors, hyperemesis, anaemia, hypertensive disorders (PIH, pre-eclampsia, eclampsia, HELLP), GDM, heart disease, Rh isoimmunisation, infections (UTI, TORCH, HIV, syphilis, Hep B), multiple pregnancy, IUGR, IUD

12 **Bleeding in Pregnancy – Abortion, Ectopic, Mole & APH**
Types of abortion, septic abortion, MTP Act, ectopic pregnancy, hydatidiform mole, cervical incompetence, placenta praevia, abruptio placentae, vasa praevia

13 **Abnormal Labour – Malpresentations, Malpositions & Dystocia**
OP position, breech, face, brow, shoulder/ transverse lie, compound; CPD; prolonged & obstructed labour; uterine dysfunction; preterm labour; PROM/ PPROM; post-term; induction of labour; Bishop score

14 **Obstetric Emergencies – PPH, Cord Prolapse, Shoulder Dystocia & More**
PPH types, 4 Ts, management steps & drugs, retained placenta, inversion, rupture uterus, cord prolapse, shoulder dystocia, amniotic fluid embolism, obstetric shock, DIC

15 **Complications of Puerperium & Newborn Problems**
Puerperal sepsis/ pyrexia, UTI, breast complications, subinvolution, thrombophlebitis, PE, urinary retention & VVF, psychiatric disorders; birth injuries, asphyxia, RDS, sepsis, hypoglycaemia, congenital anomalies

16 **Obstetric Operations & Procedures**
Caesarean section (LSCS vs classical) & nursing care, forceps, vacuum, version, ARM, MRP, D&C/ MVA, cerclage, destructive operations, amnioinfusion, analgesia & anaesthesia

17 **Drugs in Obstetrics**
Uterotonics (oxytocin, methergine, carboprost, misoprostol, dinoprostone), tocolytics, MgSO4, antihypertensives, steroids, anti-D, iron, folic acid, calcium, drugs in lactation, contraindicated drugs

18 **Family Planning & Contraception**
Population facts, eligible couple, Pearl index, spacing & terminal methods, barrier, OCP, injectables, IUCD, emergency contraception, LAM, natural methods, sterilization, national programme

19 **Gynaecological Disorders & Women's Health**
Menstrual disorders, PCOS, endometriosis, infertility, fibroid, prolapse, infections (RTI/ STI, PID), cervical & breast cancer screening, menopause, gynaec procedures

20 **Quick Revision – Master Tables & 150 One-Liners**
All OBG numbers in one place – measurements, timings, doses, signs, eponyms, programmes; 150 exam one-liners

★ **Key Points** – sabse important facts

Asha Didi ki Tip – samjhane wali baat

! Nursing Alert – galti mat karna

✎ **Exam Point** – PYQ style, baar baar aata hai

♫ **Mnemonic** – yaad karne ki trick

D Definition – exact definition

1 Introduction to Midwifery, MCH Indicators & National Programmes

1.1 Midwifery – Basics

D Definitions

- **Midwife (ICM/ WHO)**: A person who has successfully completed a recognised midwifery education programme and is qualified & licensed to practise midwifery – gives care during **pregnancy, labour, postpartum, newborn**, conducts normal deliveries on her own responsibility.
- **Obstetrics** – Latin "**obstetrix**" = **midwife** ("one who stands before" the woman). Branch dealing with **pregnancy, labour & puerperium**.
- **Gynaecology** – diseases of female reproductive organs (non-pregnant).

- **Father of Obstetrics** – **William Smellie** (British; forceps & pelvic measurements). **Hippocrates** – father of medicine. **Ignaz Semmelweis** – **puerperal fever** prevention by **hand washing (1847)**. **James Young Simpson** – chloroform in labour. **Chamberlen family** – inventors of **obstetric forceps**. **Dr. Anandibai Joshi** – first Indian woman doctor (1886).
- **ICM** (International Confederation of Midwives) – 1922; **5 May = International Day of the Midwife**.
- India – **Nurse Practitioner in Midwifery (NPM)** cadre (2018 guidelines) – midwifery-led care units; **"Midwifery Initiative" 2018**.

Important obstetric terms

Term	Meaning
Gravida	Number of **pregnancies** (including present), irrespective of outcome
Para	Number of pregnancies **delivered beyond viability (≥ 20/ 28 wks)**, alive or dead (twins = 1 para)
Primigravida	Pregnant for the **first time**
Multigravida	Pregnant **2nd time or more**
Nullipara	Never delivered beyond viability
Primipara	Delivered once
Multipara	Delivered 2–4 times
Grand multipara	**Delivered ≥ 5 times** (risk – PPH, rupture, malpresentation)
Elderly primigravida	First pregnancy at **≥ 30 yrs (FOGSI: 35)**
GTPAL	**G**ravida, **T**erm, **P**reterm, **A**bortions, **L**iving children
Viability	Fetus can survive outside uterus – India **28 wks** (MTP/ legal – 20–24 wks; WHO 22 wks / 500 g)
Term	**37–42 wks** (early term 37–38+6, full term 39–40+6, late 41–41+6, post-term ≥ 42)
Preterm	**< 37 completed weeks**

Term	Meaning
Post-term	**≥ 42 wks**
Puerperium	**6 weeks (42 days)** after delivery
Abortion	Termination **before viability (< 20 wks / < 500 g WHO; India < 28 wks)**
Stillbirth	Fetal death **≥ 28 wks** (born without life)

GTPAL example

Woman pregnant now, has 1 term baby (alive), 1 preterm (alive), 1 abortion → **G4 P2 (T1 P1 A1 L2)**. Twins count as **1 para but 2 living**.

1.2 Maternal & Child Health Indicators

Indicator	Definition	Recent India value (SRS)
MMR (Maternal Mortality Ratio)	Maternal deaths **per 1,00,000 live births**	**93 (SRS 2019–21)**; 97 (2018–20); SDG target **< 70 by 2030**
Maternal death (WHO)	Death during pregnancy or **within 42 days** of termination, from any cause related to/ aggravated by pregnancy (not accidental)	Leading causes – **PPH (most common)**, hypertensive disorders, sepsis, abortion, obstructed labour
IMR (Infant Mortality Rate)	Infant (< 1 yr) deaths **per 1000 live births**	**28 (SRS 2020)**, **27 (SRS 2021)**
NMR (Neonatal Mortality Rate)	Deaths < 28 days per 1000 live births	**~19–20** (SRS 2021: 19)
PMR (Perinatal Mortality Rate)	Stillbirths (≥ 28 wks) + early neonatal deaths (< 7 days) **per 1000 total births**	~18
U5MR	Under-5 deaths per 1000 live births	~31 (2021)
TFR (Total Fertility Rate)	Average children per woman	**2.0 (NFHS-5)** – below replacement (2.1)

- **Kerala** – lowest MMR & IMR; **Assam / MP** – highest MMR/IMR historically.
- **Gujarat MMR** ~57 (SRS 2018–20); **Gujarat IMR** ~23 (2020).
- **Leading cause of maternal death** – **haemorrhage (PPH)**; **neonatal death** – **prematurity/ LBW**, birth asphyxia, sepsis.
- **3 Delays model**: (1) delay in **deciding to seek care**, (2) delay in **reaching** facility, (3) delay in **receiving** care.

1.3 National Programmes & Schemes (very important for GSSSB)

Scheme	Year	Key points
RCH-I / RCH-II	**1997 / 2005**	Reproductive & Child Health programme
NRHM → NHM	**2005 / 2013**	ASHA (1 per 1000), JSY, VHSNC
Janani Suraksha Yojana (JSY)	**12 April 2005**	**Cash incentive for institutional delivery** (BPL/ SC/ ST); LPS (low performing states) rural mother **₹1400**, ASHA ₹600; HPS rural ₹700

Scheme	Year	Key points
Janani Shishu Suraksha Karyakram (JSSK)	**1 June 2011**	**Free & cashless** delivery incl. **C-section**, drugs, diagnostics, diet (3 days normal, 7 days CS), blood, **free transport** (home → facility → referral → drop back); **sick newborns up to 1 year**
RMNCH+A	**2013**	Reproductive, Maternal, Newborn, Child + Adolescent – **continuum of care**, "**5×5 matrix**"
Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA)	**2016**	**Free ANC by specialist/ MO on 9th of every month** (2nd/3rd trimester); stickers – **red = high risk**, green = no risk; **e-PMSMA / Extended PMSMA (2022)** – HRP tracking, ₹ incentives
Pradhan Mantri Matru Vandana Yojana (PMMVY)	**2017** (Maternity Benefit)	**₹5000** for **first live child** (now 2 instalments; ₹6000 incl. JSY); **PMMVY 2.0 (Mission Shakti, 2022)** – **2nd child girl → ₹6000**
LaQshya	**2017**	**Labour room & maternity OT quality improvement**
MusQan	2021	Child-friendly quality services in public facilities
SUMAN (Surakshit Matritva Aashwasan)	**10 Oct 2019**	**Zero tolerance for preventable maternal & newborn deaths**, assured free services, respectful care
Mission Indradhanush / IMI	**25 Dec 2014**	Full immunization (children & pregnant women)
Anaemia Mukt Bharat	**2018**	**6×6×6 strategy**; pregnant: **IFA 60 mg + 500 µg folic acid daily from 14–16 wks for 180 days** (+180 PNC)
POSHAN Abhiyaan	**8 March 2018**	Nutrition mission (Jhunjhunu, Rajasthan)
Home Based Newborn Care (HBNC)	2011	ASHA visits – **institutional delivery: 6 visits (day 3, 7, 14, 21, 28, 42)**; home delivery – **7 (day 1 extra)**
HBYC (Home Based Young Child care)	2018	ASHA visits at 3, 6, 9, 12, 15 months
Navjaat Shishu Suraksha Karyakram (NSSK)	2009	Basic newborn care & resuscitation training
Facility-based newborn care	–	**SNCU** (district), **NBSU** (FRU), **NBCC** (every delivery point)
MAA (Mothers' Absolute Affection)	**2016**	**Breastfeeding promotion**
Rashtriya Bal Swasthya Karyakram (RBSK)	**2013**	**4 Ds** – Defects at birth, Diseases, Deficiencies, Development delays (0–18 yrs)
RKSK	**2014**	Adolescent health (10–19 yrs)
Mission Parivar Vikas	2016	Family planning in high TFR districts
Kilkari / Mobile Academy	2016	Weekly voice messages to pregnant women
MCP card (Mother & Child Protection card)	–	Joint MoHFW–MWCD card; ANC, delivery, immunization, growth chart
RCH portal / ANMOL	–	Name-based tracking of mothers & children

- **Gujarat schemes**: **Chiranjeevi Yojana (2005)** – PPP model, private obstetricians deliver BPL women free (**UN award**); **Kasturba Poshan Sahay Yojana**; **Bal Sakha Yojana** (newborn care in private hospitals); **Khilkhilat** – free drop-back van service (2012); **108 ambulance** (2007); **Mamta Divas** –

VHND (Village Health & Nutrition Day); **Poshan Sudha**; **Mukhyamantri Matrushakti Yojana (2022)** – take-home ration for pregnant & lactating mothers.

- **Days**: **International Day of the Midwife – 5 May**, **National Safe Motherhood Day – 11 April** (Kasturba Gandhi's birthday; White Ribbon Alliance), **World Breastfeeding Week – 1–7 Aug**, **Newborn Care Week – 15–21 Nov**, **World Prematurity Day – 17 Nov**, **World Population Day – 11 July**, **World Contraception Day – 26 Sept**.

PYQ style

- MMR expressed per – **1,00,000 live births**; IMR per – **1000 live births**.
- Commonest cause of maternal death – **PPH**.
- JSY launched – **12 April 2005**; JSSK – **1 June 2011**; PMSMA – ANC on **9th of every month**.
- Free transport + free C-section – **JSSK**.
- Safe Motherhood Day – **11 April**.
- Puerperium – **6 weeks**; term – **37–42 wks**.
- Gujarat PPP delivery scheme – **Chiranjeevi Yojana**.

2 Female Reproductive Anatomy, Pelvis & Fetal Skull

2.1 External Genitalia (Vulva/ Pudendum)

- **Mons pubis**, **labia majora** (homologous to scrotum), **labia minora**, **clitoris** (homologous to penis – erectile), **vestibule**, **urethral meatus**, **vaginal orifice/ introitus**, **hymen**, **Bartholin's glands** (greater vestibular – lubrication; **Bartholin cyst/ abscess**), **Skene's glands** (paraurethral), **fourchette**, **perineum**.
- **Perineum (perineal body)** – pyramid-shaped tissue between **vagina & anus**, ~**4 cm**; muscles meeting – **bulbospongiosus, superficial & deep transverse perinei, levator ani (pubococcygeus), external anal sphincter**. **Levator ani** = main pelvic floor muscle (pubococcygeus, iliococcygeus, puborectalis) – supports pelvic organs.
- **Episiotomy** cuts perineal muscles (mediolateral – bulbospongiosus & transverse perinei).

2.2 Internal Genital Organs



Uterus, tubes, ovaries, cervix aur vagina

Vagina

- **Fibromuscular canal** from vulva to cervix; anterior wall **7.5 cm**, posterior wall **9 cm** (longer); directed **upward & backward**.
- **Fornices** – anterior, posterior (**deepest – culdocentesis**), 2 lateral.
- **pH 4–5 (acidic)** – **Döderlein's bacilli (lactobacilli)** convert glycogen → lactic acid → protection. pH alkaline in newborn (maternal oestrogen – acidic briefly), childhood & post-menopause.
- **No glands** in vaginal wall.

Uterus

- **Hollow, pear-shaped, muscular** organ in pelvis between bladder (front) & rectum (behind).
- Size **7.5 × 5 × 2.5 cm** ("3 × 2 × 1 inch"); weight **50–60 g**; cavity **6.5–7 cm** (sound length); capacity non-pregnant ~2–5 ml → term **~5000 ml**; weight at term **~900–1000 g**.
- Position – **anteverted (angle with vagina ~90°) & anteflexed** (body with cervix ~120°–170°).
- Parts: **Fundus** (above tubes), **Body (corpus)**, **Isthmus** (0.5 cm – forms **lower uterine segment** in pregnancy – site of LSCS incision), **Cervix** (2.5 cm – supravaginal & vaginal parts; **internal os & external os**; nulliparous external os **round**, parous **transverse slit**). Body : cervix ratio – **birth/ childhood 1 : 2**, puberty 1 : 1, **adult 2 : 1** (multipara 3 : 1).
- Layers: **Perimetrium** (peritoneum), **Myometrium** – **3 layers**: outer longitudinal, **middle oblique/ interlacing ("living ligatures" – figure-of-8 fibres compress vessels after placental separation → haemostasis)**, inner circular; **Endometrium** – functional (sheds) & basal layer.
- **Supports (ligaments)**: **Transverse cervical (Cardinal/ Mackenrodt's) – main support**, **uterosacral**, **pubocervical**, **round ligament** (maintains **anteversion**; homologous to gubernaculum; through inguinal canal to labia majora), **broad ligament** (peritoneal fold – not a real support), **levator ani**

(pelvic floor). **Pouch of Douglas (rectouterine)** – lowest point of peritoneal cavity (pus/ blood collect); uterovesical pouch – anterior.

- Blood supply – **uterine artery** (branch of **internal iliac**) – crosses **above ureter** ("water under the bridge") near cervix (ureter injury risk in hysterectomy); **ovarian artery** (from **aorta**).

Fallopian (uterine) tubes

- **10 cm** each; parts: **Interstitial** (intramural – narrowest, 1 mm) → **Isthmus** (site of **tubectomy**) → **Ampulla** (widest, longest – **site of fertilisation & commonest ectopic**) → **Infundibulum** with **fimbriae** (ovarian fimbria).
- Ovum transport by **cilia & peristalsis**; ovum reaches uterus in **3–4 days**.

Ovaries

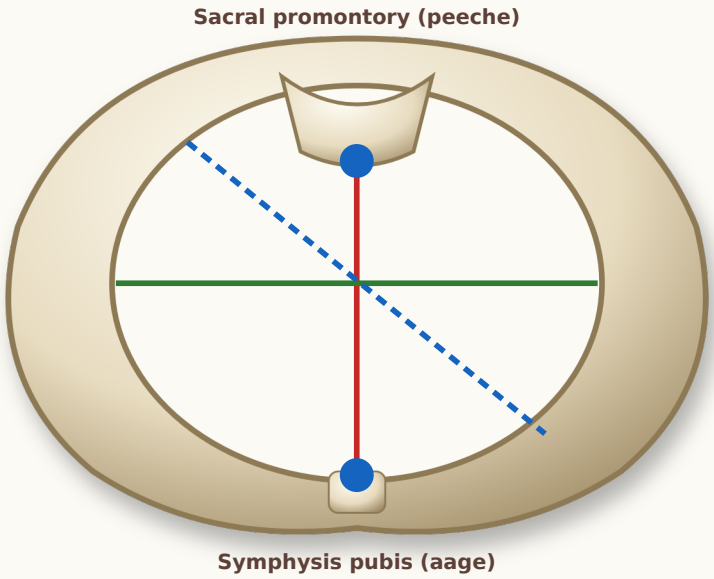
- **Almond-shaped**, **3 × 2 × 1 cm**, 5–8 g; in **ovarian fossa**; attached by **mesovarium**, **ovarian ligament** (to uterus), **infundibulopelvic (suspensory) ligament** (carries ovarian vessels).
- At birth ~**1–2 million** primordial follicles; puberty ~**3–4 lakh**; only ~**400–500 ovulate** in lifetime.
- Hormones: **oestrogen** (follicle – granulosa/ theca), **progesterone** (corpus luteum), inhibin, small androgens, relaxin.

Breast

- Modified sweat gland; **15–20 lobes**, lactiferous ducts → **lactiferous sinuses** (behind areola) → nipple; **Montgomery's tubercles** (sebaceous glands on areola – lubrication; prominent in pregnancy); **tail of Spence** (upper outer – axillary); supported by **Cooper's ligaments**.

2.3 The Female Pelvis

Female Pelvis – Inlet (Brim) Diameters



AP / True conjugate 11 cm
Transverse 13 cm (inlet ka largest)
Oblique 12 cm (dotted)

Pelvic diameters (cm)

Plane	AP	Oblique	Trans
Inlet	11	12	**13**
Cavity	12	12	12
Outlet	**13**	12	11

Inlet – transverse largest
Outlet – AP largest
Diagonal conjugate 12.5–13 cm
Obstetric conjugate 10–10.5 cm

Caldwell-Moloy pelvis types

Gynaecoid – round, ~50 %, ideal
Android – heart-shaped (male)
Anthropoid – oval AP long (apes)
Platypelloid – flat, transverse

Sub-pubic angle: female ≥ 90°, male 60–70°

Pelvic inlet diameters aur pelvis ke types

- **Bones**: **2 innominate (hip) bones** (each = **ilium + ischium + pubis** – fuse at **acetabulum**), **sacrum** (5 fused vertebrae; **sacral promontory**), **coccyx** (4 fused).
- **Joints**: **2 sacroiliac**, **symphysis pubis**, **sacrococcygeal** – relaxin/ progesterone → slight mobility in pregnancy (coccyx moves back 2–3 cm in 2nd stage).
- **False pelvis** (above brim – supports uterus) vs **True pelvis** (below brim – **birth canal** – obstetric importance).
- **True pelvis planes**: **Inlet (brim)** – bounded by sacral promontory, alae of sacrum, iliopectineal lines, superior pubic rami, upper border of symphysis; **Cavity**; **Outlet** – lower border of symphysis, ischial tuberosities, sacrotuberous ligaments, tip of coccyx. **Plane of least pelvic dimensions** – at **ischial spines** (**Station 0** – engagement reference).
- **Axis of birth canal** – **curve of Carus** (J-shaped) – downward & backward then forward.

Plane	AP (cm)	Oblique	Transverse
Inlet	**11 (true conjugate)**	12	**13 (largest)**
Cavity	12	12	12 (round)
Outlet	**13 (largest)**	12	**11 (bi-ischial/ intertuberous)**

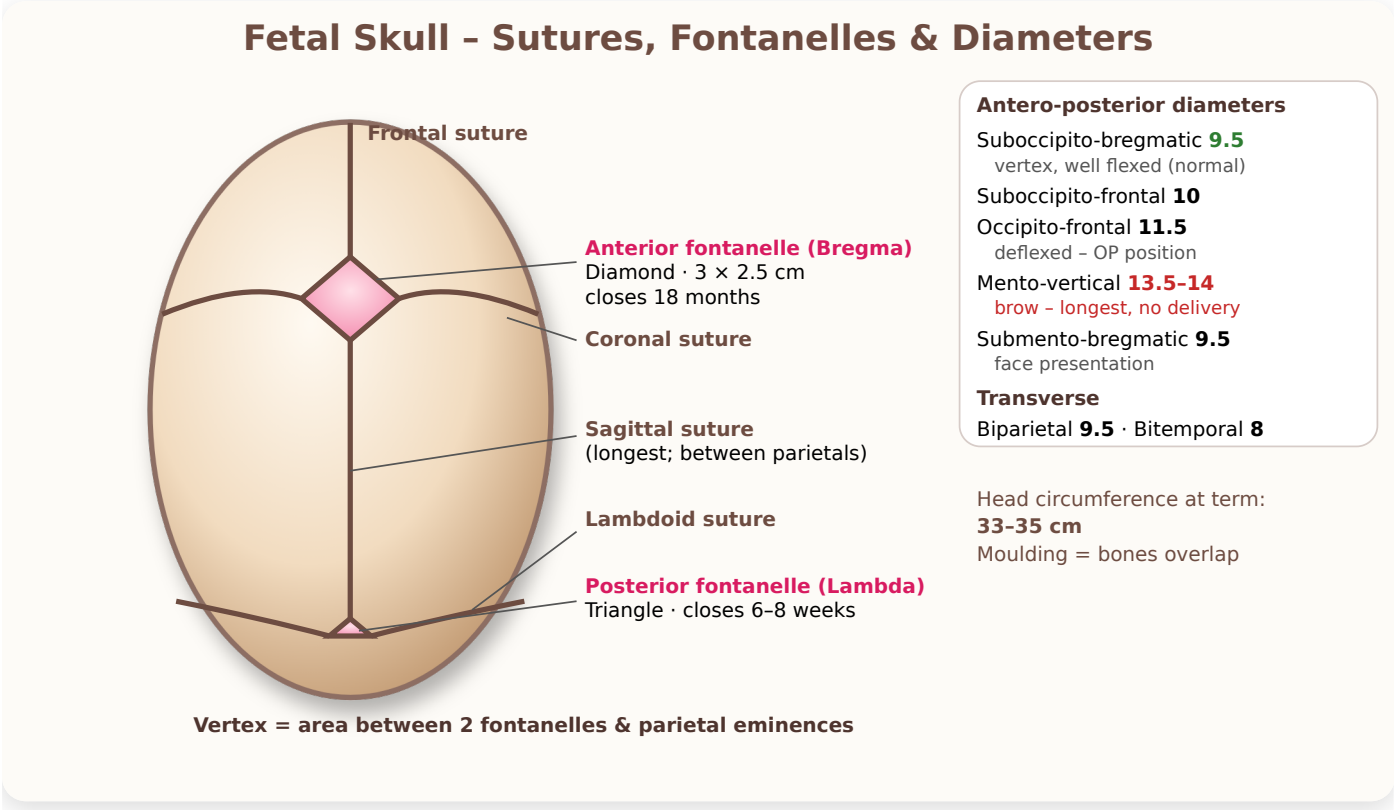
- **Conjugates**: **True/ anatomical conjugate** (promontory → upper border of symphysis) **11 cm**; **Obstetric conjugate** (promontory → **posterior surface** of symphysis – shortest AP, **10–10.5 cm**; clinical importance); **Diagonal conjugate** (promontory → **lower border** of symphysis – **12–12.5 cm** – **measured clinically by PV**; **obstetric conjugate = diagonal − 1.5–2 cm**).

- **Interspinous diameter** – **10.5 cm** (narrowest transverse in mid-pelvis). **Sub-pubic angle** – **90°** (gynaecoid) – "two fingers/ fist"; **intertuberous (bi-ischial) ≥ 10–11 cm** (admit fist – 4 knuckles).
- Head **enters inlet in transverse/ oblique** (widest transverse diameter) and **exits outlet AP** (widest AP) → **internal rotation**.

Type (Caldwell-Moloy, 1933)	Inlet shape	%	Labour
Gynaecoid	**Round/ slightly oval**	**~50 %**	**Best – normal female**
Android	**Heart/ wedge-shaped** (male type)	~20 %	Deep transverse arrest, OP, difficult
Anthropoid	**AP oval** (long AP)	~25 %	**Direct OP delivery** (face-to-pubis)
Platypelloid	**Transverse oval (flat)**	~5 %	Transverse arrest at inlet, **asynclitism**

- **Pelvic assessment** at **36–37 wks** in primigravida (CPD exclusion). **Contracted pelvis** – one or more diameters reduced ≥ 1 cm; obstetric conjugate < 10 cm. **Rickets, osteomalacia, polio, TB spine, fractures** → pelvic deformity.

2.4 Fetal Skull



Fetal skull – sutures, fontanelles aur diameters

- **Vault bones**: 2 **frontal**, 2 **parietal**, 1 **occipital** (+ parts of 2 temporal) – **membranous ossification** → not fused at birth → **moulding** possible.
- **Sutures**: **Sagittal** (between parietals – **most important, felt PV, longest**), **Coronal** (frontal–parietal), **Lambdoid** (parietal–occipital), **Frontal** (between frontal bones).
- **Fontanelles**:

- **Anterior (Bregma)** – **diamond-shaped**, **3 × 2.5 cm**, 4 sutures meet (sagittal, frontal, 2 coronal) → **closes 18 months (12–18)**; **bulging – ↑ICP/ meningitis/ hydrocephalus**; **sunken – dehydration**.
- **Posterior (Lambda)** – **triangular/ Y-shaped**, small, 3 sutures meet → **closes 6–8 weeks** (by 2–3 months).

- **Regions**: **Vertex** (between 2 fontanelles & parietal eminences), **Sinciput (brow)** – anterior fontanelle to root of nose, **Face** (root of nose to chin), **Occiput** (behind posterior fontanelle).
- **Landmarks**: **Glabella, Nasion, Mentum (chin), Occipital protuberance, Parietal eminences, Biparietal**.

Diameters

AP diameter	cm	Presentation/ attitude
Suboccipito-bregmatic	**9.5**	**Vertex, well flexed** (normal – smallest)
Suboccipito-frontal	10	Vertex, partially flexed
Occipito-frontal	**11.5**	Vertex deflexed – **OP position**
Mento-vertical	**13.5–14 (largest)**	**Brow** – **no vaginal delivery** (average size)
Submento-vertical	11.5	Face, partially extended
Submento-bregmatic	**9.5**	**Face, fully extended**

- Transverse: **Biparietal 9.5 cm** (largest transverse; **engagement** when BPD passes inlet), **Bitemporal 8 cm**, **Bimastoid 7.5 cm**. **Bisacromial (shoulders) 12 cm** – adducts to 9.5; bitrochanteric 10 cm.
- **Circumference**: suboccipito-bregmatic ~**33 cm**; occipito-frontal ~**34–35 cm** (head circumference at birth **33–35 cm**).
- **Moulding** – change in skull shape by **overlapping of bones** during labour (parietal over occipital & frontal); grades **0 (separated), +1 (touching), +2 (overlapping, reducible), +3 (overlapping, non-reducible)**; disappears in **24–48 hrs**. Excess → **intracranial haemorrhage (tentorial tear)**.
- **Caput succedaneum** – **oedema of scalp**, **crosses suture lines**, present at birth, disappears in **2–3 days**. **Cephalhaematoma** – **subperiosteal blood**, **does NOT cross sutures**, appears after 12–24 hrs, resolves in **weeks–months** (jaundice risk).

Caput vs Cephalhaematoma

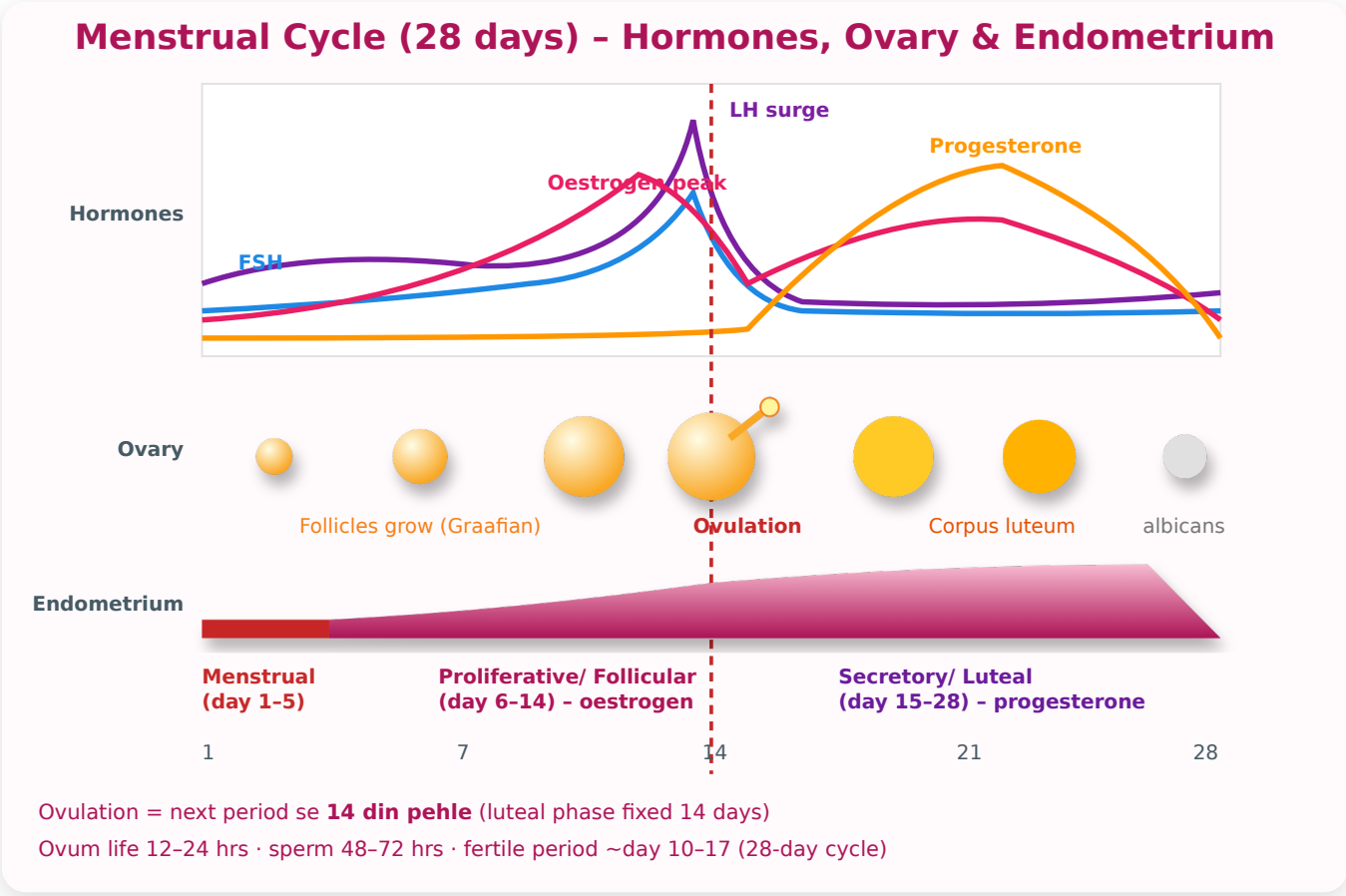
Caput **C**rosses sutures, **C**omes at birth, **C**lears in days. **Cephalhaematoma** – **sutures se bandha** (bone ka periosteum), late appear, slow disappear.

PYQ style

- Largest diameter of inlet – **transverse (13 cm)**; outlet – **AP (13 cm)**.
- Clinically measurable conjugate – **diagonal conjugate**; shortest – **obstetric conjugate**.
- Ideal female pelvis – **gynaecoid**.
- Main support of uterus – **transverse cervical (cardinal) ligament**.
- Site of fertilisation – **ampulla**; tubectomy – **isthmus**.
- Anterior fontanelle closes – **18 months**; posterior – **6–8 weeks**.
- Diameter in well-flexed vertex – **suboccipito-bregmatic 9.5 cm**.
- Largest diameter – **mento-vertical (brow) 13.5–14 cm**.
- Caput – **crosses suture lines**.
- "Living ligatures" – **middle oblique layer of myometrium**.

3 Menstrual Cycle, Fertilisation, Implantation & Fetal Development

3.1 Menstrual Cycle



Menstrual cycle – hormones, ovary aur endometrium

- **Menarche** – first period (avg **12–13 yrs** India); **menopause** – permanent cessation (avg **45–50 yrs**; confirmed after **12 months** amenorrhoea).
- Cycle **21–35 days (avg 28)**; flow **2–7 days (avg 4–5)**; blood loss **~35–80 ml** (> 80 ml = menorrhagia). Menstrual blood **does not clot** (fibrinolysins).
- **Hypothalamo-pituitary-ovarian axis**: **GnRH** (hypothalamus) → **FSH & LH** (anterior pituitary) → ovary → **oestrogen & progesterone** → endometrium.

Phase	Days	Ovary	Endometrium	Hormone
Menstrual	1–5	–	Functional layer **sheds**	↓ oestrogen & progesterone
Proliferative (Follicular)	**6–14**	Follicle growth → **Graafian follicle**	Regeneration, thickening	**Oestrogen** (FSH)
Ovulation	**~14**	Ovum released (**LH surge** 24–36 hrs before)	–	**LH peak**

Phase	Days	Ovary	Endometrium	Hormone
Secretory (Luteal)	**15–28**	**Corpus luteum**	Glands tortuous, secretion, **decidual changes** – ready for implantation	**Progesterone** (+ oestrogen)
Ischaemic (premenstrual)	27–28	Corpus luteum degenerates (no pregnancy) → **corpus albicans**	**Spiral arteries constrict**	Both fall

- **Luteal phase constant 14 days** → **ovulation = cycle length − 14** (35-day cycle → day 21).
- **Signs of ovulation**: **BBT rise 0.3–0.5 °C** (progesterone – thermogenic) after ovulation, **spinnbarkeit** (cervical mucus clear, stretchy, ≥ 10 cm), **fern pattern** (oestrogen), **mittelschmerz** (ovulation pain), LH kit, USG follicular monitoring.
- Corpus luteum – if pregnancy, maintained by **hCG** up to **10–12 wks** (placenta takes over – "luteo-placental shift").

3.2 Gametogenesis, Fertilisation & Implantation

- **Spermatogenesis** – ~**74 days**; sperm count normal **≥ 15 million/ml (WHO 2010; 16 M – 2021)**, volume ≥ 1.5 ml; sperm survive in female tract **48–72 hrs (up to 5 days)**. **Capacitation** in female tract.
- **Oogenesis** – ovum (secondary oocyte) arrested in meiosis II, completes after fertilisation; ovum life **12–24 hrs**.
- **Fertilisation** – fusion of sperm & ovum in **ampulla** of tube → **zygote (46 chromosomes)**; **sex determined by sperm** (X – female, Y – male); within **24 hrs** of ovulation. Acrosome reaction → zona reaction (prevents polyspermy).
- **Cleavage** → **morula (16 cells, day 3–4)** enters uterus → **blastocyst (day 4–5)** – inner cell mass (**embryoblast** → embryo) + outer **trophoblast** (→ placenta).
- **Implantation** – **day 6–7 after fertilisation (≈ day 20–21 of cycle)**, complete by day 10–12; usual site – **upper posterior wall of uterus (fundus)**; **implantation bleeding** (Hartman's sign) may occur. **Abnormal site** – tube (ectopic), lower segment (**placenta praevia**).
- **Decidua** = pregnant endometrium: **Decidua basalis** (under embryo → **maternal part of placenta**), **Decidua capsularis** (covers embryo), **Decidua vera/ parietalis** (rest of uterus). Capsularis & vera fuse by **12–14 wks**.
- **Germ layers** (week 3 – gastrulation): **Ectoderm** – skin, nervous system, eye lens, enamel; **Mesoderm** – muscles, bones, heart, blood, kidneys, gonads; **Endoderm** – GI lining, liver, pancreas, lungs, thyroid, bladder lining.

3.3 Stages of Development

Stage	Period
Ovum/ pre-embryonic (germinal)	**0–2 weeks** after fertilisation
Embryonic	**3rd–8th week** – **organogenesis – most vulnerable to teratogens**
Fetal	**9th week to birth** – growth & maturation

Fetal development – month-wise highlights

Gestation	Development
4 weeks	Heart starts beating (**~day 22**), neural tube closes (**day 26–28**) – **folic acid before conception** prevents NTD
6–7 weeks	**Cardiac activity on TVS USG** (5.5–6 wks), limb buds
8 weeks	All major organs formed; **embryo → fetus**; length ~3 cm
12 weeks	**External genitalia distinguishable**, fetal heart by **Doppler (10–12 wks)**, kidneys produce urine, uterus palpable above symphysis; ~6–7 cm, 14–45 g
16 weeks	Sex visible on USG, **lanugo** appears, quickening (multigravida 16–18)
20 weeks	**Quickening (primigravida 18–20 wks)**, **FHS by stethoscope (Pinard) 18–20 wks**, **vernix caseosa**, ~25 cm, ~300 g; fundus at umbilicus (22–24)
24 weeks	**Surfactant production begins (type II pneumocytes)**, eyelids separate; ~600–700 g; borderline viability (NICU)
28 weeks	**Viability (India)**; eyes open; ~1000–1100 g; 35 cm
32 weeks	Subcutaneous fat, ~1600–1800 g, toenails
36 weeks	~2500 g; lanugo disappearing; **L/S ratio ≥ 2 (lung maturity) ~35 wks**
40 weeks	**~3000–3200 g (India avg ~2.8–3 kg), 50 cm length**, testes in scrotum, nails beyond fingertips, sole creases full

- **Haase's rule** (fetal length): **1st 5 months – $(month)^2$ cm**; **6–10 months – month × 5 cm**. (e.g. 3 months = 9 cm, 7 months = 35 cm).
- **Lanugo** – fine body hair (16–20 wks). **Vernix caseosa** – cheesy white protective coating (sebaceous secretion + desquamated cells). **Meconium** – in fetal gut from **16 wks**; passage in utero = **fetal distress** (or post-term).
- **Fetal Hb (HbF)** – higher O_2 affinity; fetal blood Hb ~**17–20 g/dL**.

Teratogens

- **Drugs**: **Thalidomide** (phocomelia), **warfarin** (fetal warfarin syndrome – nasal hypoplasia), **isotretinoin**, **valproate** (NTD), **phenytoin** (fetal hydantoin syndrome), **tetracycline** (teeth discolouration), **ACE inhibitors**, **methotrexate**, **lithium** (Ebstein anomaly), **DES** (vaginal adenocarcinoma), **misoprostol**, **streptomycin** (deafness), **alcohol (fetal alcohol syndrome – most common preventable cause of mental retardation)**, smoking (LBW).
- **Infections – TORCH**: **T**oxoplasmosis, **O**thers (syphilis, varicella, parvovirus, HIV, Zika), **R**ubella (**cataract, PDA, deafness – Gregg's triad**), **C**MV (most common congenital infection), **H**erpes simplex.
- **Radiation**, maternal diabetes (caudal regression, cardiac anomalies), hyperthermia.
- **FDA pregnancy categories (old)**: A (safe) → B → C → D → **X (contraindicated)** – replaced by PLLR (2015) but still asked.

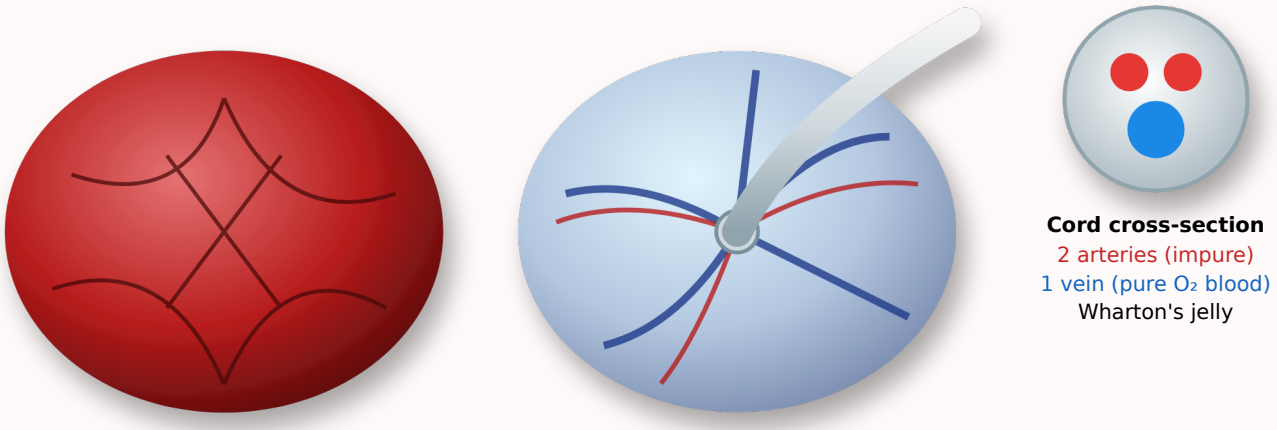
PYQ style

- Ovulation in 28-day cycle – **day 14**; 32-day cycle – **day 18**.
- Hormone causing ovulation – **LH surge**.
- Hormone of secretory phase – **progesterone**; proliferative – **oestrogen**.
- BBT rise after ovulation due to – **progesterone**.
- Site of fertilisation – **ampulla**; implantation – **upper posterior uterine wall**, day **6–7** after fertilisation.
- Organogenesis period – **3–8 weeks** (embryonic).
- Fetal heart starts beating – **~day 22 (4th week)**.
- Quickening primigravida – **18–20 wks**; multigravida – **16–18 wks**.
- Viability (India) – **28 wks**.
- Haase's rule – 5th month length = **25 cm**.
- Rubella – **cataract, deafness, PDA**.

4 Placenta, Membranes, Amniotic Fluid, Cord & Fetal Circulation

4.1 Placenta

Placenta & Umbilical Cord at Term



MATERNAL surface
Dark red, rough, 15–20 cotyledons
"Dirty Duncan" – ise check karo

FETAL surface
Shiny, smooth, bluish-grey (amnion)
"Shiny Schultze" – cord centrally

Cord cross-section
2 arteries (impure)
1 vein (pure O_2 blood)
Wharton's jelly

Placenta: 15–20 cm diameter, 2.5–3 cm thick, ~500 g (baby weight ka ~1/6)
Functions: respiration, nutrition, excretion, barrier, hormones (hCG, hPL, oestrogen, progesterone)

Cord: ~50 cm (30–100), 1–2 cm thick · Short < 30 cm, long > 100 cm
Single umbilical artery → renal/ cardiac anomalies check karo

Placenta ki maternal & fetal surface aur cord

- Develops from **trophoblast (chorion frondosum – fetal part)** + **decidua basalis (maternal part)**; **fully formed by 12 weeks**; **functional from ~3rd week**.
- At term: **discoid**, **15–20 cm diameter**, **2.5–3 cm thick** (centre), weight **~500 g** (**ratio placenta : baby ≈ 1 : 6**), **15–20 cotyledons** on maternal surface.
- **Maternal surface** – rough, dark red, **cotyledons** → check for **completeness** (missing cotyledon → **PPH, infection**). **Fetal surface** – smooth, shiny, covered by **amnion**, umbilical vessels radiating.
- **Separation**: **Schultze** (central – **fetal surface first, "Shiny Schultze"**, more common ~80 %, less bleeding, retroplacental clot inside) vs **Matthews Duncan** (marginal – **maternal surface first, "Dirty Duncan"**, more bleeding, membranes may be retained).
- **Utero-placental blood flow at term ~500–750 ml/min**. Exchange in **intervillous space** – maternal & fetal blood **do not mix** (placental barrier).

Functions of placenta

1. **Respiratory** – O_2 & CO_2 exchange (simple diffusion)
2. **Nutritive** – glucose (facilitated diffusion – main fuel), amino acids (active transport), vitamins, minerals, iron

3. **Excretory** – urea, uric acid, creatinine
4. **Protective/ barrier** – blocks many microorganisms & drugs (but **viruses, TORCH, alcohol, nicotine, many drugs cross**); **IgG crosses** (passive immunity – **only IgG**)
5. **Endocrine** – hormones:

Hormone	Function / Point
hCG (human chorionic gonadotropin)	**Maintains corpus luteum**; basis of **pregnancy test**; detectable in serum **8–10 days after ovulation**; **peak 60–70 days (8–10 wks)**; very high in **molar pregnancy, twins**; low in ectopic/ abortion; hyperemesis linked
hPL (human placental lactogen)/ hCS	**Anti-insulin (diabetogenic)** – ↑ maternal blood glucose for fetus; ↑ with placental mass
Oestrogen (mainly **oestriol** – needs fetal adrenal & liver)	Uterine growth, breast ducts; oestriol = **fetal wellbeing marker**
Progesterone	**Maintains pregnancy** – uterine relaxation (quietening), breast alveoli; produced by placenta from ~10–12 wks
Relaxin, CRH, PAPP-A	Relaxin – pelvic joints soften; PAPP-A – first trimester screening

Placental abnormalities

Abnormality	Description
Succenturiate lobe	Accessory lobe connected by vessels → **retained → PPH** (check membranes for torn vessels)
Bipartite/ bilobed	Two lobes
Circumvallate	Fetal surface with thick white ring (folded membranes) – APH, preterm
Battledore placenta	**Cord attached at margin** (edge)
Velamentous insertion	**Cord inserted into membranes**; vessels run in membranes → **vasa praevia** (fetal bleeding on ROM – fetal death)
Placenta accreta / increta / percreta	Abnormal adherence – villi into myometrium (**accreta** – superficial, **increta** – into myometrium, **percreta** – through serosa/ bladder); **previous LSCS + placenta praevia** = major risk → PPH, hysterectomy
Placenta membranacea	Thin, covering most of chorion
Infarcts, calcification	Post-term, pre-eclampsia

4.2 Membranes & Amniotic Fluid

- **Amnion** (inner, fetal side; tough, avascular, no nerves) & **Chorion** (outer, adjacent to decidua). **Amniotic sac** = "bag of waters".
- **Amniotic fluid (liquor amnii)**: clear, pale straw, **alkaline (pH 7–7.5)**; **98–99 % water**; volume **~1000 ml at 36–38 wks (max)**, **~800 ml at term**, **50 ml at 12 wks**, 400 ml at 20 wks.
- Source – early: **amnion/ maternal transudate**; after 20 wks: **fetal urine (main)** + lung fluid. Removed by **fetal swallowing**.
- **Functions**: protection from trauma, **constant temperature**, free fetal movement (musculoskeletal development), **lung development**, prevents adhesions & cord compression, during labour –

forewaters dilate cervix, flushes birth canal (antiseptic).

- **Test for ROM**: **Nitrazine paper** (alkaline liquor → **blue**), **fern test** (microscope – arborization), pooling on speculum, AmniSure.
- **Colour**: **clear** (normal), **green/ meconium-stained** (fetal distress, post-term, breech – may be normal), **golden yellow** (Rh isoimmunisation – bilirubin), **red/ blood-stained** (APH), **dark brown/ tobacco juice** (IUD – old), **greenish-yellow & foul** (infection – chorioamnionitis).

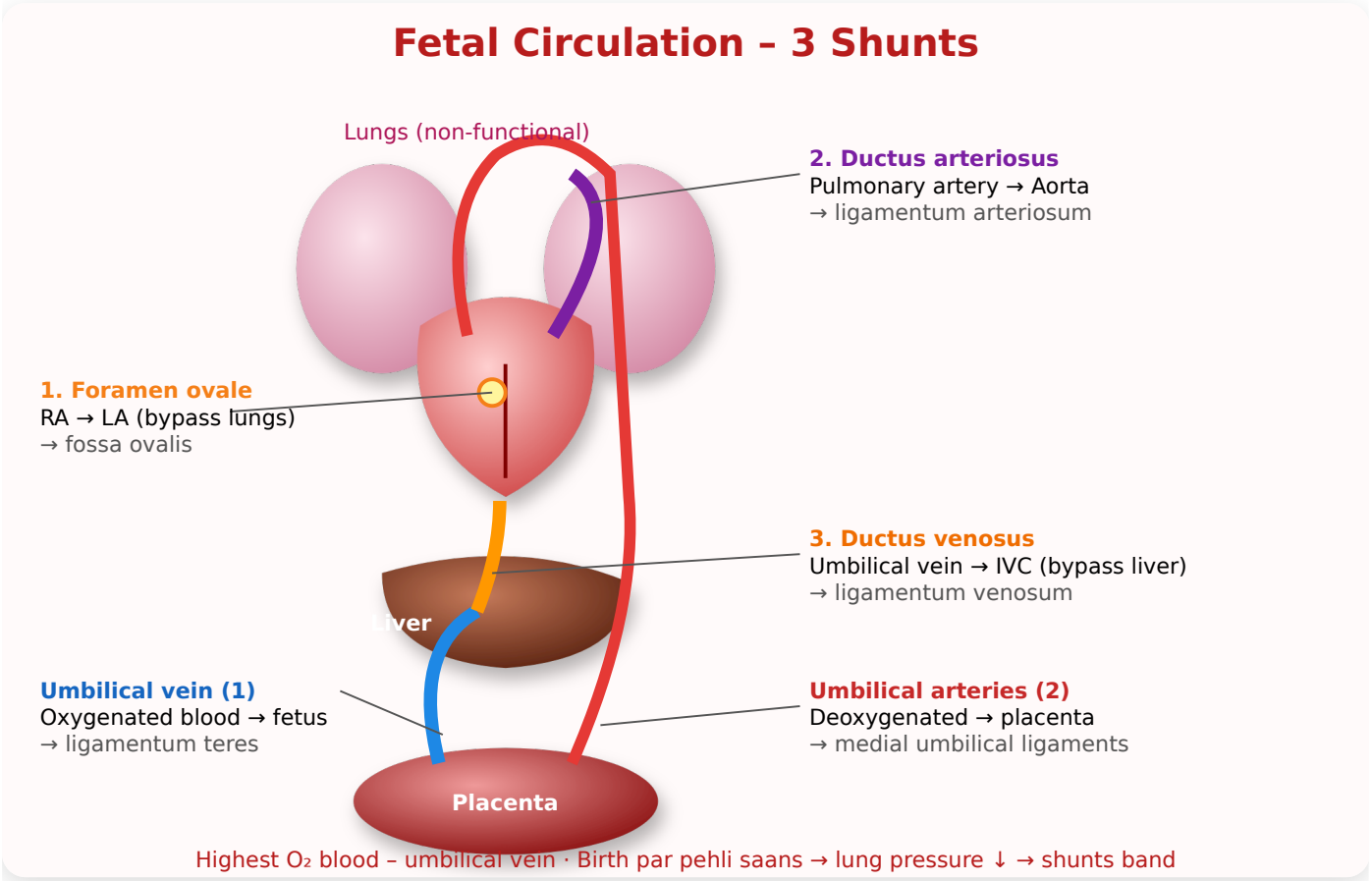
	Polyhydramnios	Oligohydramnios
Definition	**> 2000 ml**; **AFI > 24–25 cm**; SDP > 8 cm	**< 200–500 ml**; **AFI < 5 cm**; SDP < 2 cm
Causes	**Fetal: oesophageal/ duodenal atresia (can't swallow)**, **anencephaly/ NTD**, twins (TTTS); **maternal diabetes**, Rh isoimmunisation, chorioangioma, idiopathic	**Renal agenesis (Potter's syndrome)**, **posterior urethral valve**, PROM, **IUGR/ placental insufficiency**, post-term
Effects	Malpresentation, **cord prolapse**, PROM, preterm labour, **PPH (overdistension)**, abruption (sudden decompression), dyspnoea	**Pulmonary hypoplasia**, limb deformities (club foot), cord compression, fetal distress, Potter's facies
Management	**Amniocentesis/ amnioreduction**, **indomethacin** (↓ fetal urine), treat cause; ARM slowly with care	Hydration, **amnioinfusion** (in labour), deliver if term

- **AFI (Amniotic Fluid Index)** – sum of deepest vertical pockets in **4 quadrants** by USG; **normal 5–25 cm (8–18 optimal)**.

4.3 Umbilical Cord (Funis)

- Length **~50 cm (30–100)**; **short cord < 30 cm** (abruption, inversion), **long cord > 100 cm** (prolapse, knots, nuchal cord).
- **2 umbilical arteries** (deoxygenated – fetus → placenta) + **1 umbilical vein** (oxygenated – placenta → fetus) embedded in **Wharton's jelly** (protects from compression); **allantois** remnant.
- **Single umbilical artery** (~1 %) → associated with **renal & cardiac anomalies**, trisomy.
- **True knot** (danger – fetal death) vs **false knot** (vessel loop – harmless).
- Cord **cut** – **delayed cord clamping 1–3 min** (WHO; ↑ iron stores) unless asphyxiated baby needing resuscitation.
- **Cord blood** – **blood group, Rh, Coombs**, bilirubin, stem cell banking.
- Cord stump falls in **5–10 days** (dry cord care – nothing applied; chlorhexidine 7.1 % where high NMR).

4.4 Fetal Circulation



Fetal circulation – teeno shunts aur unke adult remnants

Fetal circulation – key points

- **Placenta = fetal lung**; oxygenated blood via **umbilical vein** (SaO_2 ~80 % – **highest O_2 in fetus**).
- **3 shunts**:
  1. **Ductus venosus** – umbilical vein → **IVC** (**bypasses liver**) → closes in few days → **ligamentum venosum**
  2. **Foramen ovale** – **right atrium → left atrium** (**bypasses lungs**) → closes functionally at birth (↑ LA pressure) → **fossa ovalis**
  3. **Ductus arteriosus** – **pulmonary artery → aorta** (**bypasses lungs**) → functional closure **10–15 hrs**, anatomical **2–3 weeks** (↑ O_2, ↓ prostaglandins) → **ligamentum arteriosum**
- **Umbilical vein → ligamentum teres** (round ligament of liver); **umbilical arteries → medial umbilical ligaments**; hypogastric arteries proximal part → superior vesical arteries.
- Lungs fluid-filled, **high pulmonary vascular resistance** → blood bypasses lungs.

- **Changes at birth**: first breath → lungs expand → **pulmonary resistance ↓** → more blood to lungs → **LA pressure ↑** → **foramen ovale closes**; cord clamped → placental circulation stops → **SVR ↑**; ↑ PaO_2 → **ductus arteriosus constricts**.
- **PDA** (patent ductus arteriosus) – preterm, rubella → **machinery murmur**; Rx **indomethacin/ ibuprofen/ paracetamol** (prostaglandin inhibitors). **Prostaglandin E1** keeps ductus open (duct-dependent heart disease).
- **Fetal heart rate** – normal **110–160 bpm** (older 120–160).

PYQ style

- Placenta weight at term – **500 g (1/6 of baby)**; cotyledons – **15–20**.
- Shiny surface delivered first – **Schultze**; maternal first – **Matthews Duncan**.
- Pregnancy test hormone – **hCG**; diabetogenic hormone – **hPL**.
- Only immunoglobulin crossing placenta – **IgG**.
- Amniotic fluid maximum – **~1000 ml at 36–38 wks**; pH – **alkaline**.
- Polyhydramnios – **oesophageal atresia, anencephaly, diabetes**; oligohydramnios – **renal agenesis**.
- Nitrazine paper with liquor – **blue**.
- Cord contains – **2 arteries + 1 vein**; highest O_2 – **umbilical vein**.
- Ductus arteriosus connects – **pulmonary artery to aorta**; remnant – **ligamentum arteriosum**.
- Cord inserted on membranes – **velamentous**; at edge – **battledore**.

5 Physiological Changes in Pregnancy

5.1 Reproductive Organs

- **Uterus**: weight **50–60 g → ~1000 g**; size 7.5 cm → **35 cm**; capacity ~5 L; growth by **hypertrophy (main) + hyperplasia** (oestrogen) + stretching. Shape: pear → globular (12 wks) → ovoid (term). **Lower uterine segment** forms from isthmus by **32 wks** (thin, passive – **LSCS incision**); upper segment – thick, active, contractile.
- **Braxton Hicks contractions** – **painless, irregular, non-dilating** contractions from **early pregnancy (felt ~20 wks onward)**; **false labour pain** near term.
- **Cervix** – **Goodell's sign (softening, ~6 wks)**, **operculum (mucus plug)** – prevents ascending infection; expelled as **"show"** at labour onset; bluish (Chadwick).
- **Vagina** – **Chadwick's (Jacquemier's) sign** – **bluish/ purplish discolouration** of vagina & cervix (~8 wks) – vascularity; ↑ secretions (**leucorrhoea**), pH ↓ (more acidic – 3.5–6) → **candidiasis common**.
- **Ovaries** – no ovulation; **corpus luteum** active till 10–12 wks.
- **Perineum & vulva** – ↑ vascularity, varicosities.

5.2 Breasts

- Early sign (**6–8 wks**): **fullness, tingling, tenderness**; **enlargement**, **prominent veins**; **primary areola** darker; **Montgomery's tubercles** (sebaceous glands) prominent (**8 wks**); **secondary areola** (mottled, **20 wks**); **colostrum** expressible from **12–16 wks**; striae.
- Hormones – **oestrogen (ducts)**, **progesterone (alveoli)**, prolactin, hPL.

5.3 Cardiovascular & Blood

Parameter	Change
Blood volume	**↑ 40–50 % (~1.5 L)** – peak **30–34 wks**
Plasma volume	↑ **50 %** (more than RBC)
RBC mass	↑ 20–30 %
Haemoglobin	**↓ → physiological (dilutional) anaemia** – normal **≥ 11 g/dL** (2nd trimester ≥ 10.5)
Haematocrit	↓ (~33–35 %)
Cardiac output	**↑ 40–50 %** – starts 5 wks, **peak 28–32 wks**; **highest immediately after delivery (~80 % ↑) → heart disease patients decompensate after delivery**
Heart rate	↑ **10–15 bpm**
Stroke volume	↑
BP	**↓ in 2nd trimester (diastolic ↓ 5–10 mmHg)**, back to normal at term
Peripheral vascular resistance	↓ (progesterone – vasodilation)
Heart	**Pushed up & to the left** (apex beat 4th ICS), **systolic flow murmur (ejection)**, **S3** may be normal

Parameter	Change
WBC	**↑ (up to 15,000; labour 25,000)** – physiological leucocytosis
Platelets	Slight ↓ (gestational thrombocytopenia)
Coagulation factors	**↑ fibrinogen (400–650 mg/dL), VII, VIII, IX, X → hypercoagulable state → DVT risk ↑** (protects from PPH)
ESR	**↑↑ (not useful)**
Iron requirement	Total ~**1000 mg** for pregnancy

- **Supine hypotension syndrome (aortocaval compression)** – late pregnancy, supine → gravid uterus compresses **IVC → ↓ venous return → hypotension, dizziness, pallor, sweating, fetal distress → turn to LEFT lateral position** (or wedge under right hip).
- **Oedema of feet** (physiological – venous pressure ↑) – common, relieved by rest & elevation; **varicose veins, haemorrhoids**.

5.4 Respiratory System

- **Tidal volume ↑ 30–40 %, minute ventilation ↑ 40–50 %** (progesterone stimulates respiratory centre) → **$PaCO_2$ ↓ (~30 mmHg) → mild compensated respiratory alkalosis** (helps CO_2 transfer from fetus).
- RR unchanged/ slight ↑; **residual volume & FRC ↓ 20 %**; vital capacity unchanged; **diaphragm elevated 4 cm**; **subcostal angle widens (68° → 103°)**; **dyspnoea** common (physiological); **O_2 consumption ↑ 20 %**.
- Nasal congestion, epistaxis (oestrogen – vascular).

5.5 Urinary System

- **Kidneys enlarge**; **GFR ↑ 50 %**, renal plasma flow ↑ → **serum urea & creatinine ↓** (creatinine > 0.8 suspicious).
- **Ureters dilate** (progesterone + compression) – **right > left (dextrorotation of uterus) → hydroureter/ hydronephrosis → UTI/ pyelonephritis risk ↑** (right sided).
- **Frequency of micturition** – **1st trimester** (bladder pressure by uterus) & **last weeks** (presenting part engages); 2nd trimester relief.
- **Glycosuria** (↓ renal threshold) – common, not necessarily diabetes; mild proteinuria (< 300 mg/day).

5.6 GI, Skin, Endocrine & Metabolic

- **GI**: **nausea & vomiting (morning sickness – 4–14/16 wks; hCG, oestrogen)**, **ptyalism (excess saliva)**, **pica** (craving non-food – clay, chalk – iron deficiency), **heartburn** (progesterone relaxes cardiac sphincter + pressure), **constipation** (↓ motility – progesterone, iron), **gum hypertrophy/ bleeding (epulis/ pregnancy gingivitis)**, **gallstones** (bile stasis), haemorrhoids.
- **Skin** (melanocyte-stimulating hormone ↑):
  - **Chloasma/ melasma ("mask of pregnancy")** – brown patches on face
  - **Linea nigra** – dark midline on abdomen
  - **Striae gravidarum** – **stretch marks** (pink → silvery **striae albicans** after delivery)
  - Spider naevi, **palmar erythema**, pigmentation of areola, vulva

- **Endocrine**: **pituitary enlarges** (prolactin ↑ – Sheehan's risk with PPH), **thyroid enlarges** (total T4 ↑ – TBG ↑; free T4 normal), **parathyroid ↑**, **adrenal cortisol ↑**, **insulin resistance (hPL, cortisol)** – fasting glucose ↓, post-meal ↑.
- **Musculoskeletal**: **lordosis** (centre of gravity shifts forward) → **backache**; **relaxin** – ligament laxity, pelvic joints mobile (waddling gait); **leg cramps** (calcium ↓/ phosphate ↑).
- **Metabolic**: **BMR ↑ 15–20 %**; **water retention ~6–8 L**; protein, fat storage ↑; **calcium – fetus needs ~30 g (mostly 3rd trimester)**.

Weight gain in pregnancy

- Average total **10–12 kg (11 kg)** – **1 kg in 1st trimester**, **~5 kg each in 2nd & 3rd trimester**; **~0.5 kg/week (2 kg/month)** after 20 wks.
- IOM 2009 by BMI: underweight **12.5–18 kg**, normal **11.5–16 kg**, overweight **7–11.5 kg**, obese **5–9 kg**.
- **Sudden excessive weight gain (> 1 kg/week or > 3 kg/month)** → **pre-eclampsia (oedema)** suspicion; **poor gain** → IUGR.
- Distribution at term: **fetus ~3.3 kg**, placenta 0.6, liquor 0.8, uterus 0.9, breasts 0.4, blood 1.5, fluid 1.5–2, fat 3–3.5 kg.

PYQ style

- Blood volume increases by – **40–50 %**, peak at **30–34 wks**.
- Physiological anaemia due to – **haemodilution (plasma > RBC)**.
- Cardiac output maximum – **immediately after delivery**.
- Supine hypotension relief – **left lateral position**.
- Chadwick's sign – **bluish discolouration of vagina/ cervix**; Goodell's – **cervical softening**.
- Mask of pregnancy – **chloasma**; dark midline – **linea nigra**.
- Ureteric dilatation more on – **right side**.
- Pregnancy is a – **hypercoagulable state**.
- Respiratory change – **compensated respiratory alkalosis (↓ $PaCO_2$)**.
- Average weight gain – **11 kg (10–12)**.

6 Diagnosis of Pregnancy & Antenatal Care

6.1 Signs of Pregnancy

Presumptive (subjective)	Probable (objective – examiner)	Positive (diagnostic – fetus)
Amenorrhoea (first & most common symptom)	**Enlargement of abdomen/ uterus**	**Fetal heart sounds** – Doppler **10–12 wks**, stethoscope **18–20 wks**
Morning sickness (4–14 wks)	**Hegar's sign** (6–10 wks – softening of **isthmus**; bimanual – fingers almost meet)	**Fetal movements felt by examiner** (~20 wks)
Breast changes (tenderness, Montgomery's)	**Goodell's sign** (cervical softening, 6 wks)	**USG – gestational sac (TVS 4.5–5 wks; TAS 6 wks)**, **fetal pole & cardiac activity (TVS 6 wks)**
Frequency of micturition	**Chadwick's/ Jacquemier's sign** (bluish vagina, 8 wks)	**Palpation of fetal parts** (~20–24 wks)
Quickening (first fetal movements felt by mother – 18–20 wks primi, 16–18 multi)	**Osiander's sign** (pulsation in lateral fornix, 8 wks)	**X-ray fetal skeleton** (16 wks – not used)
Fatigue, skin pigmentation	**Piskacek's sign** (asymmetric enlargement at implantation site)	
	Palmer's sign (rhythmic uterine contractions, 4–8 wks)	
	Braxton Hicks contractions	
	Ballottement (internal 16–20 wks/ external)	
	Positive pregnancy test (hCG)	

- **Pregnancy tests (hCG)**: **urine card test (immunological – Nischay kit, ASHA kit)** – positive ~**1 week after missed period** (urine hCG 25 mIU/ml); **early morning urine** best (concentrated); **serum β-hCG – 8–11 days after conception** (most sensitive); **doubles every 48 hrs** in early normal pregnancy (not doubling → ectopic/ abortion).

6.2 Calculation of Dates

Naegele's Rule (EDD)

EDD = LMP + 9 months + 7 days (or LMP − 3 months + 7 days + 1 year) → **280 days (40 wks)** from LMP.

- Example: **LMP 10 March 2026** → EDD **17 December 2026**.
- Example: **LMP 25 Oct 2025** → 25 + 7 = 32 → **1 Aug 2026** (Oct − 3 = July; 32 July = 1 Aug).
- Assumes **28-day regular cycle**; longer cycle → add extra days (cycle length − 28).
- **Only ~4–5 % deliver on EDD**; most within ± 2 wks.
- Gestational age by **first trimester USG (CRL – crown-rump length, 7–13 wks)** – **most accurate dating**.
- Other: date of quickening + 22 wks (primi)/ 24 wks (multi); fundal height.

6.3 Antenatal Care (ANC)

- **Aims**: healthy mother & baby, **early detection of high-risk** cases, health education, birth preparedness, prevent complications.
- **First visit/ registration** – **as early as possible, within 12 wks (1st trimester)**.
- **Visits** – **India (MoHFW): minimum 4 ANC visits** – **1st: within 12 wks**, **2nd: 14–26 wks**, **3rd: 28–34 wks**, **4th: 36 wks–term** (+ PMSMA on 9th). **WHO 2016: minimum 8 contacts** (12, 20, 26, 30, 34, 36, 38, 40 wks). Traditional: monthly till 28 wks, fortnightly 28–36, weekly after 36.

History & examination at first visit

- **History**: LMP (EDD), obstetric history (GTPAL), previous complications (PPH, LSCS, stillbirth, abortions), medical history (DM, HTN, heart disease, TB, epilepsy), family history (twins, DM), drugs, allergies.
- **General exam**: height (**< 140/ 145 cm – CPD risk**), weight, BMI, BP, pulse, **pallor**, oedema, thyroid, breasts & nipples (flat/ inverted), heart & lungs, spine.
- **Obstetric exam** – abdominal palpation (from 2nd trimester), FHS.

Routine investigations

Test	Purpose
Hb (each trimester)	Anaemia (< 11 g/dL)
Blood group & Rh	**Rh –ve → ICT (indirect Coombs) at 28 wks**, anti-D
Urine – albumin & sugar (every visit)	**Pre-eclampsia (albumin)**, diabetes
Blood sugar – OGTT 75 g (DIPSI)	**GDM screening** – first visit & **24–28 wks**; **2-hr ≥ 140 mg/dL = GDM** (non-fasting, single step – India)
VDRL/ RPR	Syphilis
HIV (with counselling)	PPTCT
HBsAg	Hepatitis B → baby HBIG + vaccine
Urine routine/ culture	Asymptomatic bacteriuria

Test	Purpose
TSH (high-risk/ universal in some)	Hypothyroidism
Ultrasound	**Dating (1st trim)**, **NT scan 11–14 wks**, **anomaly scan (TIFFA) 18–20 wks**, growth & placenta (3rd trim)
Double marker (11–14 wks)/ Triple/ Quadruple (15–20 wks)	**Down syndrome & NTD screening** (MSAFP ↑ – NTD; ↓ – Down)
Sickle cell, thalassaemia (HbA2) screening	Endemic areas/ Gujarat tribal

Supplements & immunization

- **Td (Tetanus-diphtheria)**: **2 doses – Td1 early in pregnancy, Td2 4 weeks after Td1**; **booster (1 dose)** if previously vaccinated within 3 yrs; (**TT replaced by Td – 2018**). IM deltoid, 0.5 ml.
- **IFA**: **1 tablet (60 mg elemental iron + 500 µg folic acid) daily from 14–16 wks (2nd trimester) for 180 days**, continue **180 days postpartum** (Anaemia Mukt Bharat); anaemic (Hb 7–10.9) – **2 tablets daily**; severe – IV iron/ blood.
- **Folic acid 400 µg (0.4 mg)** daily – **preconception (3 months before) & first trimester** – NTD prevention; previous NTD baby → **4 mg/day**.
- **Calcium**: **1 g elemental (2 × 500 mg tabs) daily from 14 wks** after meals (not with iron – separate by 2 hrs) – pre-eclampsia prevention.
- **Albendazole 400 mg single dose in 2nd trimester** (deworming) – not in 1st trimester.
- **Iodized salt**; Vit D as needed. **Influenza vaccine** (WHO), **COVID**; **live vaccines (MMR, varicella, BCG, yellow fever) contraindicated**.

Nutrition in pregnancy (ICMR)

- **Extra +350 kcal/day in 2nd & 3rd trimester**; protein **+8–10 g (2nd), +18–23 g (3rd)**; iron rich foods, **folate**, calcium (milk, ragi), vitamins; **small frequent meals**; avoid alcohol, smoking, raw/ undercooked food, excess caffeine (< 200 mg/day).
- **Lactation**: **+600 kcal (0–6 months)**, +520 (6–12 months); protein +13–19 g.

6.4 Minor Ailments of Pregnancy & Relief

Ailment	Cause	Relief
Morning sickness	hCG, oestrogen	**Dry toast/ biscuits before getting out of bed**, small frequent meals, avoid fatty/ spicy foods, ginger, fluids between meals (not with meals), **doxylamine + pyridoxine**
Heartburn	Progesterone (sphincter relax) + pressure	Small meals, **avoid lying down 1–2 hrs after meals**, head end elevated, antacids
Constipation	Progesterone, iron, pressure	**Fibre, fluids (2–3 L), exercise**, isabgol; no strong purgatives
Backache	**Lordosis**, relaxin	**Good posture, low-heeled shoes**, firm mattress, pelvic tilt exercise, proper body mechanics

Ailment	Cause	Relief
Leg cramps	**↓ Calcium/ ↑ phosphorus**, pressure	**Dorsiflex the foot (stretch calf)**, massage, warmth, calcium
Varicose veins	Venous pressure, progesterone	Elevate legs, elastic stockings, avoid long standing, don't cross legs
Oedema of feet (dependent)	Venous pressure	Rest with legs elevated, **left lateral**; rule out pre-eclampsia (face/ hands, BP)
Haemorrhoids	Constipation, pressure	Fibre, sitz bath, ointments
Frequency of micturition	Bladder pressure	Reassure; rule out UTI (burning); reduce fluids at bedtime
Leucorrhoea	↑ vascularity, oestrogen	Hygiene, cotton undergarments; foul/ itchy → treat infection
Fainting/ supine hypotension	Vasodilation, IVC compression	Left lateral, rise slowly
Insomnia	Discomfort, movements	Left lateral with pillows, warm milk, relaxation
Ptyalism, **pica**, **gum bleeding**, **carpal tunnel**, **round ligament pain**, **dyspnoea**		Reassurance, specific measures

6.5 Danger Signs & Birth Preparedness

! Danger signs in pregnancy – turant hospital

- **Vaginal bleeding** (any amount) · **Leaking of fluid (PROM)**
- **Severe headache, blurring of vision, epigastric pain, swelling of face/ hands** (pre-eclampsia)
- **Convulsions/ fits** (eclampsia) · High fever · Severe abdominal pain
- **Decreased/ absent fetal movements** (< 10 in 12 hrs/ DFMC) · Breathlessness, palpitations
- Severe vomiting · Burning micturition · Labour pains before 37 wks

- **Birth Preparedness & Complication Readiness (BPCR)**: identify **place of delivery** (institution), **skilled birth attendant**, **transport (108/ 102 – Janani Express)**, **money**, **blood donor**, companion, documents (MCP card), know danger signs.
- **Daily fetal movement count (DFMC/ Cardiff "count to 10")** – from **28 wks**; **≥ 10 movements in 12 hrs** normal (or ≥ 3 per hour after meals – Sadovsky); < 10 → report.
- **Health education topics**: diet, rest (**2 hrs day + 8 hrs night, left lateral**), hygiene, exercise (walking), **avoid heavy work & long travel**, **sexual intercourse – avoid in first trimester & last 6 wks** (and in threatened abortion, praevia), clothing (loose, low heels), breast care (nipple – correct flat/ inverted nipples – Hoffman's exercise/ syringe method – not routinely), **drugs only on advice**, X-ray avoid, preparation for breastfeeding, family planning, birth spacing.

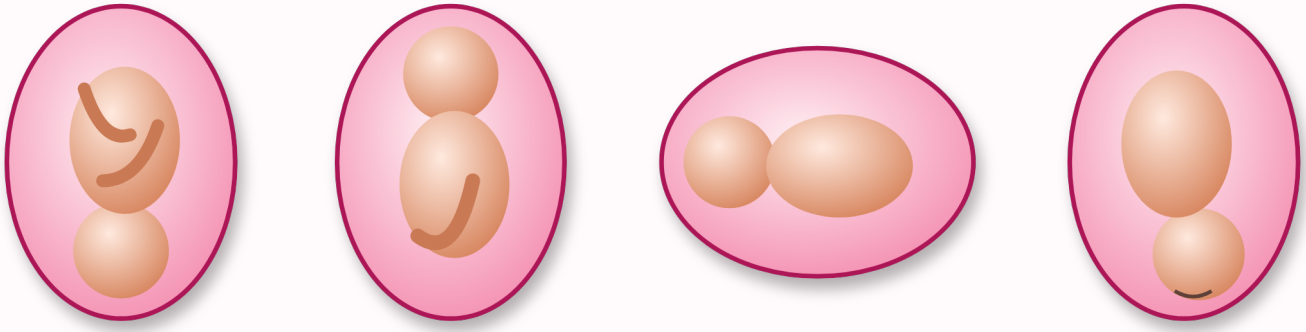
PYQ style

- First symptom of pregnancy – **amenorrhoea**.
- Positive sign of pregnancy – **FHS/ fetal parts/ USG fetus**.
- Hegar's sign – **softening of isthmus (6–10 wks)**.
- Quickening primi – **18–20 wks**.
- Naegele's rule – **LMP + 9 months + 7 days**.
- Minimum ANC visits (India) – **4**; first visit – **within 12 wks**.
- IFA prophylaxis – **60 mg iron + 500 µg FA, 180 days**.
- Td doses – **2 (4 weeks apart)**.
- GDM screening (DIPSI) – **75 g glucose, 2-hr ≥ 140 mg/dL**.
- Leg cramps relief – **dorsiflexion of foot**.
- Fetal kick count normal – **≥ 10 in 12 hrs**.
- Morning sickness relief – **dry carbohydrate before getting up**.

7 Obstetric Abdominal Examination & Fetal Wellbeing

7.1 Obstetric Terms

Lie & Presentation of Fetus



Cephalic - Vertex
~95–96 % · most common
well-flexed head
LOA – commonest position

Breech
3–4 % at term
Frank (commonest) –
legs extended at knees

Transverse lie
Shoulder presentation
Vaginal delivery NOT
possible → LSCS

Face (extended)
Denominator – mentum
Mento-anterior → vaginal
Mento-posterior → LSCS

Lie = fetus ki long axis vs mother ki long axis (longitudinal / transverse / oblique)
Presentation = part jo pelvis me pehle aaye · **Attitude** = body parts ka relation (flexion)
Denominator: vertex – occiput, face – mentum, breech – sacrum, shoulder – acromion

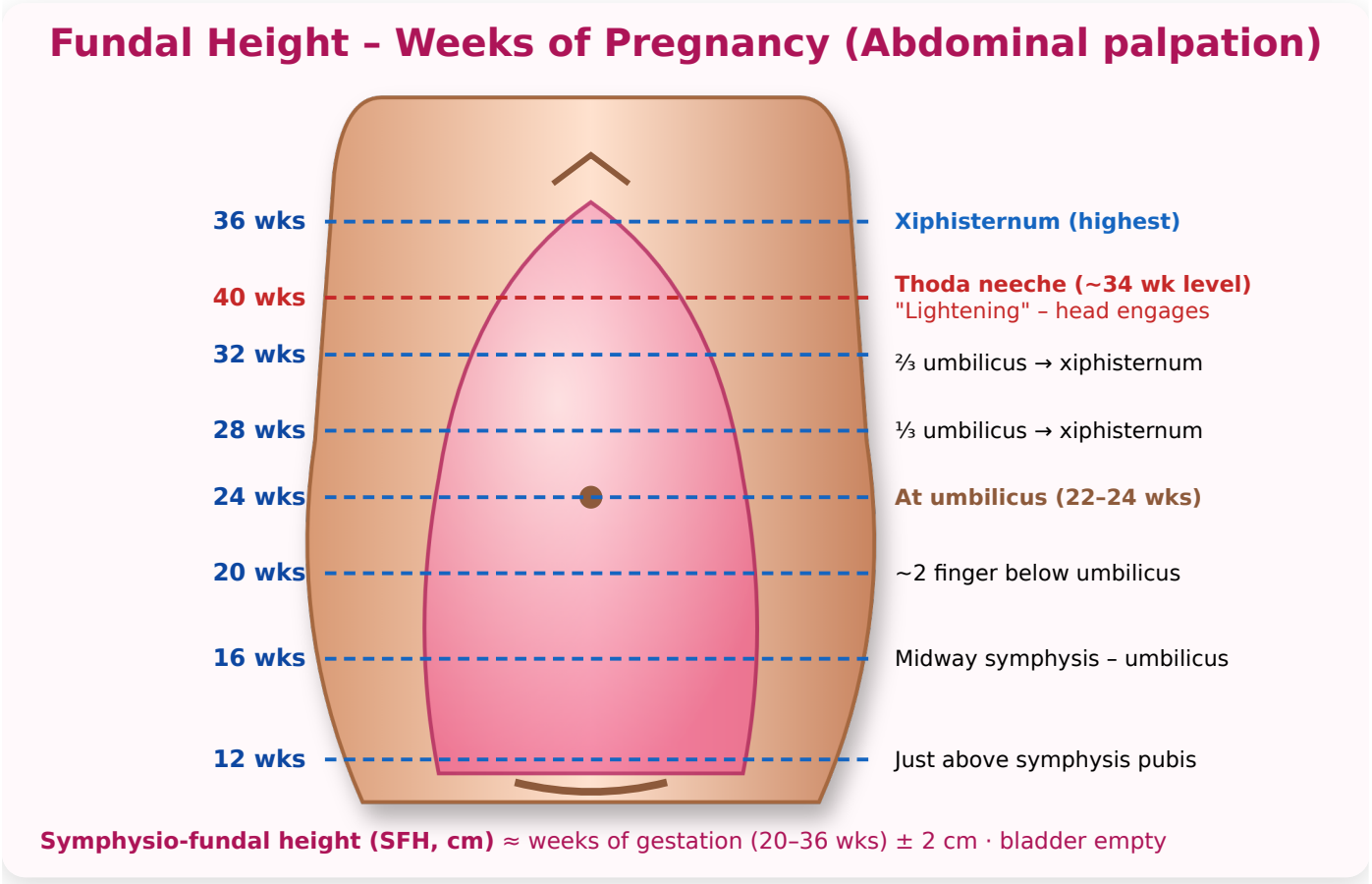
Lie, presentation aur denominator

Term	Meaning
Lie	Relation of **long axis of fetus to long axis of mother/ uterus** – **longitudinal (99 %)**, transverse, oblique (unstable)
Presentation	**Part of fetus lying over the pelvic inlet** (first to enter) – **cephalic (~96 %: vertex, brow, face)**, breech, shoulder, compound
Presenting part	Part felt first on PV (e.g. **vertex – posterior part of parietal bone**)
Attitude	Relation of fetal parts to each other – normal **universal flexion** (head flexed, limbs flexed)
Denominator	Arbitrary bony point to name position – **vertex: occiput**, **face: mentum (chin)**, **brow: frontal eminence**, **breech: sacrum**, **shoulder: acromion**
Position	Relation of **denominator to maternal pelvis** (left/ right, anterior/ posterior/ transverse)
Engagement	**Widest presenting diameter (BPD in vertex) passes through pelvic inlet** – PV: **station 0 (at ischial spines)**; abdominal: **≤ 2/5 palpable**; primigravida **~36–38 wks**, multigravida in labour
Station	Level of presenting part relative to **ischial spines (0)** – **−5 to +5 cm** (+ below spines)
Lightening	Descent of uterus/ head into pelvis **2–3 wks before labour (primi)** → breathing easier, frequency ↑

- **Positions of vertex** (6): **LOA (left occiput anterior – most common)**, ROA, LOT, ROT, **LOP, ROP (occipito-posterior – ~10–20 %, prolonged labour)**. **Direct OA/ OP** – rare.
- **Fifths of head palpable (Crichton)**: 5/5 – floating above brim; 3/5; **2/5 – engaged**; 0/5 – deep in pelvis.

7.2 Abdominal Palpation

- **Preparation**: explain, **empty bladder**, privacy, **supine with knees slightly flexed (dorsal)**, warm hands, abdomen exposed (xiphisternum to symphysis); avoid prolonged supine (left tilt).
- **Inspection**: size, shape (ovoid – longitudinal; **broad – transverse**), linea nigra, striae, scars (previous LSCS), fetal movements, **flattening below umbilicus – OP**, **pendulous abdomen (grand multipara)**.
- **Fundal height** – with ulnar border of hand (weeks) & **symphysio-fundal height (SFH) tape (cm) ≈ gestational weeks ± 2** (20–36 wks).



Fundal height – kitne hafte me kahan

- **Fundal height > dates**: wrong dates, **multiple pregnancy**, **polyhydramnios**, macrosomia (diabetes), **molar pregnancy**, fibroids. **< dates**: wrong dates, **IUGR**, **oligohydramnios**, IUD, transverse lie.



Leopold ke 4 grips

Leopold's manoeuvres (4 grips)

1. **Fundal grip** – what occupies fundus → **lie & presentation** (**breech – broad, soft, irregular, not ballotable**; head – hard, round, ballotable).
2. **Lateral (umbilical) grip** – back (smooth, continuous resistance) vs limbs (small knobs) → **position**; **FHS best heard over back**.
3. **Pawlik's grip (first pelvic grip)** – lower pole grasped with **one hand (thumb & fingers)** → **presenting part & engagement** (head mobile = not engaged). **Painful – done gently**.
4. **Pelvic grip (second pelvic)** – **examiner faces patient's feet**, both hands → **flexion (attitude), descent, engagement** (cephalic prominence/ occiput).

- **Auscultation of FHS**: **Pinard fetoscope**/ stethoscope/ **Doppler (handheld)**; **count 1 full minute**; **normal 110–160 bpm (120–160)**, regular; distinguish from maternal pulse (feel radial simultaneously) & **uterine souffle** (maternal – synchronous with maternal pulse) vs **funic souffle** (cord – synchronous with FHS).
  - **Location**: **vertex anterior – below umbilicus on side of back (spinoumbilical line)**; **OP – flank, lateral**; **breech – at/ above umbilicus**; face – on side of limbs (chest).
  - **Fetal bradycardia < 110**, **tachycardia > 160** → fetal distress (hypoxia, maternal fever).

7.3 Tests of Fetal Wellbeing

Test	Details	Interpretation
Daily Fetal Movement Count (DFMC)	From **28 wks**; **Cardiff "count to 10"**	**< 10 movements in 12 hrs** → NST/ report; **fetal movements stop ~12–48 hrs before IUD** ("movement alarm signal")
NST (Non-Stress Test)	CTG for **20 min (extend to 40)** – FHR accelerations with fetal movement; after 28–32 wks	**Reactive (normal)** – **≥ 2 accelerations of ≥ 15 bpm lasting ≥ 15 sec in 20 min** ("15 × 15"); **Non-reactive** → BPP/ CST
CST/ OCT (Contraction Stress Test – oxytocin challenge)	**3 contractions in 10 min** (oxytocin/ nipple stimulation)	**Negative (normal)** – no late decelerations; **Positive** – **late decelerations with ≥ 50 % contractions** → uteroplacental insufficiency
BPP (Biophysical Profile – Manning)	USG 30 min, **5 parameters × 2 = 10**: **NST, Fetal breathing movements, Gross body movements, Fetal tone, Amniotic fluid (AFI/ SDP)**	**8–10 normal**; **6 equivocal** (repeat 24 hrs); **≤ 4 abnormal – deliver**. **Modified BPP = NST + AFI**
Doppler velocimetry	**Umbilical artery S/D ratio, MCA, ductus venosus**	**Absent/ reversed end-diastolic flow (AEDF/ REDF)** → IUGR, deliver; **MCA PSV ↑ → fetal anaemia (Rh)**
USG	Dating (CRL), NT (11–14 wks; > 3 mm – Down), **anomaly scan 18–20 wks**, growth (BPD, HC, AC, FL), AFI, placenta	**AC (abdominal circumference) – best for IUGR**

Test	Details	Interpretation
Amniocentesis	**15–18 wks** (genetic – karyotype), later – lung maturity, Rh (ΔOD 450); **USG-guided**, **empty bladder (> 20 wks)**/ full (< 20)	Risk – abortion **0.5–1 %**, infection, Rh sensitization (give **anti-D** if Rh –ve)
CVS (Chorionic villus sampling)	**10–13 wks** – earliest prenatal genetic diagnosis	**Limb reduction defects if < 10 wks**, abortion 1 %
Cordocentesis (PUBS)	> 18 wks – fetal blood	Fetal anaemia, karyotype
Lung maturity	**L/S (lecithin/sphingomyelin) ratio ≥ 2** = mature (~35 wks); **phosphatidylglycerol (PG)** present (best) ; shake/ foam test	Steroids if immature
Maternal serum screening	Double marker (**PAPP-A, free β-hCG** – 11–14 wks), Triple (**AFP, hCG, uE3** – 15–20), Quadruple (+ **inhibin A**)	**Down: AFP ↓, uE3 ↓, hCG ↑, inhibin ↑; NTD: AFP ↑**
NIPT (cell-free fetal DNA)	From 10 wks	Screening for trisomies

CTG – intrapartum FHR patterns

- **Baseline 110–160**, **variability 5–25 bpm** (most important sign of fetal wellbeing), accelerations present.
- **Early deceleration** – mirrors contraction (**head compression – benign**).
- **Variable deceleration** – V/U-shaped, abrupt, variable timing (**cord compression**) → **change position**, amnioinfusion.
- **Late deceleration** – begins after peak of contraction, slow return (**uteroplacental insufficiency – ominous**) → **left lateral, O_2, stop oxytocin, IV fluids, inform – expedite delivery**.
- **Sinusoidal pattern** – severe fetal anaemia (Rh), hypoxia.

VEAL CHOP

Variable – **C**ord compression · **E**arly – **H**ead compression · **A**ccelerations – **O**K (oxygenated) · **L**ate – **P**lacental insufficiency

PYQ style

- Most common lie – **longitudinal**; presentation – **cephalic (vertex)**; position – **LOA**.
- Denominator of face – **mentum**; breech – **sacrum**.
- Engagement means – **BPD has passed the inlet**; station **0** = at ischial spines.
- Pawlik's grip – **presenting part & engagement**.
- 4th Leopold – examiner faces **patient's feet**.
- FHS heard best over – **fetal back**; normal **110–160 bpm**.
- Reactive NST – **2 accelerations of 15 bpm × 15 sec in 20 min**.
- BPP parameters – **5** (NST, breathing, movement, tone, liquor).
- Late deceleration – **uteroplacental insufficiency**; early – **head compression**.
- Lung maturity – **L/S ratio ≥ 2**.
- CVS done at – **10–13 wks**; amniocentesis – **15–18 wks**.

8 Normal Labour – Stages, Mechanism & Management

8.1 Labour – Basics

D Normal labour (eutocia)

Labour with **spontaneous onset at term (37–42 wks)**, **vertex presentation**, **without undue prolongation**, natural termination with **minimal aid**, **without complications** to mother & baby. Labour = series of events by which **products of conception (fetus, placenta, membranes)** are expelled from uterus. **Dystocia** = difficult labour.

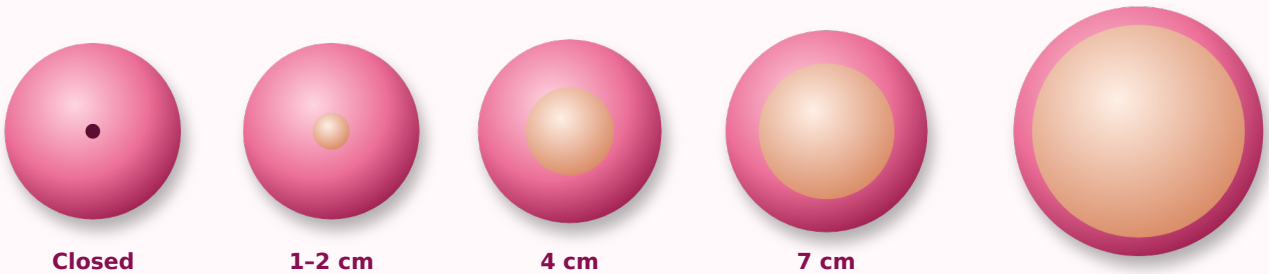
- **Causes/ theories of onset**: **oxytocin theory** (receptors ↑), **progesterone withdrawal**, **oestrogen ↑** (↑ gap junctions & oxytocin receptors), **prostaglandins (PGE2, PGF2α)**, **fetal cortisol (fetal HPA axis)**, **uterine distension/ stretch**, Ferguson reflex (cervix stretch → oxytocin).
- **Premonitory signs (pre-labour)**: **lightening** (2–3 wks before in primi), **frequency of micturition**, **show (blood-stained mucus plug)**, **false pains**, **cervical ripening**, slight weight loss, burst of energy ("nesting").

True labour	False labour
Regular, painful contractions, ↑ frequency, duration & intensity	**Irregular**, no progressive change
Pain **back → front (abdomen)**, radiates to thighs	Pain **lower abdomen** only
Not relieved by rest/ sedation/ enema	**Relieved by rest/ sedation/ walking change**, enema
Cervical effacement & dilatation (progressive)	**No cervical change**
Show present	No show
Formation of **bag of waters**	No

- **3 P's (+ more)**: **Powers** (uterine contractions + voluntary bearing down), **Passage** (bony pelvis + soft tissues), **Passenger** (fetus + placenta); + **Psyche**, **Position**.
- **Uterine contractions**: begin at **fundus near cornua (pacemaker)**, **fundal dominance**, spread downward; **retraction** (fibres don't return to original length – permanent shortening of upper segment) → **physiological retraction ring**; **pathological (Bandl's) ring** – obstructed labour.
- Contraction: **intensity 40–60 mmHg** (2nd stage up to 100); frequency – early 10–15 min apart → active **every 2–3 min**, duration **45–60 sec (up to 90)**. **Montevideo units > 200** = adequate.

8.2 Stages of Labour

Cervical Effacement & Dilatation (1st stage of labour)



Closed
long, thick

1–2 cm
effacing

4 cm
active phase

7 cm
active

10 cm = FULL
2nd stage starts

Latent phase: 0 → 4 cm (WHO 2018: active starts at 5 cm) · slow
Active phase: ~1 cm/hr (primi) · 1.5 cm/hr (multi) – partograph
Effacement: cervix ka chhota/ patla hona (%) – primi: effacement pehle, phir dilatation; multi: dono saath
Stages: I – onset → full dilatation · II – full dilatation → baby birth · III – baby → placenta · IV – 1 hr after placenta
Duration: 1st stage primi 12 hrs, multi 6 hrs · 2nd stage primi ~1 hr (max 2), multi 30 min · 3rd stage 15–30 min

Cervix ka effacement aur dilatation

Stage	From → To	Primigravida	Multigravida
1st stage (cervical stage)	Onset of true labour → **full dilatation (10 cm)**	**~12 hrs** (latent up to 20 hrs)	**~6 hrs** (latent up to 14 hrs)
2nd stage (stage of expulsion)	Full dilatation → **birth of baby**	**~1 hr (max 2 hrs; 3 with epidural)**	**~30 min (max 1 hr; 2 with epidural)**
3rd stage (placental stage)	Birth of baby → **expulsion of placenta & membranes**	**15–30 min** (5–15 with AMTSL)	Same
4th stage (observation)	**1 hr (to 2 hrs) after placenta – PPH watch**		

- **1st stage phases**: **Latent** – slow, up to **4 cm** (WHO 2018/ LCG – **5 cm**); **Active** – **4 (5) cm → 10 cm**, **~1 cm/hr** (primi), 1.5 cm/hr (multi); (Friedman – acceleration, maximum slope, deceleration).
- **Rupture of membranes** – usually end of 1st stage; **early ROM** (before full dilatation, in labour), **PROM** (before labour onset).
- **2nd stage signs**: **urge to push (bear down)**, **rectal pressure/ bulging perineum**, **anal gaping**, crowning, show ↑, vomiting; **confirm by PV (full dilatation)**.
- **Crowning** – **biparietal diameter distends vulva, head no longer recedes between contractions**.

8.3 Mechanism of Labour (Vertex – LOA)

Cardinal movements – "Every Darn Fool In Egypt Eats Raw Eggs"

Engagement → **D**escent → **F**lexion → **I**nternal rotation → **E**xtension → **R**estitution → **E**xternal rotation → **E**xpulsion

(Descent throughout labour; flexion → smallest diameter suboccipito-bregmatic 9.5 cm presents.)

Movement	What happens
Engagement	Sagittal suture in **transverse/ oblique** diameter of inlet (widest); primi before labour
Descent	Throughout – contractions, bearing down
Flexion	Head meets resistance (cervix, pelvic floor) → chin to chest → **suboccipito-bregmatic 9.5 cm**
Internal rotation	Occiput rotates **1/8 of circle (45°) anteriorly** (LOA) at **pelvic floor (levator ani)** → **sagittal suture in AP diameter** of outlet; occiput under symphysis
Crowning & Extension	Occiput escapes under symphysis (**nape of neck pivots – suboccipital region**) → head born by **extension** – vertex, brow, face, chin sweep perineum
Restitution	Head **untwists 1/8 circle** (45°) to realign with shoulders (opposite direction of internal rotation)
External rotation	Shoulders rotate internally to AP of outlet → head turns further **1/8 circle** (occiput towards **mother's thigh of the side of original position – left in LOA**)
Expulsion	**Anterior shoulder** under symphysis → **posterior shoulder** over perineum (lateral flexion) → trunk

8.4 Management of 1st Stage

- **Admission**: history, vitals, abdominal exam, FHS, **PV examination** (dilatation, effacement, station, membranes, presenting part, position, moulding, pelvis) – **4-hourly PV** (avoid frequent – infection), **strict asepsis**.
- **Partograph** – started at **active phase (4 cm/ 5 cm)**.



Partograph – alert line & action line

- **Monitoring**: **FHS every 30 min (1st stage; 15 min in active high-risk) & after contraction for 1 min**; **contractions every 30 min** (frequency/ 10 min, duration); **pulse ½ hourly**; **BP & temperature 4 hourly**; **urine output, protein, acetone** – **empty bladder every 2 hrs** (full bladder delays descent); **PV 4 hourly**; liquor colour.
- **Care**: **ambulation & upright positions** (encouraged in latent & early active), **left lateral** when lying; **oral fluids/ light diet** (clear fluids – low-risk); IV fluids if needed; **birth companion** (LaQshya/ respectful maternity care); **no routine enema, no routine shaving** (WHO); **pain relief** – breathing, massage, back rub, warm compress, **epidural (gold standard)**, pethidine (avoid near delivery – neonatal respiratory depression); emotional support, privacy.
- **Avoid**: routine ARM/ oxytocin, frequent PV, **fundal pressure**, lithotomy for long.

8.5 Management of 2nd Stage

- **Position** – **upright/ semi-recumbent/ lateral/ squatting** (WHO – woman's choice); traditional lithotomy/ dorsal.
- **Bearing down** – with contractions, **open glottis pushing** preferred; avoid prolonged Valsalva.
- **FHS every 5 min (after each contraction)**.
- **Perineal care** – **perineal support (Ritgen's manoeuvre)** – control head extension (**slow delivery of head between contractions**), **flex head**, protect perineum; **warm compress**.
- **Episiotomy** – **restrictive (not routine)** – indications below.

- After head delivered: **wipe/ clear airway** (suction only if needed – not routine), **feel for cord round neck** (loose → slip over head; tight → **double clamp & cut**), wait for **restitution & external rotation** → **deliver anterior shoulder (gentle downward traction)** → posterior shoulder (upward) → body.
- **Immediate newborn care** – dry, warm, **skin-to-skin**, **delayed cord clamping (1–3 min)**, assess breathing; note **time of birth & sex**.

8.6 Management of 3rd Stage

- **Signs of placental separation**: (1) **Gush of blood**, (2) **lengthening of cord** (permanent), (3) **uterus rises & becomes globular, firm, smaller** (fundus rises above umbilicus), (4) **suprapubic bulge**; (Küstner's/ Strassmann's/ Klein's signs – cord doesn't recede on suprapubic pressure).

AMTSL – Active Management of Third Stage of Labour (PPH prevention – most effective)

1. **Uterotonic within 1 minute of birth** (after ruling out 2nd baby): **Inj. Oxytocin 10 IU IM** (drug of choice); alternatives – **misoprostol 600 µg oral**, carbetocin, ergometrine (not in HTN/ heart disease)
2. **Controlled Cord Traction (CCT)** with **counter-traction** (Brandt-Andrews – hand above symphysis pushes uterus up) – **only with contraction**; never without counter-traction (**inversion**)
3. **Uterine massage** after placenta delivery (**every 15 min for 2 hrs**) + check uterus remains contracted
   - **Delayed cord clamping (1–3 min)** – now part of essential care (not "early clamping").
   - Examine **placenta & membranes for completeness**; inspect **perineum & vagina** for tears; estimate blood loss (**normal < 500 ml vaginal**).

- **Delivery of placenta**: **Brandt-Andrews** (CCT), Crede's method (fundal squeezing – old, not recommended), **manual removal** (retained > 30 min with AMTSL/ bleeding – under anaesthesia).

8.7 Fourth Stage & Immediate Postpartum

- Monitor **every 15 min for 1 hr**, then every 30 min for next hour: **pulse, BP, uterine tone (fundus firm, at/ below umbilicus), vaginal bleeding (pads), bladder (full bladder → atony)**, perineum.
- **Breastfeeding within 1 hour**, skin-to-skin, warmth, vitamin K & vaccines for baby, record.

8.8 Episiotomy & Perineal Tears

- **Episiotomy (perineotomy)** – surgical incision of perineum to **enlarge the outlet**.
- **Indications** (restrictive): **rigid perineum** (elderly primi), anticipated tear, **instrumental delivery (forceps/ vacuum)**, **breech delivery**, **shoulder dystocia**, **fetal distress (expedite)**, face-to-pubis, previous perineal repair/ 3rd degree tear, preterm (protect head).
- **Timing** – when perineum **thinned out & bulging, ~3–4 cm of head visible** during contraction (**just before crowning**); local anaesthesia (**1 % lignocaine** – pudendal block / local infiltration).
- **Types**: **Mediolateral (most common, preferred in India)** – from midpoint of fourchette **45° (60° at crowning) laterally** – **less risk of extension to anal sphincter**; **Median/ midline** – easy repair, less blood loss, better healing, **but risk of 3rd/4th degree extension**; lateral (obsolete); J-shaped.
- **Repair** (**episiorrhaphy**) – in 3 layers: **vaginal mucosa (continuous, starting 1 cm above apex)** → **perineal muscles (interrupted)** → **skin (subcuticular/ interrupted)**; **absorbable suture – Vicryl rapide/ chromic catgut 2-0/ 3-0**; check **rectum after repair** (stitch through rectal mucosa).

- **Care**: perineal hygiene (front to back), **sitz bath/ warm water**, **ice packs first 24 hrs**, analgesics, change pads, **Kegel exercises**, high fibre (avoid straining), watch for haematoma (severe pain), infection, dehiscence; sutures absorb in 2–3 wks.
- **Perineal tears**:
  - **1st degree** – vaginal mucosa & perineal skin only
  - **2nd degree** – + **perineal muscles** (≈ episiotomy)
  - **3rd degree** – + **anal sphincter** (3a < 50 % EAS, 3b > 50 % EAS, 3c IAS)
  - **4th degree** – + **anal/ rectal mucosa** → repair in OT; stool softeners, antibiotics; risk of **faecal incontinence, rectovaginal fistula**

PYQ style

- 1st stage of labour ends at – **full cervical dilatation (10 cm)**.
- 2nd stage primi – **~1 hr (max 2)**; 3rd stage – **15–30 min**; 4th stage – **1 hr after placenta**.
- True labour pain – **back to front, not relieved by rest, cervical dilatation**.
- Internal rotation takes place at – **pelvic floor**; occiput rotates – **1/8 of circle** (LOA).
- Smallest presenting diameter – **suboccipito-bregmatic 9.5 cm**.
- Partograph alert line – **1 cm/hr**; action line – **4 hrs to the right**.
- FHS in 2nd stage – **every 5 min**; 1st stage – **every 30 min**.
- AMTSL drug – **oxytocin 10 IU IM within 1 min**.
- Sign of placental separation – **lengthening of cord, gush of blood, uterus globular & rises**.
- Commonest episiotomy – **mediolateral**; timing – **at crowning/ head visible 3–4 cm**.
- 3rd degree tear involves – **anal sphincter**.

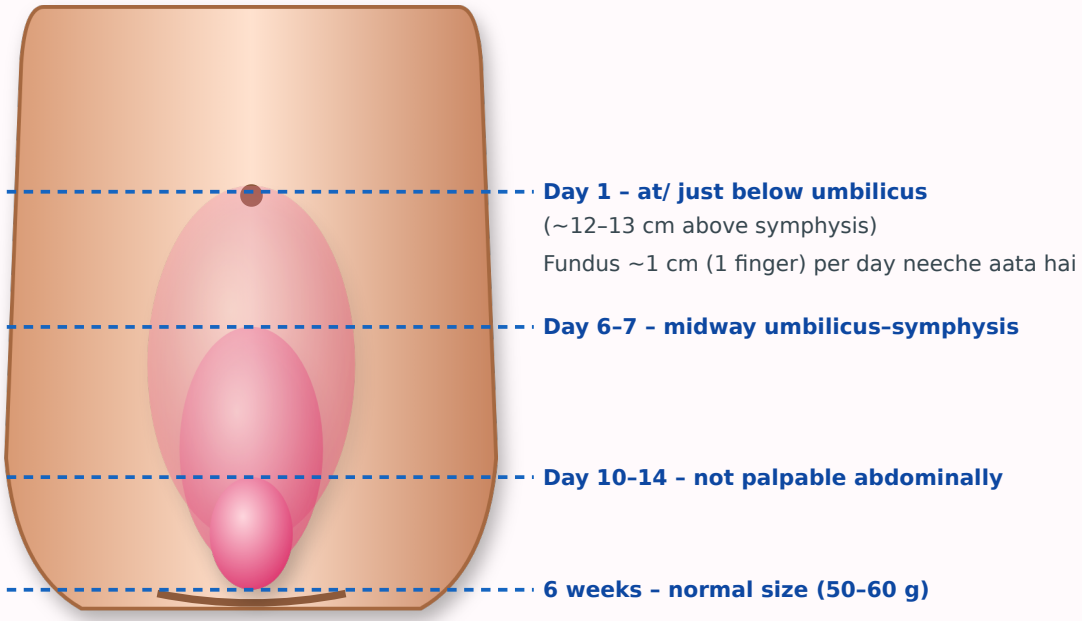
9 Normal Puerperium, Postnatal Care & Breastfeeding

9.1 Puerperium

D Puerperium

Period **following delivery (expulsion of placenta)** during which the **reproductive organs return to pre-pregnant state** – **6 weeks (42 days)**. **Immediate** (first 24 hrs), **Early** (up to 7 days), **Remote/ late** (up to 6 weeks).

Involution of Uterus & Lochia (Puerperium – 6 weeks)



Lochia rubra (red) day 1–3/4 · **Lochia serosa** (pink-brown) day 4–10
Lochia alba (yellow-white) day 10 – 6 weeks
Foul smell → infection · Rubra persist / bright red clots → subinvolution / retained bits

Uterus ka involution aur lochia ke types

Involution of uterus

- Immediately after delivery – fundus **~at/ just below umbilicus (13.5 cm above symphysis)**, weight **~1000 g**, firm.
- Descends **~1 cm (1 finger breadth)/ day**; **not palpable abdominally by 10th–14th day**; **normal size (50–60 g) by 6 weeks**.
- Weight: 1000 g (birth) → **500 g (1 week)** → **300 g (2 weeks)** → **50–60 g (6 weeks)**.
- Mechanism: **autolysis** (proteolytic enzymes – muscle fibres shrink), **ischaemia** (living ligatures), **phagocytosis**; **after-pains** (contractions – multipara, during breastfeeding – oxytocin).
- **Subinvolution** – slow involution → causes: **retained bits**, **infection (endometritis)**, full bladder, fibroids, no breastfeeding, overdistension (twins); signs – fundus higher than expected, boggy, **persistent lochia rubra**, PPH (secondary).

- **Placental site** heals by exfoliation (no scar) by ~6 wks; **endometrium regenerates** – 3 weeks (except placental site).
- **Cervix** – external os **closes by ~1 week (admits 1 finger)**, becomes **transverse slit** (parous); **vagina** – rugae reappear by 3 wks.

Lochia

Type	Days	Colour & content
Lochia rubra	**1–4**	**Red** – blood, shreds of decidua, vernix, lanugo, meconium
Lochia serosa	**5–9 (4–10)**	**Pinkish-brown/ yellowish** – less RBC, more serum, leucocytes, cervical mucus
Lochia alba	**10 – 15 days (up to 6 wks)**	**Yellowish-white/ creamy** – leucocytes, decidual cells, mucus

- Normal lochia – **fleshy, earthy (stale) odour**, alkaline, total ~250 ml. **Foul smell → infection**; **persistent rubra/ return of red lochia → subinvolution/ retained products**; **scanty/ absent with fever → infection (lochiometra)**.

Other physiological changes

- **Vital signs** – temp slightly ↑ (< 38 °C in first 24 hrs – reactionary); **> 38 °C on 2 occasions (24 hrs apart) after first 24 hrs within 10 days = puerperal pyrexia** → investigate; pulse **normal/ slight bradycardia (50–70) first days** – tachycardia → PPH/ infection.
- **Diuresis** (day 2–5) – fluid excretion; urinary retention possible (oedema, epidural, trauma) – **void within 6 hrs**.
- **Weight loss** ~5–6 kg at delivery, further 2–3 kg in puerperium.
- **Blood** – leucocytosis, ↑ fibrinogen → **DVT risk (first 2 wks – highest)**.
- **Menstruation return** – non-lactating **6–8 weeks**; lactating – delayed (**LAM**); **ovulation may precede first period** → contraception.
- **Postpartum blues ("baby blues")** – **day 3–5**, 50–80 % mothers, weepy, mood swings, **self-limiting (≤ 2 wks)** → reassurance. **Postpartum depression** (10–15 %, > 2 wks) & **postpartum psychosis** (1–2/1000, first 2 wks – emergency) – see complications chapter.

9.2 Postnatal Care

- **Postnatal check-ups (India – MoHFW)**: **1st – within 24 hrs (at facility; stay 48 hrs after normal delivery)**, **2nd – day 3**, **3rd – day 7**, **4th – 6 weeks** (+ HBNC ASHA visits 3, 7, 14, 21, 28, 42).
- **WHO**: at least **4 PNC contacts** – first 24 hrs, day 3, day 7–14, 6 weeks.

Postnatal nursing care – "BUBBLE-HE"

Breasts (engorgement, nipples, feeding) · **U**terus (fundal height, tone) · **B**ladder (voiding within 6–8 hrs, output) · **B**owels (constipation – fibre; first BM by day 2–3) · **L**ochia (amount, colour, odour) · **E**pisiotomy/ perineum (**REEDA** – Redness, Oedema, Ecchymosis, Discharge, Approximation) · **H**omans' sign/ legs (DVT) · **E**motions (bonding, blues)

- **Vital signs** – first hour every 15 min, then 4 hrly first 24 hrs, then BD.

- **Early ambulation – within 6–8 hrs** (normal delivery) – prevents DVT, ↑ bladder & bowel function, involution.
- **Diet** – normal, nutritious, **+600 kcal**, fluids ≥ 3 L, iron-folic acid (**180 days**), **calcium (6 months)**; no food taboos.
- **Perineal hygiene**, **pads changed every 4–6 hrs**, wash front to back; sitz bath.
- **Rest** – sleep when baby sleeps; **Postnatal exercises** – **deep breathing, Kegel (pelvic floor), leg exercises (DVT)** from day 1; **abdominal exercises (head lift, pelvic tilt)** after 24 hrs/ 2nd day (normal delivery); after LSCS – gradually.
- **Family planning counselling** – **PPIUCD (within 48 hrs/ at LSCS)**, LAM, POP (progestin-only pills – from 6 wks), condoms, **tubectomy (within 7 days postpartum – minilap)**; **combined OCP avoid in lactation < 6 months**.
- **Immunization** of baby, **MCP card**; **danger signs** (heavy bleeding, foul lochia, fever, severe headache/ fits, breast pain with fever, calf pain, breathlessness, depression); **JSY payment**, birth registration (within 21 days).
- **Postnatal sexual activity** – after lochia stops & perineum heals (~6 wks).

9.3 Lactation & Breastfeeding

Physiology

- **Mammogenesis** (breast growth – pregnancy) → **Lactogenesis I** (colostrum from 2nd trimester) → **Lactogenesis II** (copious milk **day 2–4** after placenta delivery – progesterone ↓, prolactin acts) → **Galactopoiesis** (maintenance – **suckling**).
- **Prolactin reflex (milk secretion/ production reflex)** – suckling → **anterior pituitary → prolactin** → milk production (more at **night**; **more suckling = more milk** – "demand & supply").
- **Let-down reflex (milk ejection/ oxytocin reflex)** – suckling/ baby's cry/ sight → **posterior pituitary → oxytocin → myoepithelial cells contract** → milk ejection; **inhibited by stress, pain, anxiety, embarrassment**; also causes **uterine contraction (after-pains) → involution**.

Milk	Time	Features
Colostrum	**First 2–4 days** (10–40 ml/day)	**Yellow, thick**, **high protein, Vit A, IgA (secretory), lactoferrin, leucocytes**, **less fat & lactose**; **"first immunisation"**; laxative (meconium passage, ↓ jaundice)
Transitional milk	Day 4–14	
Mature milk	After 2 weeks (~600–800 ml/day)	**Foremilk** – watery, thirst-quenching (lactose, water); **Hindmilk** – **rich in fat**, satiety, weight gain → **empty one breast fully before switching**

- Breast milk composition (per 100 ml): **energy ~67–70 kcal**, protein **1.1 g** (whey : casein = **60 : 40**), fat 3.5–4 g, lactose 7 g, iron low but **50 % absorbed (lactoferrin)**; **low in Vit K & D**.
- Advantages – **baby**: ideal nutrition, **immunity (IgA, lactoferrin, lysozyme, bifidus factor)**, ↓ diarrhoea/ pneumonia/ otitis/ allergy/ NEC, ↓ SIDS, better IQ, bonding, always available, clean & warm; **mother**: **↓ PPH (oxytocin)**, faster involution, **LAM contraception**, ↓ **breast & ovarian cancer**, weight loss, economical.

Breastfeeding recommendations (WHO/ IYCF)

- **Initiate within 1 hour of birth** (normal & LSCS – as soon as possible) – **skin-to-skin contact**.
- **Exclusive breastfeeding for 6 months** – **no water, honey, ghutti, prelacteal feeds** (prelacteals harmful).
- **Complementary feeding from 6 months (180 days)** + **continue breastfeeding up to 2 years or beyond**.
- **On demand (baby-led)** – **8–12 times/ 24 hrs** (at least every 2–3 hrs), day & night; **rooming-in**.
- **Baby-Friendly Hospital Initiative (BFHI – 1991, WHO/UNICEF) – 10 steps**; **MAA programme (2016)**; **IMS Act 1992 (amended 2003)** – bans promotion of infant milk substitutes & feeding bottles.
- **World Breastfeeding Week 1–7 August**.

Positioning & attachment

- **Position**: baby's **head & body in straight line**, **body close to mother (tummy-to-tummy)**, **facing breast – nose opposite nipple**, **whole body supported** (not just head/ shoulders). Holds – **cradle, cross-cradle (preterm), football/ underarm (LSCS, twins)**, side-lying (night, after CS).
- **Good attachment (latch) – 4 signs**: (1) **Chin touching breast**, (2) **mouth wide open**, (3) **lower lip turned outward (everted)**, (4) **more areola visible above than below** the mouth. + **slow deep sucks with pauses, audible swallowing**, no pain.
- **Adequacy of milk**: **urine ≥ 6 times/ day** (after day 5), **weight gain (regains birth weight by 10–14 days; ~20–30 g/day)**, baby satisfied, sleeps 2–3 hrs.
- **Burping** after feeds; breast **not washed with soap** before each feed (normal bathing ok); **apply hindmilk on nipples** after feed (healing).

Breastfeeding problems

Problem	Cause	Management
Sore/ cracked nipples	**Poor attachment (most common)**	**Correct attachment**, start with less sore side, apply **hindmilk**, air dry; continue feeding
Breast engorgement	Milk stasis (day 3–5), infrequent feeds	**Frequent feeding**, **warm compress before feed**, **express milk**, **cold compress after**, supportive bra, analgesics
Blocked duct	Localized stasis	Massage toward nipple, frequent feeding
Mastitis	Infection (Staph aureus) via cracks	**Continue breastfeeding (from both breasts)**, antibiotics (cloxacillin/ flucloxacillin), analgesics, rest, fluids
Breast abscess	Untreated mastitis	**Incision & drainage/ needle aspiration**, antibiotics; feed from healthy side, express affected
Flat/ inverted nipples		Attachment (baby takes areola, not nipple); **syringe method** (inverted syringe suction) after birth; **Hoffman's exercises (antenatal) not recommended routinely**
"Not enough milk"	Perceived; poor attachment/ infrequent feeds	Frequent feeding, correct latch, rest, reassurance, ↑ fluids; galactagogues (no strong evidence)

- **Expressed breast milk (EBM)**: **hand expression** (thumb & index at areolar edge, press back toward chest, compress, release); store **room temp (≤ 25 °C) – 6–8 hrs (4 hrs per some)**, **refrigerator – 24**

hrs (up to 3–5 days), **freezer – 3 months**; feed with **katori-spoon/ paladai** (not bottle).

- **Contraindications to breastfeeding**: **galactosemia (classic – baby)**, maternal **HIV (where safe replacement available; in India – breastfeed with ART)**, **active untreated TB** (may breastfeed after 2 wks of Rx/ with mask; baby INH prophylaxis), **anticancer drugs, radioactive iodine**, lithium, amiodarone; **NOT contraindicated**: hepatitis B (after HBIG/ vaccine), mastitis, common cold, maternal fever.
- **Suppression of lactation** (stillbirth/ medical): tight binder, ice packs, avoid stimulation, **cabergoline** (preferred) – bromocriptine (not routinely – stroke/ MI risk).

PYQ style

- Duration of puerperium – **6 weeks**.
- Uterus not palpable abdominally by – **10–14 days**; normal size by – **6 weeks**.
- Lochia rubra – **1–4 days**; serosa – **5–9**; alba – **10 days to 6 wks**.
- Puerperal pyrexia – **≥ 38 °C** on 2 occasions after 24 hrs within 10 days.
- Postpartum blues – **day 3–5**, self-limiting.
- Hormone for milk production – **prolactin**; ejection – **oxytocin**.
- Colostrum rich in – **protein, IgA, Vit A** (less fat).
- Breastfeeding initiation – **within 1 hr**; exclusive – **6 months**.
- Sign of good attachment – **chin touching breast, more areola above**.
- Most common cause of sore nipple – **poor attachment**.
- Mastitis – **continue breastfeeding**.
- EBM in refrigerator – **24 hrs**.
- Absolute contraindication (baby) – **galactosemia**.

10 Care of the Newborn

10.1 Definitions

Term	Meaning
Neonate	Birth to **28 days**
Early neonatal	First **7 days**
Term	37–42 wks
LBW (Low birth weight)	**< 2500 g** (irrespective of gestation)
VLBW	**< 1500 g**
ELBW	**< 1000 g**
SGA	Weight **< 10th percentile** for gestational age
LGA	> 90th percentile (e.g. diabetic mother)
IUGR/ FGR	Fetus failed to achieve growth potential – **symmetrical** (early – genetic/ infection; head & body both small) vs **asymmetrical** (late – placental insufficiency; **head spared**, AC small – more common)

- **Average Indian newborn**: weight **~2.8–3 kg**, length **50 cm**, **head circumference 33–35 cm** (> chest by 1–2 cm), chest 31–33 cm. **Weight loss up to 10 % in first week**, **regains birth weight by 10–14 days**; **doubles by 5 months**, **triples by 1 year**.

10.2 Immediate / Essential Newborn Care (ENBC)

Immediate care at birth (sequence)

1. **Note time of birth & sex**; call out
2. **Dry the baby immediately with warm towel** (head first), **discard wet towel** – stimulates breathing & prevents hypothermia
3. **Assess breathing/ crying** – if breathing well → **skin-to-skin on mother's abdomen**, cover with dry cloth & cap
4. **Delayed cord clamping 1–3 min** (or when pulsation stops); cut with sterile blade **2–3 cm from abdomen**, clamp/ tie; **dry cord care** (nothing applied)
5. **Initiate breastfeeding within 1 hour**
6. **Eye care** (clean from inner to outer canthus) – prophylaxis per policy
7. **Vitamin K 1 mg IM** (anterolateral thigh) – **0.5 mg if < 1500 g** – prevents **haemorrhagic disease of newborn**
8. **Weigh** (after first feed), **identification band**, examine for anomalies, **APGAR at 1 & 5 min**
9. **Birth dose vaccines** – **BCG** (intradermal, left upper arm, **0.05 ml** if < 1 month; 0.1 ml after 1 month), **OPV-0** (oral 2 drops), **Hepatitis B birth dose** (**within 24 hrs**, IM thigh, 0.5 ml)
10. **Delay bathing – at least 24 hrs** (WHO; Indian – after 24 hrs/ 48 hrs; LBW – later); **don't apply anything on cord/ eyes/ ears**; **no prelacteal feeds**

APGAR score (Virginia Apgar, 1952)

Sign	0	1	2
Appearance (colour)	Blue/ pale all over	Body pink, **extremities blue (acrocyanosis)**	Completely pink
Pulse (heart rate)	Absent	**< 100**	**≥ 100**
Grimace (reflex irritability)	No response	Grimace	Cry/ cough/ sneeze
Activity (muscle tone)	Limp	Some flexion	Active movement
Respiration	Absent	Slow, irregular, weak cry	Good, strong cry

- Done at **1 min & 5 min** (and 10 min if < 7); **7–10 normal**, **4–6 moderate depression**, **0–3 severe asphyxia**. **Resuscitation is NOT delayed for APGAR**. Most babies score 1 for colour (acrocyanosis) → **9 typical**.
- **Heart rate is the most important** sign during resuscitation.

Neonatal resuscitation (NRP/ NSSK – "Golden Minute")

- **Initial steps (within 30 sec)**: **provide warmth (radiant warmer)**, **position – neck slightly extended ("sniffing")**, **clear airway if needed** (mouth before nose – "**M before N**"), **dry**, **stimulate** (flick soles/ rub back), reposition.
- **Not breathing/ gasping or HR < 100 → positive pressure ventilation (PPV) with bag & mask within 1 minute (golden minute)** – rate **40–60 breaths/min**, room air (term) / 21–30 % O_2 (preterm); check chest rise (**MR SOPA** for corrections).
- **HR < 60 after 30 sec effective PPV → chest compressions** (2-thumb technique, lower third sternum, **3 : 1 ratio – 90 compressions + 30 breaths/min**) + **100 % O_2**.
- **HR < 60 despite → adrenaline (epinephrine) 1 : 10,000 – 0.1–0.3 ml/kg IV (umbilical vein)/ 0.5–1 ml/kg ET**.
- **Meconium-stained, non-vigorous baby → no routine intubation & tracheal suction** (current guidelines) – start PPV.
- Bag size – **240–250 ml (self-inflating)**; mask covers chin, mouth, nose.

10.3 Normal Newborn – Features & Physiological Phenomena

- **Vital signs**: **HR 120–160 bpm** (apical), **RR 30–60/min** (abdominal, irregular, periodic breathing pauses < 15–20 sec normal; **apnoea ≥ 20 sec**), **temperature 36.5–37.5 °C** (axillary), BP ~70/45.
- **Skin**: **vernix caseosa** (don't rub off), **lanugo**, **milia** (white spots on nose – sebaceous), **Mongolian spots** (bluish – sacral; fade), **erythema toxicum** (red rash – day 2–3, benign), **stork bites** (naevus simplex), **desquamation** (post-term), **acrocyanosis** (first 24–48 hrs normal), **physiological jaundice**.
- **Head**: moulding, caput, cephalhaematoma, **fontanelles**.
- **Eyes**: subconjunctival haemorrhage (benign), no tears for first weeks.
- **Mouth**: **Epstein's pearls** (white cysts on palate), natal teeth (rare), tongue-tie.
- **Breast engorgement** (both sexes – maternal hormones, day 3–5) – **don't squeeze** ("witch's milk"); **pseudomenstruation** (vaginal bleeding in girls – day 3–7, oestrogen withdrawal), **white vaginal discharge**.
- **Umbilical cord** – **2 arteries, 1 vein**; falls **5–10 days**.
- **Meconium** – first stool, **greenish-black, tarry, odourless** – passed **within 24 hrs (90 %), all by 48 hrs** (delay → Hirschsprung's, meconium ileus – cystic fibrosis, imperforate anus); **transitional stools**

(day 3–5, greenish-yellow); **breastfed – golden yellow, soft, frequent**.

- **Urine** – first void **within 24 hrs** (most by 48 hrs).

Newborn reflexes

Reflex	Elicitation / Response	Disappears
Moro (startle)	Sudden drop of head/ loud noise → **arms abduct & extend, then adduct (embrace)**, cry	**3–6 months**; **asymmetric → brachial plexus palsy (Erb's), clavicle fracture**
Rooting	Stroke cheek → turns head toward stimulus, opens mouth	3–4 months
Sucking	Object in mouth → sucks	~4 months (becomes voluntary)
Palmar grasp	Finger in palm → grasps	**3–4 months**
Plantar grasp	Pressure on sole → toes curl	**8–12 months**
Babinski	Stroke sole → **toes fan, big toe dorsiflexes** (positive normal)	**12–24 months**
Tonic neck (fencing)	Head turned → arm & leg extend on that side, flex opposite	4–6 months
Stepping/ walking	Held upright, feet touching surface → stepping	2 months
Gallant (trunk incurvation), **Blink**, **Swallowing**, **Gag**, **Sneeze**		

10.4 Thermal Protection

- Newborn loses heat fast (large surface area, thin skin, little fat) – **evaporation (wet), conduction (cold surface), convection (draught), radiation (cold walls)**; heat produced by **brown fat (non-shivering thermogenesis)** – newborns **can't shiver**.
- **Warm chain (WHO 10 steps)**: warm delivery room (**25–28 °C**), immediate drying, **skin-to-skin**, breastfeeding, **postpone bathing & weighing**, appropriate clothing & bedding (cap, socks), mother & baby together (rooming-in), warm transport, warm resuscitation, training.
- **Hypothermia**: **cold stress 36–36.4 °C**, **moderate 32–35.9 °C**, **severe < 32 °C** → skin-to-skin (KMC), radiant warmer, warm room; signs – cold extremities, lethargy, poor feeding, **hypoglycaemia**, apnoea, **sclerema**.
- **Temperature check**: axillary **3 min** (digital as per device); feel **abdomen & feet** (warm abdomen + cold feet = cold stress).

10.5 Neonatal Jaundice

	Physiological	Pathological
Onset	**After 24 hrs (day 2–3)**	**Within first 24 hrs**
Peak	**Day 3–5 (term)**; day 5–7 (preterm)	Any

	Physiological	Pathological
Serum bilirubin	**< 12–15 mg/dL (term)**; rise < 5 mg/dL/day	**> 15 mg/dL**, rise > 5 mg/dL/day, **conjugated > 2 mg/dL**
Duration	**Disappears by 10–14 days (term)**, 3 wks (preterm)	**> 2 weeks (term)**, > 3 wks (preterm)
Cause	↑ RBC breakdown (HbF, short RBC life 70–90 days), **immature liver (glucuronyl transferase ↓)**, ↑ enterohepatic circulation	**Rh/ ABO incompatibility (haemolysis)**, **G6PD deficiency**, sepsis, hypothyroidism, **biliary atresia (clay stools, dark urine)**, breast milk jaundice

- **Kramer's rule** (cephalocaudal spread): face (~5 mg) → chest (~10) → abdomen/ thigh (~12) → legs (~15) → **palms & soles (> 15 mg/dL)**.
- **Treatment**: **adequate breastfeeding (8–12/day)**, **phototherapy (blue light 425–475 nm)** – converts bilirubin to water-soluble **lumirubin** (excreted); **exchange transfusion** (severe – > 20–25 mg/dL, rising; **double volume – 160–170 ml/kg**).
- **Phototherapy nursing care**: **eyes covered (eye shields – retinal damage)**, **genitals covered (diaper – gonads)**, baby **naked** (maximum skin exposed), **change position every 2–3 hrs**, distance **30–45 cm**, **monitor temperature** (hyper/ hypothermia), **↑ fluids/ breastfeeding (insensible loss ↑ 20 %)**, daily weight, stool (green, loose – normal), skin rash (normal), **bronze baby syndrome** (conjugated hyperbilirubinaemia – stop), check bilirubin 12–24 hrly; **sunlight exposure not recommended**.
- **Kernicterus** – unconjugated bilirubin deposits in **basal ganglia** → lethargy, poor suck, **opisthotonus, high-pitched cry, seizures**, sequelae – **athetoid cerebral palsy, deafness**, gaze palsy.

10.6 Preterm & LBW Baby

- **Preterm features**: **< 37 wks**, weight < 2500 g, **large head relative to body**, thin shiny red/ gelatinous skin, **plenty of lanugo & vernix**, **soft ear cartilage (no recoil)**, **flat/ absent breast nodule (< 5 mm)**, **few sole creases (anterior ⅓ only)**, **testes undescended**/ **labia majora don't cover minora**, **poor tone (extended limbs – "frog position")**, weak cry & suck, **poor reflexes**. **New Ballard score** – gestational age assessment.
- **Problems**: **hypothermia**, **hypoglycaemia (< 45 mg/dL)**, **RDS (hyaline membrane disease – surfactant deficiency)**, **apnoea of prematurity (caffeine)**, infections, **jaundice**, feeding difficulty, **NEC**, **IVH**, **ROP (retinopathy of prematurity – O_2 toxicity → screen at 4 wks / < 34 wks or < 2000 g)**, PDA, anaemia, BPD.
- **Antenatal steroids** (mother – preterm labour 24–34 wks) – **dexamethasone 6 mg IM 12 hrly × 4 doses** or **betamethasone 12 mg IM 24 hrly × 2** → **↓ RDS, IVH, NEC**.
- **Feeding**: **breast milk best**; can't suck → **expressed breast milk by paladai/ katori-spoon** (≥ 32 wks) or **orogastric tube** (< 32 wks); start **60–80 ml/kg/day day 1**, ↑ by 15–30 ml/kg/day to **150–180 ml/kg/day**.

Kangaroo Mother Care (KMC)

- Developed **Bogotá, Colombia (1978 – Rey & Martinez)**; for **LBW (< 2000 g – India), stable** babies.
- Components: (1) **Skin-to-skin contact** – baby **naked except nappy, cap & socks**, upright between mother's breasts, **frog position**, head turned to one side, slightly extended (airway), tied with cloth/ binder; (2) **Exclusive breastfeeding**; (3) **Early discharge & follow-up**.
- Duration – **as long as possible (≥ 1 hr per session; ideally 24 hrs – continuous)**; father/ relatives can also give.
- Benefits – **thermal control**, ↓ infection, ↑ breastfeeding & weight gain, bonding, ↓ mortality (~40 %), earlier discharge.
- **WHO 2022 – immediate KMC** right after birth for LBW babies.

- **Newborn screening** – hypothyroidism (TSH), G6PD, CAH, PKU, galactosaemia, hearing (OAE), CCHD (pulse oximetry), hip (Ortolani & Barlow – DDH).
- **Danger signs in newborn (IMNCI)**: **not feeding well**, **convulsions**, **fast breathing (≥ 60/min)**, **severe chest indrawing**, **temperature ≥ 37.5 or < 35.5 °C**, **movement only when stimulated/ no movement**, jaundice on palms & soles/ first 24 hrs, umbilical redness/ pus, bulging fontanelle.

PYQ style

- LBW – **< 2500 g**; VLBW – **< 1500 g**.
- APGAR components – **5**; done at **1 & 5 min**; normal **7–10**.
- Most important sign in resuscitation – **heart rate**.
- Golden minute – **start PPV within 1 minute** if not breathing.
- Neonatal CPR ratio – **3 : 1**.
- Vitamin K dose term – **1 mg IM**.
- Hep B birth dose within – **24 hrs**; BCG – **intradermal left upper arm**.
- Physiological jaundice appears – **after 24 hrs**, peaks **day 3–5**.
- Jaundice within 24 hrs – **pathological** (haemolytic).
- Phototherapy – **cover eyes & genitals**, extra fluids, turn 2–3 hrly.
- Moro reflex disappears – **3–6 months**; asymmetric – **Erb's palsy/ clavicle fracture**.
- Babinski disappears – **1–2 years**.
- Meconium passed within – **24–48 hrs**.
- KMC – **skin-to-skin, exclusive breastfeeding**, for LBW.
- Brown fat – **non-shivering thermogenesis**.

11 High-Risk Pregnancy & Medical Disorders in Pregnancy

11.1 High-Risk Pregnancy

- Pregnancy in which **mother/ fetus/ newborn** is at **↑ risk of morbidity or mortality**. ~20–30 % of pregnancies; contribute ~70–80 % of perinatal deaths.
- **Risk factors**: **age < 18 or > 35**, **height < 140/ 145 cm**, weight < 40/ > 90 kg, **grand multipara (≥ 5)**, **short birth interval (< 18–24 months)**, **bad obstetric history (BOH)** – previous stillbirth, abortions, PPH, LSCS, preterm; **medical diseases** (HTN, DM, heart, renal, anaemia Hb < 7, TB, epilepsy, thyroid), **Rh –ve**, multiple pregnancy, malpresentation, APH, pre-eclampsia, **elderly primi**, infertility treatment, previous congenital anomaly, poor socioeconomic, smoking/ alcohol.
- **PMSMA red sticker = high risk**; refer to **FRU (First Referral Unit – CEmONC)**. **BEmONC 7 signal functions**: parenteral antibiotics, parenteral oxytocics, **parenteral anticonvulsants ($MgSO_4$)**, manual removal of placenta, removal of retained products (MVA), **assisted vaginal delivery (vacuum)**, **newborn resuscitation** (+ CEmONC: **C-section & blood transfusion**).

11.2 Hyperemesis Gravidarum

- **Severe vomiting** in pregnancy → **dehydration, ketosis, electrolyte imbalance (hypokalaemia, hyponatraemia, metabolic alkalosis/ ketoacidosis), weight loss > 5 %**; typically **first trimester**.
- More in **primigravida, multiple pregnancy, H. mole** (↑ hCG), young, psychological.
- Complications – **Wernicke's encephalopathy (thiamine B1 deficiency)**, Mallory-Weiss tear, IUGR.
- **Management**: **admission**, **NPO 24–48 hrs**, **IV fluids (NS/ RL; dextrose after thiamine)**, **thiamine 100 mg** before dextrose, **antiemetics (doxylamine-pyridoxine, promethazine, metoclopramide, ondansetron)**, correct electrolytes, then **small frequent dry carbohydrate meals**; I/O, urine ketones, weight, psychological support.

11.3 Anaemia in Pregnancy

- **Definition (WHO)**: **Hb < 11 g/dL** (1st & 3rd trimester), **< 10.5 g/dL (2nd)**. **Mild 10–10.9**, **moderate 7–9.9**, **severe < 7**, **very severe < 4**. India – **~52 % pregnant women anaemic (NFHS-5)**.
- **Most common – iron deficiency** (then folate/ B12 – megaloblastic, haemoglobinopathies – **sickle cell, thalassaemia** in Gujarat tribal).
- **Effects** – mother: fatigue, **PPH intolerance**, infections, **heart failure (Hb < 5 – especially in labour & immediately postpartum)**, preterm labour, maternal death; fetus: **LBW, IUGR, preterm**, stillbirth.
- **Management**: **prophylaxis – IFA 1 tab daily × 180 days** (from 2nd trimester) + 180 PNC; **treatment** – Hb 9–10.9: **2 IFA tabs daily (120 mg iron)**; moderate/ intolerant/ late (> 30 wks) → **IV iron sucrose / ferric carboxymaltose (FCM)**; **severe after 34 wks or Hb < 5–7 → packed RBC transfusion**; deworming, diet (green leafy vegetables, jaggery, dates, meat; **Vit C ↑ absorption**, tea/ coffee/ calcium ↓).
- Labour in severe anaemia: **left lateral/ propped up**, O_2, **cut short 2nd stage (forceps/ vacuum)**, AMTSL, **avoid fluid overload**, diuretics, antibiotics.

11.4 Hypertensive Disorders of Pregnancy

Disorder	Definition
Gestational hypertension (PIH)	**BP ≥ 140/90 (2 readings 4 hrs apart) after 20 wks**, **no proteinuria**, resolves by 12 wks postpartum
Pre-eclampsia	**HTN after 20 wks + proteinuria (≥ 300 mg/24 hr, dipstick ≥ 1+)** or HTN + end-organ dysfunction (even without proteinuria)
Severe pre-eclampsia	**BP ≥ 160/110**, **proteinuria ≥ 2+/ 5 g**, **headache, visual disturbance, epigastric/ RUQ pain**, oliguria (< 400 ml/24 h), **platelets < 1 lakh**, ↑ liver enzymes, pulmonary oedema, creatinine > 1.1, IUGR
Eclampsia	**Pre-eclampsia + convulsions (generalised tonic-clonic)** not due to other cause – antepartum (~50 %), intrapartum, postpartum (within 48 hrs mostly)
HELLP syndrome	**H**aemolysis, **E**levated **L**iver enzymes, **L**ow **P**latelets – severe pre-eclampsia variant
Chronic HTN	Before pregnancy/ **before 20 wks**; superimposed pre-eclampsia possible

- **Risk factors**: **primigravida (most common)**, extremes of age, **multiple pregnancy, H. mole (pre-eclampsia < 20 wks)**, diabetes, chronic HTN, renal disease, obesity, family/ previous history, polyhydramnios, APLA.
- **Pathology**: **failure of 2nd wave of trophoblastic invasion of spiral arteries** → placental ischaemia → **endothelial dysfunction, vasospasm**.
- **Triad (classic)**: **Hypertension + Proteinuria + Oedema** (oedema no longer required). **Earliest sign – rapid weight gain/ oedema**; **most important – hypertension**.
- **Imminent eclampsia signs**: **severe headache (frontal/ occipital), blurring of vision/ flashes (scotomas), epigastric pain**, vomiting, oliguria, **brisk reflexes/ clonus**.
- **Prevention** (high risk): **low-dose aspirin 75–150 mg from 12–16 wks till 36 wks**, **calcium 1–1.5 g/day**.
- **Antihypertensives**: **Labetalol** (oral/ IV – first-line), **Nifedipine** (oral, not sublingual), **Methyldopa** (oral – safe), **Hydralazine** (IV). **Avoid ACE inhibitors, ARBs** (fetotoxic), diuretics (↓ plasma volume). **Aim BP < 150/100 (target ~135/85)**.

Eclampsia – management (ABC + $MgSO_4$ + delivery)

- **During fit**: **don't leave patient, call for help**, **protect from injury** (padded side rails, **no restraint**), **left lateral position**, **airway – mouth gag/ airway (between fits, not forcibly)**, **suction**, **O_2 4–6 L**; note duration.
- **Anticonvulsant of choice – Magnesium sulphate ($MgSO_4$)** – **Pritchard regimen**:
  - **Loading dose**: **4 g IV (20 % solution – 20 ml, slowly over 5–10 min) + 10 g deep IM (5 g in each buttock – 50 % solution, 10 ml each, with 1 ml 2 % lignocaine)**
  - **Maintenance**: **5 g IM (50 %) every 4 hrs in alternate buttock**, continued **24 hrs after delivery or last fit** (whichever later)
  - **Zuspan regimen (IV)**: 4–6 g IV loading → **1–2 g/hr infusion**
- **Before each maintenance dose check**: **patellar (knee) reflex present**, **RR ≥ 16/min (≥ 12)**, **urine output ≥ 30 ml/hr (100 ml in 4 hrs)**.
- **Therapeutic serum Mg 4–7 mEq/L**; loss of patellar reflex – 8–10; respiratory depression 12–15; cardiac arrest > 25.
- **Toxicity antidote**: **10 % calcium gluconate 10 ml IV slowly (over 10 min)** – keep at bedside.
- **Recurrent fit** – extra **2 g IV $MgSO_4$**.
- **BP control** (labetalol/ nifedipine/ hydralazine), **Foley catheter** (hourly urine), **I/O chart** (restrict fluids ~**80 ml/hr** – pulmonary oedema risk), **quiet dark room**, lab (LFT, platelets, RFT, coagulation).
- **Definitive treatment – DELIVERY** (within 6–12 hrs of stabilization, irrespective of gestation) – vaginal (ARM + oxytocin) preferred if favourable, LSCS for obstetric indications; **avoid ergometrine/ methergine** (HTN).
- **Complications**: cerebral haemorrhage (most common cause of death), pulmonary oedema, aspiration pneumonia, HELLP, DIC, acute renal failure, **abruptio placentae**, PRES, retinal detachment; fetus – IUGR, prematurity, IUD.

11.5 Diabetes in Pregnancy

- **GDM** – glucose intolerance first detected in pregnancy. **India – DIPSI single-step test**: **75 g oral glucose irrespective of last meal → 2-hr plasma glucose ≥ 140 mg/dL = GDM** (at first ANC & **24–28 wks**). **IADPSG/ WHO**: fasting ≥ 92, 1-hr ≥ 180, 2-hr ≥ 153.
- **Risk factors** – obesity, family history, **previous macrosomic baby (> 4 kg)**, previous GDM, unexplained stillbirth, PCOS, age > 25–30, Asian ethnicity, polyhydramnios.
- **Effects** – mother: **polyhydramnios**, pre-eclampsia, infections (UTI, candida), **shoulder dystocia**, LSCS, PPH, future **Type 2 DM (~50 %)**; fetus: **macrosomia (> 4 kg)**, **congenital anomalies (pre-gestational DM – caudal regression syndrome, cardiac, NTD)**, **unexplained IUD (after 36 wks)**, RDS (delayed lung maturity), birth injury; neonate: **hypoglycaemia (most common – fetal hyperinsulinaemia)**, hypocalcaemia, **polycythaemia, hyperbilirubinaemia**.
- **Management**: **MNT (diet) + exercise** first (2 weeks); targets **fasting < 95, 1-hr PP < 140, 2-hr PP < 120 mg/dL**; **insulin (drug of choice)** / **metformin** (now accepted); **oral sulfonylureas avoided**; SMBG; fetal surveillance; **deliver at 38–39 wks** (well controlled 39–40); during labour **insulin + dextrose infusion**, glucose 1–2 hrly; **neonatal blood sugar** within 1 hr; **insulin requirement falls sharply after delivery** (stop for GDM); **OGTT at 6–12 wks postpartum**.

11.6 Heart Disease in Pregnancy

- **Most common – rheumatic heart disease (India) – mitral stenosis**; congenital in west.
- **NYHA classes I–IV**; **critical periods (maximum risk of failure)**: **28–32 wks** (peak blood volume/ CO), **during labour (2nd stage)**, and **immediately after delivery (most dangerous – autotransfusion ~500 ml)**.
- **Management**: rest, iron, treat anaemia/ infection, **penicillin prophylaxis (RHD)**, **avoid fluid overload**, digoxin/ diuretics/ beta-blockers as needed, **anticoagulants** (mechanical valve – heparin/ LMWH 1st trimester & near term, warfarin 2nd), endocarditis prophylaxis.
- **Labour**: **propped-up (semi-Fowler's)/ left lateral**, O_2, **epidural analgesia** (↓ stress), **cut short 2nd stage (forceps/ vacuum – avoid bearing down)**; **3rd stage – oxytocin (slow); AVOID ERGOMETRINE (methergine)** – sudden ↑ venous return → pulmonary oedema; **furosemide after delivery**; monitor closely **first 24–48 hrs**. **LSCS only for obstetric indications**.
- **Breastfeeding allowed**; contraception – barrier/ progestin; **tubectomy** after family complete (preferably after puerperium); **Eisenmenger syndrome/ severe pulmonary HTN → pregnancy contraindicated (MTP)**.

11.7 Rh Isoimmunisation

- **Rh –ve mother + Rh +ve fetus** → fetal RBCs enter maternal circulation (**delivery, abortion, ectopic, amniocentesis, APH, ECV, trauma**) → mother forms **anti-D (IgG) antibodies** → in **next pregnancy** cross placenta → **fetal haemolysis** → **haemolytic disease of fetus & newborn (HDFN)** – anaemia, jaundice, **hydrops fetalis**, kernicterus. **First pregnancy usually not affected**.
- **ABO incompatibility** (mother O, baby A/B) – can affect **first pregnancy**, milder.
- **Tests**: **Indirect Coombs test (ICT) – mother** (at first visit & **28 wks**) – detects anti-D in maternal serum; **Direct Coombs test (DCT) – baby's cord blood** – antibodies on fetal RBCs. Severity – **MCA Doppler PSV** (fetal anaemia), amniotic fluid bilirubin (ΔOD 450 – Liley chart), cordocentesis.
- **Prevention – Anti-D immunoglobulin (RhoGAM)**:
  - **Antenatal prophylaxis – 300 µg (1500 IU) IM at 28 wks** (non-sensitized, ICT –ve)
  - **Postnatal – 300 µg IM within 72 hrs of delivery** (if baby Rh +ve)
  - **After abortion/ ectopic/ MTP**: **< 12 wks – 50 µg**, **> 12 wks – 300 µg** within 72 hrs; also after amniocentesis, CVS, APH, ECV
  - **300 µg neutralizes ~30 ml fetal whole blood (15 ml RBCs)**; **Kleihauer-Betke test** – quantify fetomaternal haemorrhage
  - **Not given if mother already sensitized (ICT +ve)**
- **Treatment of affected fetus** – **intrauterine transfusion (O –ve)**, early delivery; newborn – **phototherapy, exchange transfusion (O –ve, cross-matched with mother)**.

11.8 Infections in Pregnancy

- **UTI & asymptomatic bacteriuria** (> 10^5 CFU/ml) – screen & treat (nitrofurantoin, cephalexin, amoxicillin) → prevents **pyelonephritis (right side common)**, preterm labour.
- **TORCH** – see Chapter 3. **Rubella** in 1st trimester – **congenital rubella syndrome** (cataract, PDA, deafness, microcephaly) → MTP option; **MR vaccine not in pregnancy** (avoid pregnancy 1 month after).

- **Syphilis** – **VDRL/ RPR at first visit**; crosses placenta (any stage – mostly after 16 wks) → **congenital syphilis** (snuffles, rash, hepatosplenomegaly; late – Hutchinson's teeth, saddle nose); Rx – **Benzathine penicillin 2.4 MU IM** (mother & partner; single dose for early syphilis); Jarisch-Herxheimer reaction.
- **HIV (PPTCT)** – **lifelong ART (TLD) for all pregnant positive women irrespective of CD4** (Test & Treat); viral load monitoring; **safe institutional delivery – avoid ARM, episiotomy, instrumental, prolonged labour** (vaginal delivery fine if suppressed); baby – **Nevirapine syrup 6 weeks** (high risk – AZT + NVP for 12 wks), **cotrimoxazole from 6 wks**, **DNA PCR at 6 wks**; **exclusive breastfeeding + maternal ART** (no mixed feeding).
- **Hepatitis B** – HBsAg screening; baby **HBIG (0.5 ml) + Hep B vaccine within 12 hrs** (different thighs); breastfeeding allowed; **tenofovir** to mother in 3rd trimester if high viral load.
- **Malaria** – more severe in pregnancy (anaemia, abortion, IUGR, hypoglycaemia); Rx – **chloroquine (vivax)**, **ACT in 2nd/ 3rd trimester**, **quinine in 1st trimester (falciparum)**; **primaquine contraindicated**.
- **TB** – ATT safe (except **streptomycin – ototoxic**); breastfeeding ok; baby INH prophylaxis, BCG.
- **Group B Streptococcus** – intrapartum penicillin (west).

11.9 Multiple Pregnancy, IUGR & IUD

- **Twins**: incidence ~**1 in 80–90** (Hellin's rule – twins 1 : 80); **dizygotic (binovular, ~2/3 – heredity, ↑ maternal age, ovulation induction)** – 2 placentas, different sex possible; **monozygotic (identical)** – same sex. **Chorionicity by USG at 11–14 wks** – **lambda (twin peak) sign – dichorionic**, **T-sign – monochorionic** (**TTTS** risk).
  - Diagnosis – **fundal height > dates**, **multiple fetal parts**, **2 FHS (difference ≥ 10 bpm, heard simultaneously by 2 persons)**, USG.
  - Complications – **anaemia, pre-eclampsia, hyperemesis, polyhydramnios, preterm labour (most common)**, APH, malpresentation, **PPH (atonic – most common postpartum)**, cord prolapse, **locked twins** (first breech, second vertex).
  - Delivery – most common presentation **both vertex (~50 %)**; after first baby: **clamp cord (both ends – in case monozygotic)**, check lie of 2nd (USG/ palpation), FHS, **ARM after 2nd presenting part fixed**, deliver 2nd within ~30 min; **uterotonic only after last baby** (oxytocin infusion after delivery – PPH prophylaxis).
- **IUGR** – causes: maternal (**hypertension – most common**, anaemia, malnutrition, smoking, chronic disease), placental insufficiency, fetal (chromosomal, infections, multiple). Monitoring – **AC, Doppler (umbilical artery)**, BPP; deliver timely; neonate – hypoglycaemia, hypothermia, polycythaemia.
- **Intrauterine death (IUD)** – fetal death after viability before labour; signs – **absent fetal movements, absent FHS**, uterus smaller/ regressing, **Spalding's sign** (overlapping skull bones – X-ray ~7 days), **Robert's sign** (gas in fetal great vessels), USG – no cardiac activity (confirmatory). Complication – **DIC (if retained > 4 wks – hypofibrinogenaemia)**. Management – induction (**misoprostol**/ mifepristone + misoprostol), psychological support, lactation suppression, investigate cause, **Rh anti-D**.

PYQ style

- Anaemia in pregnancy – **Hb < 11 g/dL**; severe **< 7**.
- Pre-eclampsia – **HTN ≥ 140/90 after 20 wks + proteinuria**.
- Eclampsia drug of choice – **$MgSO_4$** (Pritchard: 4 g IV + 10 g IM loading; 5 g IM 4-hrly).
- $MgSO_4$ check – **knee jerk, RR ≥ 16, urine ≥ 30 ml/hr**; antidote – **10 % calcium gluconate**.
- Methergine contraindicated in – **pre-eclampsia, heart disease**.
- Antihypertensive first-line – **labetalol, nifedipine, methyldopa**; avoid **ACE inhibitors**.
- GDM (DIPSI) – **75 g OGTT 2-hr ≥ 140 mg/dL**; drug of choice – **insulin**.
- Commonest neonatal problem in GDM – **hypoglycaemia**.
- Heart disease – most critical period – **immediately after delivery**.
- Anti-D dose after delivery – **300 µg IM within 72 hrs**; antenatal **28 wks**.
- Test for maternal antibodies – **indirect Coombs**; baby – **direct Coombs**.
- Twins – most common complication – **preterm labour**; dichorionic sign – **lambda/ twin peak**.
- Spalding's sign – **IUD (overlapping skull bones)**.

12 Bleeding in Pregnancy – Abortion, Ectopic, Mole & APH

12.1 Early Pregnancy Bleeding – Causes

Abortion (most common), **ectopic pregnancy**, **hydatidiform mole**, implantation bleed, cervical lesions (polyp, ectropion, cancer).

12.2 Abortion (Miscarriage)

D Abortion

Expulsion/ extraction of products of conception **before viability** – **WHO: < 20 wks or < 500 g**; India (traditional) **< 28 wks**. **Spontaneous abortion** – ~10–15 % of clinical pregnancies; mostly **first trimester (~80 %)**; **most common cause – chromosomal (genetic) anomalies (~50 %)** (trisomy 16 commonest). Second trimester – **cervical incompetence**, uterine anomalies, APLA.

Type	Bleeding	Pain	Os	Uterus size	Management
Threatened	**Slight**	Mild/ none	**Closed**	= dates	**Rest**, avoid intercourse, progesterone (selected), reassurance, USG – pregnancy **continues (~80 %)**
Inevitable	More	**Painful cramps**	**Open** (products felt/ not yet expelled)	= dates	**Evacuation** (MVA/ suction), oxytocin/ misoprostol
Incomplete	**Heavy, continuous**	Present	**Open**, products partly passed	**< dates**	**Evacuation (MVA/ D&E)**, misoprostol, blood, antibiotics
Complete	**Little/ stopped**	Subsides	**Closed**	**Smaller, firm**	USG confirm; no evacuation; anti-D
Missed (silent)	Brownish discharge/ none	None	Closed	**< dates**, symptoms of pregnancy regress	**Mifepristone + misoprostol** / evacuation; **DIC risk (retained > 4 wks)** – check coagulation
Septic	Foul discharge	+ **fever, tenderness**	Open	–	**Antibiotics (broad-spectrum)**, evacuation after 6 hrs of antibiotics, IV fluids, tetanus; complications – **septic shock, DIC, AKI, peritonitis**

Type	Bleeding	Pain	Os	Uterus size	Management
Recurrent (habitual)	**≥ 2–3 consecutive** spontaneous abortions				Investigate – karyotype, APLA (aspirin + heparin), thyroid, diabetes, uterine anomaly, **cervical incompetence (cerclage)**

- **Rh-negative** mother – **anti-D** after any abortion (< 12 wks 50 µg; > 12 wks 300 µg).
- **Cervical incompetence (insufficiency)** – **painless dilatation** in **2nd trimester** → recurrent mid-trimester abortion/ preterm (cause – previous D&C, cone biopsy, congenital) → **cervical cerclage at 12–14 wks** – **McDonald's (purse-string – most common, removed at 37 wks)** / **Shirodkar's** (submucosal – may be permanent, LSCS); remove earlier if labour starts/ ROM/ infection.

MTP Act (Medical Termination of Pregnancy)

- **MTP Act 1971** (effective 1 April 1972); **MTP (Amendment) Act 2021** (rules 2021).

Gestation	Opinion required
Up to 20 weeks	**1 Registered Medical Practitioner (RMP)**
20–24 weeks	**2 RMPs** – for **special categories** (rape survivors, incest, **minors**, change of marital status – widowhood/ divorce, disability, mentally ill, **fetal malformation**, humanitarian/ disaster)
Beyond 24 weeks	Only for **substantial fetal abnormalities** – **State-level Medical Board** decision
Any time	To **save life of the woman** – **1 RMP (immediate)**

- Grounds: risk to life/ grave injury to **physical or mental health**, substantial fetal abnormality, rape (mental anguish), **contraceptive failure – now for any woman (married or unmarried) – 2021**.
- **Consent – only woman's** (if **≥ 18 yrs & sane**); **minor (< 18) or mentally ill – written consent of guardian**. **Husband's consent NOT required**.
- **Confidentiality** – name of woman can't be revealed (except to authorised person) – punishable.
- Place – **government/ approved facility**; **medical abortion (MMA) up to 9 weeks (63 days)** – **Mifepristone 200 mg oral (day 1) → Misoprostol 800 µg buccal/ vaginal/ sublingual after 24–48 hrs (day 3)** – by RMP; follow-up day 15.
- **Surgical**: **MVA (Manual Vacuum Aspiration – up to 12 wks)**, **suction evacuation & curettage (EVA)**, **D&E (2nd trimester)**; 2nd trimester – **misoprostol/ mifepristone**, extra-amniotic ethacridine (old), oxytocin.
- **PCPNDT Act 1994** (amended 2003) – **prohibits sex determination**.

12.3 Ectopic Pregnancy

- **Implantation outside uterine cavity**; **most common site – fallopian tube (~95–97 %)** – **ampulla (most common, ~70–80 %)** > isthmus (ruptures earliest – 6–8 wks) > fimbrial > **interstitial (cornual – latest & most dangerous rupture ~12–16 wks, massive bleeding)**; others – ovary, cervix, abdomen, **caesarean scar**.
- **Risk factors**: **previous ectopic (strongest)**, **PID (Chlamydia/ gonorrhoea – most common cause)**, **tubal surgery/ tubectomy failure**, **IUCD in situ** (if pregnancy occurs), **ART/ IVF**, smoking,

endometriosis, progestin-only pills.

- **Classical triad**: **amenorrhoea (6–8 wks) + abdominal pain (most common symptom) + vaginal bleeding (spotting)**.
- **Ruptured ectopic** – **sudden severe lower abdominal pain**, **shoulder tip pain (Kehr's sign – diaphragmatic irritation by blood)**, fainting, **haemorrhagic shock (pallor, tachycardia, hypotension)**, **cervical motion tenderness (excitation)**, **tender fullness in pouch of Douglas**, abdominal distension, **Cullen's sign** (periumbilical bluish – rare).
- **Diagnosis**: **serum β-hCG + TVS** – **empty uterus with β-hCG above discriminatory zone (1500–2000 mIU/ml – TVS)** → ectopic; **β-hCG fails to double in 48 hrs**; **culdocentesis** (non-clotting blood from POD – old); **laparoscopy – gold standard**.
- **Management**:
  - **Ruptured – EMERGENCY**: **treat shock (2 large-bore IV, blood, cross-match)** + **immediate laparotomy/ laparoscopy – salpingectomy**; consent, NPO.
  - **Unruptured, stable** – **medical: Methotrexate 50 mg/m² IM** (criteria – haemodynamically stable, **sac < 3.5–4 cm**, **no cardiac activity**, **β-hCG < 5000**) – follow β-hCG day 4 & 7 (≥ 15 % fall); **avoid folic acid, NSAIDs, alcohol, intercourse, sun**; laparoscopic **salpingostomy**; expectant (low hCG, declining).
  - **Anti-D** if Rh –ve.

12.4 Gestational Trophoblastic Disease – Hydatidiform Mole

- **Abnormal proliferation of trophoblast**; villi become **grape-like vesicles** ("**bunch of grapes**").
- **Complete mole** – **46 XX (paternal origin – empty ovum fertilized by sperm)**, **no fetus**, **higher malignancy risk (15–20 %)**. **Partial mole** – **triploid (69 XXY)**, fetal parts present, lower risk.
- Risk – **extremes of age (< 20, > 35–40)**, previous mole, Asian, low protein/ carotene diet.
- **Clinical**: **amenorrhoea + vaginal bleeding (most common)** with **passage of grape-like vesicles (pathognomonic)**, **uterus larger than dates (doughy/ soft)**, **absent FHS & fetal parts**, **hyperemesis**, **early pre-eclampsia (< 20 wks)**, **hyperthyroidism**, **theca-lutein cysts** (bilateral ovarian), anaemia.
- **Investigations**: **very high β-hCG** (> 1,00,000 mIU/ml), **USG – "snowstorm appearance" / "bunch of grapes/ honeycomb"**; chest X-ray (metastasis).
- **Management**: **suction evacuation (treatment of choice)** + oxytocin drip after start of evacuation, blood ready, **histopathology**; hysterectomy (older women, family complete); **anti-D**.
- **Follow-up (very important)**: **serum β-hCG weekly till negative × 3 consecutive weeks**, then **monthly for 6 months** (complete mole); **contraception – OCPs/ barrier for 6–12 months (avoid pregnancy & IUCD till hCG normal)**. Persistent/ rising hCG → **GTN (invasive mole/ choriocarcinoma)** → **methotrexate/ actinomycin-D (chemosensitive – high cure)**. **Choriocarcinoma** – metastasis to **lungs (most common)**, vagina, brain, liver.

12.5 Antepartum Haemorrhage (APH)

D **APH**

Bleeding from genital tract after 20 weeks (India – 28 wks) of pregnancy and before birth of baby. Causes: **placenta praevia (~1/3)**, **abruptio placentae (~1/3 – most common in some data)**, **unclassified/ indeterminate** (local lesions, vasa praevia, show).

APH – golden rule

NO PER VAGINAL (PV) EXAMINATION in APH outside OT (can cause torrential bleeding in placenta praevia) – **only USG/ speculum (gentle)**; PV only in OT with everything ready for LSCS ("**double set-up examination**"). **No rectal exam, no enema**.

Feature	Placenta Praevia	Abruptio Placentae (accidental haemorrhage)
Definition	Placenta **implanted partly/ wholly in lower uterine segment**	**Premature separation of normally situated placenta**
Bleeding	**Painless, causeless, recurrent, bright red**, "**warning haemorrhage**" (often first at ~32–34 wks)	**Painful**, dark (old) blood, may be **concealed** (retroplacental) or revealed/ mixed
Shock	**Proportionate to visible blood loss**	**Out of proportion** to visible loss (concealed)
Uterus	**Soft, relaxed, non-tender**	**Tense, tender, hard ("woody/ board-like")**, ↑ size
Fetal parts / FHS	Easily felt; **FHS usually present**	Difficult to feel; **FHS often absent/ distress**
Presentation	**Malpresentation common**, **high/ floating head** (not engaged)	Normal, may be engaged
Associated	**Multiparity, previous LSCS/ scar, multiple pregnancy, previous praevia, smoking, advanced age**	**Pre-eclampsia/ HTN (most common association)**, trauma, **sudden decompression (polyhydramnios, twin delivery)**, short cord, smoking, cocaine, folic acid deficiency
Complications	PPH (LUS poor contractility), **placenta accreta**, preterm, malpresentation	**DIC (hypofibrinogenaemia)**, **Couvelaire uterus** (blood infiltrates myometrium – bluish, may still contract), **AKI**, PPH, **IUD**
Diagnosis	**USG (TVS safe)**	Clinical; USG (retroplacental clot – may miss)

- **Degrees of placenta praevia**: **Type I (low-lying)** – lower segment but not reaching os; **Type II (marginal)** – reaches margin of internal os; **Type III (partial)** – partially covers os; **Type IV (complete/ central)** – completely covers os → **LSCS**. Types I & II anterior – vaginal delivery may be possible; **posterior type II ("dangerous placenta praevia")** – compresses presenting part/ cord.
- **Placenta praevia management**: **expectant – "Macafee & Johnson" regimen** (if bleeding not severe, < 37 wks, mother & fetus stable) – **hospitalization, bed rest, blood available (cross-matched 2 units), haematinics, steroids (< 34 wks), anti-D**, no PV/ intercourse; **deliver at 37 wks (36–37)**; **LSCS** for major degrees; **active (immediate delivery)** – heavy bleeding, labour, term, fetal distress/ death.
- **Abruption management**: **resuscitate (2 large-bore IV, blood, FFP/ cryo)**, **monitor coagulation (fibrinogen, clotting time – bedside clot test)**, **Foley catheter (urine output ≥ 30 ml/hr)**, **CVP**; **ARM (reduces pressure, speeds delivery, ↓ thromboplastin entry)** + oxytocin; **LSCS if fetus alive with distress/ not delivering quickly**; IUD → vaginal delivery preferred (if coagulopathy corrected); **watch PPH**.
- **Vasa praevia** – fetal vessels in membranes over os (velamentous cord) → **ROM → fetal bleeding** (painless bleed + **sudden fetal distress/ sinusoidal FHR**) → **emergency LSCS**; **Apt test** (fetal Hb – alkali resistant).

PYQ style

- Commonest cause of spontaneous abortion – **chromosomal anomalies**.
- Threatened abortion – **os closed**, slight bleeding; inevitable – **os open**.
- Incomplete abortion – **evacuation**; complete – **os closed, uterus small**.
- Cervical incompetence – **painless mid-trimester abortion**; Rx – **McDonald's cerclage at 12–14 wks**.
- MTP up to 20 wks – **1 RMP**; 20–24 wks (special) – **2 RMPs**; consent – **woman herself**.
- Medical abortion up to – **9 weeks** (mifepristone + misoprostol).
- Commonest site of ectopic – **ampulla**; most dangerous – **interstitial**.
- Classical triad of ectopic – **amenorrhoea, pain, bleeding**; shoulder tip pain – **Kehr's sign**.
- Unruptured ectopic medical Rx – **methotrexate**.
- Snowstorm appearance – **H. mole**; follow-up – **β-hCG**; contraception – **OCP (not IUCD)**.
- APH – **no PV examination**.
- Painless recurrent bright red bleeding – **placenta praevia**; painful with tense tender uterus – **abruptio**.
- Abruption associated with – **pre-eclampsia**; complication – **DIC, Couvelaire uterus**.

13 Abnormal Labour – Malpresentations, Malpositions & Dystocia

13.1 Malposition – Occipito-Posterior (OP)

- **Vertex with occiput in posterior quadrant** (ROP more common than LOP); ~**10–20 %** at onset of labour; **deflexed head → occipito-frontal 11.5 cm** presents → **prolonged labour**.
- Causes – **android/ anthropoid pelvis**, flat sacrum, deflexion, anterior placenta.
- **Signs**: **flattening of lower abdomen below umbilicus ("saucer-shaped")**, **fetal limbs anteriorly**, **FHS heard in flank**, **early ROM**, **backache (severe)**, early urge to push before full dilatation, slow descent, **anterior fontanelle felt easily** on PV.
- **Outcome**: **~90 % – long anterior rotation (3/8 circle → OA)** – normal delivery; **short posterior rotation → persistent OP ("face-to-pubis" delivery)**; **deep transverse arrest** (android pelvis); arrest.
- **Management**: wait (good contractions, **hands & knees/ lateral position**, back massage, hydration, empty bladder, analgesia/ epidural), oxytocin augmentation; persistent OP → **vacuum/ forceps (manual rotation/ Kielland's)**, **LSCS** for arrest/ CPD; large episiotomy (perineal tears more).

13.2 Breech Presentation

- **Buttocks/ lower limbs present**; **~3–4 % at term** (25 % at 28 wks); **most common malpresentation**.
- **Types**: **Frank (extended) – hips flexed, knees extended – most common (~65–70 %, primi)**; **Complete (flexed)** – hips & knees flexed (multipara); **Incomplete – footling/ knee** – feet/ knees present (**cord prolapse risk highest**).
- **Causes**: **prematurity (most common)**, multiparity (lax uterus), **placenta praevia**, uterine anomalies (septate/ bicornuate), polyhydramnios/ oligohydramnios, **hydrocephalus/ anencephaly**, multiple pregnancy, short cord, contracted pelvis.
- **Diagnosis**: **hard round ballotable head in fundus**, **FHS above umbilicus**, PV – soft irregular (anus, ischial tuberosities in line; genitalia; **meconium on glove – normal in breech**), USG.
- **Risks**: fetal – **cord prolapse**, **head entrapment (after-coming head)**, **birth asphyxia**, **intracranial haemorrhage (tentorial tear)**, **brachial plexus injury (Erb's)**, fractures (humerus, clavicle, femur), **visceral injury**; mother – perineal tears, PPH, LSCS.
- **Management**:
  - **External cephalic version (ECV)** – at **36–37 wks (primi 36, multi 37)** if no contraindication (**praevia, previous LSCS, PROM, twins, fetal compromise, APH, pre-eclampsia, Rh iso – give anti-D**); success ~50 %; tocolysis; CTG before & after.
  - **Mode of delivery** – **planned LSCS preferred** (Term Breech Trial 2000) especially primi, big baby > 3.5 kg, footling, hyperextended head, contracted pelvis, previous LSCS.
  - **Assisted vaginal breech delivery** (selected – frank/ complete, adequate pelvis, average size, experienced attendant): **"hands-off the breech"** till umbilicus; **episiotomy**; loop of cord pulled; **arms – Lovset's manoeuvre** (extended arms); **after-coming head** delivered by **Mauriceau-Smellie-Veit (MSV) manoeuvre (jaw flexion, shoulder traction)**, **Burns-Marshall method**, **Piper's forceps**; avoid hyperextension; **don't pull on breech**.
  - **Breech extraction** – only for 2nd twin.

13.3 Face, Brow, Shoulder & Compound Presentations

Presentation	Key facts
Face	**Complete extension of head**; ~1 in 500; **denominator – mentum**; diameter **submento-bregmatic 9.5 cm**; causes – **anencephaly** (most common fetal cause), neck tumour, cord round neck, contracted pelvis, multiparity; **mento-anterior → vaginal delivery possible (by flexion)**; **mento-posterior persistent → LSCS (impossible vaginally)**; face oedema & bruising – reassure; **no vacuum**
Brow	**Partial extension (midway)**; **mento-vertical 13.5–14 cm (largest) → no vaginal delivery** in average-size baby → **LSCS** (may convert to face/ vertex); PV – **anterior fontanelle, orbital ridges, root of nose**
Shoulder (transverse lie)	~1 in 300; causes – **multiparity (lax abdomen – most common)**, **placenta praevia**, polyhydramnios, contracted pelvis, uterine anomaly, prematurity, twins (2nd baby); **abdomen broad, fundus low**, no presenting part; **vaginal delivery impossible (term live baby) → LSCS**; danger – **cord prolapse, hand prolapse, obstructed labour, rupture uterus (neglected shoulder)**; ECV/ internal podalic version (2nd twin only)
Compound	Hand/ arm along with head – usually resolves; don't pull the hand
Unstable lie	Changing lie after 37 wks – admit, ECV, stabilizing induction

13.4 Cephalopelvic Disproportion (CPD), Prolonged & Obstructed Labour

- **CPD** – **disparity between fetal head & maternal pelvis** → vaginal delivery not possible. Causes – **contracted pelvis** (rickets, osteomalacia, short stature < 145 cm), **big baby** (diabetes, post-term), malposition/ deflexion, hydrocephalus.
- **Assessment**: **head-fitting test (Müller-Munro Kerr)**, clinical pelvimetry (diagonal conjugate, sacral curve, ischial spines, sub-pubic angle, intertuberous diameter), **trial of labour** (borderline CPD – adequate contractions, partograph, 4–6 hrs).
- **Prolonged labour**: **combined 1st & 2nd stage > 18 hrs (some: 24 hrs)**; **active phase crossing action line**; prolonged latent phase (> 20 hrs primi, > 14 multi); **prolonged 2nd stage** – primi > 2 hrs (3 with epidural), multi > 1 hr (2). **Causes – 3 Ps**: Power (inefficient contractions – most common), Passage (CPD), Passenger (malposition, big baby).
- **Obstructed labour** – **no progress (descent) despite good contractions** due to **mechanical obstruction**.

! Signs of obstructed labour

- **Mother**: exhaustion, **dehydration, ketosis**, **tachycardia, fever**, **hot dry vagina**, **oedematous vulva/ cervix**, **Bandl's (pathological retraction) ring** – rising ring between thick upper & thin lower segment (impending **rupture uterus**), **tonic uterine contraction**, distended bladder/ **haematuria** (bladder compression), **foul liquor**.
- **Fetus**: **excessive moulding (+3)**, **large caput**, head not descending, **fetal distress/ death**.
- **Complications**: **rupture uterus**, **vesicovaginal fistula (VVF – pressure necrosis)**, rectovaginal fistula, sepsis, PPH, foot drop (nerve injury), **maternal death**; fetal asphyxia/ death.
- **Management**: **resuscitate (IV fluids, correct ketosis/ electrolytes), antibiotics, catheterize (continuous drainage 10–14 days – prevent VVF)** → **immediate delivery – LSCS** (live baby); dead baby → **destructive operation (craniotomy)** in selected; **NEVER give oxytocin** in obstructed labour.

Abnormal uterine action

- **Hypotonic/ inefficient uterine contractions (inertia)** – weak, infrequent → **augmentation: ARM + oxytocin**.
- **Hypertonic/ incoordinate** – painful but ineffective; **colicky uterus**, **constriction ring** → analgesia, epidural, tocolysis, LSCS.
- **Precipitate labour** – **total labour < 3 hrs** (multipara) → **lacerations, PPH, inversion, amniotic fluid embolism**; baby – **intracranial haemorrhage, hypoxia, injury, infection (unsterile)**; history → admit near term.
- **Cervical dystocia** – rigid cervix (scarring).

13.5 Preterm Labour & PROM

- **Preterm labour** – **regular contractions with cervical changes between 20/ 28 and < 37 wks**. **Most common cause of neonatal death**.
- Causes – **idiopathic (most)**, **infection (UTI, BV, chorioamnionitis)**, **PPROM**, **multiple pregnancy, polyhydramnios**, APH, pre-eclampsia, **cervical incompetence**, uterine anomalies, previous preterm, smoking, low socioeconomic, short cervix (< 25 mm on TVS).
- **Prediction**: **fetal fibronectin (fFN)** (negative – reassuring), **cervical length < 25 mm**.
- **Management**:
  - **Tocolysis (delay 48 hrs for steroids)**: **Nifedipine (first-line)**, **atosiban**, indomethacin (< 32 wks, short course – PDA closure risk), ritodrine/ terbutaline (β-agonists – tachycardia, pulmonary oedema), $MgSO_4$; contraindications – chorioamnionitis, IUD, fetal distress, APH heavy, severe PE, PPROM with infection.
  - **Antenatal corticosteroids (24–34 wks)**: **Dexamethasone 6 mg IM 12-hrly × 4 doses** or **Betamethasone 12 mg IM 24-hrly × 2 doses** – ↓ **RDS, IVH, NEC**, neonatal death.
  - **$MgSO_4$ for neuroprotection (< 32 wks)** – ↓ cerebral palsy.
  - **Antibiotics** (GBS/ PPROM – erythromycin/ ampicillin), deliver at **higher centre with NICU** ("in-utero transfer" – best transport incubator is uterus).
  - **Progesterone (vaginal)** – prevention in short cervix/ previous preterm; **cerclage**.
- **PROM (premature/ prelabour rupture of membranes)** – **ROM before onset of labour** at any gestation; **PPROM** – **before 37 wks**; **prolonged ROM > 18–24 hrs** before delivery → infection risk.

- Diagnosis – history of **gush/ leaking**, **sterile speculum – pooling in posterior fornix**, **nitrazine (blue)**, **fern test**, USG (oligohydramnios); **avoid digital PV** (infection).
- Complications – **chorioamnionitis (fever, maternal & fetal tachycardia, foul liquor, uterine tenderness)**, **cord prolapse**, preterm, abruption, oligohydramnios → pulmonary hypoplasia.
- **Management**: **term PROM – induce with oxytocin** (or expectant ≤ 24 hrs); **PPROM < 34 wks – expectant**: admit, **sterile pad**, temperature/ pulse 4-hrly, **TLC/ CRP**, **antibiotics (erythromycin ± ampicillin – latency)**, **steroids**, **avoid PV**, deliver at **34 wks** or earlier if infection/ distress; chorioamnionitis → **deliver (any gestation)** + antibiotics.

13.6 Post-term Pregnancy & Induction of Labour

- **Post-term – ≥ 42 wks (294 days)**; risks – **placental insufficiency**, **oligohydramnios**, **meconium aspiration**, macrosomia (shoulder dystocia), **post-maturity syndrome** (dry, peeling, wrinkled skin – "**washer-woman hands**", long nails, no vernix/ lanugo, alert look), IUD. **Induce at 41 wks** (WHO – 41 completed wks).
- **Induction of labour (IOL)** – artificial initiation of labour before spontaneous onset. **Indications**: **post-term**, **pre-eclampsia**, **PROM**, **IUGR**, **diabetes**, **IUD**, Rh isoimmunisation, chorioamnionitis, abruption (mild). **Contraindications**: **placenta praevia**, transverse lie, **CPD**, **previous classical CS/ > 1 LSCS** (relative), active genital herpes, cord presentation, fetal distress.

Bishop score (cervical ripeness) – 0 to 13

Factor	0	1	2	3
Dilatation (cm)	0	1–2	3–4	≥ 5
Effacement (%)	0–30	40–50	60–70	≥ 80
Station	−3	−2	−1/ 0	+1/ +2
Consistency	Firm	Medium	**Soft**	–
Position	Posterior	Mid	**Anterior**	–

- **Score ≥ 8 (≥ 6) = favourable** → ARM + oxytocin likely successful; **< 6 unfavourable → cervical ripening first**.

- **Methods**: **Cervical ripening** – **PGE2 gel (dinoprostone 0.5 mg intracervical, 6-hrly, max 3)**, **misoprostol (PGE1) 25 µg vaginal 4-hrly / 25–50 µg oral 2–4 hrly**, **Foley catheter balloon (mechanical – 30 ml)**, membrane sweeping (stripping); **ARM (amniotomy)** – check cord prolapse, liquor colour, FHS after; **Oxytocin infusion** – **low dose start 1–2 mU/min (2.5 units in 500 ml NS/RL)**, ↑ every 30 min till **3–4 contractions in 10 min, each 40–60 sec**; max 32 mU/min; **infusion pump**.
- **Oxytocin nursing care**: **never IV bolus**, **dedicated line/ pump**, **monitor contractions & FHS every 15–30 min**, **stop if hyperstimulation (> 5 contractions/10 min, contraction > 90 sec, FHR abnormality)** → left lateral, O_2, IV fluids, tocolytic (terbutaline); watch **water intoxication** (antidiuretic effect – high dose with large dextrose), **rupture uterus**, I/O.

PYQ style

- Commonest malpresentation – **breech**; commonest type – **frank breech**.
- Commonest cause of breech – **prematurity**.
- ECV done at – **36–37 wks**.
- After-coming head delivery – **Mauriceau-Smellie-Veit manoeuvre**; extended arms – **Lovset's**.
- Face presentation denominator – **mentum**; mento-posterior – **LSCS**.
- Brow – **mento-vertical 13.5–14 cm**, LSCS.
- Transverse lie commonest cause – **multiparity**; delivery – **LSCS**.
- OP position – **flattened lower abdomen, FHS in flank, backache**.
- Bandl's ring – **obstructed labour (impending rupture)**.
- Obstructed labour – **oxytocin contraindicated**; VVF prevention – **continuous catheter drainage**.
- Precipitate labour – **< 3 hrs**.
- Tocolytic first-line – **nifedipine**; antenatal steroid – **dexamethasone 6 mg × 4 doses 12-hrly**.
- PROM – **avoid digital PV**; nitrazine **blue**.
- Bishop score favourable – **≥ 8**.
- Post-term – **≥ 42 wks**; induce at **41 wks**.
- Oxytocin hyperstimulation – **stop infusion, left lateral, O_2**.

14 Obstetric Emergencies – PPH, Cord Prolapse, Shoulder Dystocia & More

14.1 Postpartum Haemorrhage (PPH)

D PPH

Blood loss ≥ 500 ml after vaginal delivery or **≥ 1000 ml after caesarean**, **or any amount causing haemodynamic instability** (WHO/ RCOG). **Primary PPH** – within **24 hrs** of delivery (most common, mostly within first hour). **Secondary (late) PPH** – **24 hrs to 6 weeks** (commonest cause – **retained bits/ infection – subinvolution**). **Leading cause of maternal death worldwide & India.**

♫ 4 T's – causes of PPH

Tone – **uterine atony (~70–80 % – most common)** · **T**rauma – tears (cervix, vagina, perineum), rupture, inversion, haematoma (~20 %) · **T**issue – retained placenta/ bits, accreta (~10 %) · **T**hrombin – coagulopathy (DIC, abruption, HELLP, IUD, anticoagulants) (~1 %)

- **Risk factors for atony**: **overdistended uterus (twins, polyhydramnios, macrosomia)**, **grand multiparity**, **prolonged/ precipitate labour**, **oxytocin augmentation**, **APH (abruption, praevia)**, **anaemia**, general anaesthesia (halothane), chorioamnionitis, **full bladder**, fibroids, previous PPH, $MgSO_4$/ tocolytics.
- **Signs**: excessive vaginal bleeding, **boggy/ soft uterus above umbilicus**, **tachycardia (earliest sign)**, hypotension (late), pallor, sweating, restlessness, ↓ urine, shock. **Shock index (HR/SBP) > 0.9–1** → significant.
- **Bleeding with well-contracted uterus → trauma (tears)**.

PPH management – step by step

1. **Call for help** (team – "**HAEMOSTASIS**" mnemonic), note time
2. **Uterine massage (fundal rub)** – expel clots; **bimanual compression** if continues
3. **ABC, O_2, 2 large-bore IV (16–18 G)**, **blood sample – grouping, cross-match (4 units)**, Hb, coagulation; **IV crystalloids (NS/ RL)** rapidly; **legs raised**, keep warm
4. **Empty the bladder (Foley catheter)** – full bladder prevents contraction; hourly urine
5. **Uterotonics**:
   - **Oxytocin 10 IU IM/ slow IV** → **infusion 20–40 IU in 500–1000 ml NS/RL** (first-line)
   - **Methylergometrine (Methergine) 0.2 mg IM/ slow IV** – repeat every 15 min (max 5 doses / 1 mg); **contraindicated – hypertension, pre-eclampsia, heart disease**
   - **Carboprost (15-methyl PGF2α) 250 µg IM (or intramyometrial)** every 15–20 min, **max 8 doses (2 mg)**; **contraindicated – asthma**
   - **Misoprostol 800 µg sublingual/ rectal** (600 µg oral)
   - **Tranexamic acid 1 g IV** (slow over 10 min) **within 3 hrs of birth** – repeat after 30 min if bleeding (WOMAN trial)
6. **Examine placenta & membranes**; explore for **tears** (repair) and **retained tissue** (manual removal/ evacuation)
7. **Non-pneumatic anti-shock garment (NASG)**, **uterine balloon tamponade (UBT – condom catheter / Bakri balloon)**, **aortic compression**
8. **Surgical**: **B-Lynch compression suture**, **uterine artery ligation**, **internal iliac (hypogastric) artery ligation**, **hysterectomy (last resort – life-saving)**
9. **Blood & products** – PRBC, FFP, platelets, cryo (massive transfusion); monitor vitals every 15 min, urine output, Hb, DIC

- **Prevention – AMTSL (Oxytocin 10 IU IM within 1 min)**, correct anaemia, **identify high-risk & deliver at FRU**, avoid prolonged labour, **empty bladder**, examine placenta, observe **4th stage closely (1–2 hrs)**, **Misoprostol 600 µg oral by ASHA/ ANM for home deliveries**.
- **Sheehan's syndrome** – **postpartum pituitary necrosis** after severe PPH/ shock → **failure of lactation (first sign)**, amenorrhoea, loss of axillary/ pubic hair, hypothyroidism, adrenal insufficiency.
- **Blood loss estimation**: soaked pad ~50–100 ml; **visual estimation underestimates** → weigh pads/ drapes (1 g = 1 ml), calibrated drapes (**BRASSS-V**).

14.2 Retained Placenta

- **Placenta not delivered within 30 min** (with AMTSL; 60 min physiological). Causes – **atony**, **trapped (cervix closed – constriction ring)**, **adherent (accreta)**.
- Management: **empty bladder**, oxytocin, **CCT** again; if fails/ bleeding → **manual removal of placenta (MRP)** under GA/ sedation with **IV antibiotics**, uterotonics after; **accreta** → hysterectomy/ conservative.

14.3 Acute Uterine Inversion

- **Uterus turns inside out** (fundus through cervix) – mostly **3rd stage** due to **mismanagement – excess cord traction on fundal placenta without counter-traction**, **fundal pressure (Crede's)** with relaxed uterus, short cord, adherent placenta.

- Signs: **severe shock (neurogenic – out of proportion to bleeding)**, haemorrhage, **fundus not palpable / cup-shaped depression**, mass in vagina/ outside.
- Management: **immediate manual replacement (Johnson's manoeuvre – push fundus up through cervix, last-out part first in) without removing placenta (if attached)**, treat shock, **tocolytics (terbutaline/ $MgSO_4$/ GA)** to relax; **O'Sullivan's hydrostatic method** (warm saline into vagina); after replacement – **oxytocin** to maintain, remove placenta; surgical (Huntington's, Haultain's).

14.4 Rupture of Uterus

- **Most common cause (today) – previous caesarean scar** (especially **classical CS – upper segment**); others – **obstructed labour**, **injudicious oxytocin/ prostaglandins (in multipara/ scarred uterus)**, **grand multiparity**, trauma, instrumental (internal podalic version, destructive ops), placenta percreta.
- **Signs**: **scar tenderness**, **sudden severe abdominal pain** (may "tearing" then **cessation of contractions**), **fetal distress/ loss of FHS (earliest & most consistent sign – abnormal CTG)**, **vaginal bleeding**, **shock**, **fetal parts palpable superficially**, **recession of presenting part**, haematuria, Bandl's ring before rupture.
- Management: **resuscitation + immediate laparotomy** – **repair** (young, small tear) or **hysterectomy**; baby usually dead if complete rupture.
- **VBAC (Vaginal Birth After Caesarean)/ TOLAC** – allowed after **1 previous LSCS (lower segment, transverse)** with non-recurring indication, spontaneous labour, institutional with emergency CS facility; success ~60–80 %; **rupture risk ~0.5–1 %**; **contraindicated – previous classical/ T-shaped incision, previous rupture, ≥ 2–3 LSCS, praevia**; avoid prostaglandins.

14.5 Cord Prolapse & Presentation

- **Cord presentation** – cord below presenting part, **membranes intact**; **cord prolapse** – cord **below presenting part after ROM** (**occult** – beside presenting part).
- **Risk**: **malpresentation (footling breech, transverse lie)**, **prematurity/ LBW**, **polyhydramnios**, **ARM with high/ unengaged head**, twins (2nd), long cord, multiparity, CPD.
- **Diagnosis**: cord felt/ seen in vagina; **sudden fetal bradycardia/ variable decelerations after ROM**.

! Cord prolapse – EMERGENCY ("Push up, lift off, call")

1. **Call for help**; note time; **don't handle cord** (vasospasm) – if outside, **wrap in warm saline gauze/ replace into vagina gently**
2. **Relieve pressure on cord**: **gloved hand in vagina – push presenting part up** (keep hand there till CS)
3. **Position**: **knee-chest** or **exaggerated Sims' (left lateral with hips elevated)** or **steep Trendelenburg**
4. **Bladder filling – 500–750 ml NS via Foley** (lifts presenting part) + clamp
5. **Tocolysis** (terbutaline 0.25 mg SC) to ↓ contractions
6. **O_2 to mother**, IV line, continuous FHS
7. **Immediate delivery – LSCS** (fastest) unless **fully dilated & delivery imminent → vacuum/ forceps** (vertex) / breech extraction
8. Cord **pulsatile = baby alive**; neonatal team ready

14.6 Shoulder Dystocia

- **Failure of shoulders to deliver after head by gentle downward traction** – **anterior shoulder impacted behind symphysis pubis**. **"Turtle sign"** – head delivered, then **retracts against perineum**.
- **Risk**: **macrosomia (> 4–4.5 kg) – most important**, **maternal diabetes**, obesity, post-term, prolonged 2nd stage, instrumental delivery, previous shoulder dystocia.
- **Complications**: fetal – **brachial plexus injury (Erb's palsy – C5–C6, "waiter's tip")**, **clavicle/ humerus fracture**, hypoxia, death; maternal – **PPH, 3rd/ 4th degree tears**.

Management – "HELPERR"

H – **Call for Help** (note time; head-to-body interval) ·
E – **Evaluate for Episiotomy** (to allow hand entry) ·
L – **Legs – McRoberts manoeuvre** (hyperflex mother's thighs onto abdomen – **first & most effective; ~40–50 % success**) ·
P – **Suprapubic Pressure (Rubin I)** – on posterior aspect of anterior shoulder, **NOT fundal pressure** ·
E – **Enter** – internal rotational manoeuvres (**Rubin II, Woods' corkscrew, reverse Woods**) ·
R – **Remove posterior arm** (deliver posterior arm) ·
R – **Roll** – **all-fours (Gaskin manoeuvre)**.
Last resort – **cleidotomy** (dead fetus), **Zavanelli** (push head back + CS), symphysiotomy.
NEVER: fundal pressure, excessive lateral/ downward traction on head, pivoting → worsens impaction, brachial plexus injury.

14.7 Amniotic Fluid Embolism & Others

- **Amniotic fluid embolism (AFE)** – amniotic fluid/ fetal cells enter maternal circulation → anaphylactoid reaction; during **labour/ delivery/ immediately postpartum**; **sudden dyspnoea, cyanosis, hypoxia, hypotension/ cardiovascular collapse, seizures, DIC (bleeding)** → **very high mortality**; risk – **precipitate labour, oxytocin hyperstimulation, abruption, CS, multiparity, advanced age**. Management – **ABC, CPR (perimortem CS within 4–5 min if > 20 wks)**, O_2/ ventilation, inotropes, treat DIC (blood products).
- **Obstetric shock** – haemorrhagic (PPH, APH, ectopic, rupture), **neurogenic (inversion – disproportionate)**, septic (abortion, chorioamnionitis), cardiogenic, anaphylactic, embolic (AFE, PE). Management – **legs raised (left lateral tilt if pregnant)**, O_2, 2 IV lines, fluids/ blood, treat cause.
- **DIC in obstetrics** – **abruptio placentae (most common cause)**, **IUD (retained > 4 wks)**, **AFE**, **septic abortion**, severe pre-eclampsia/ HELLP, massive PPH. Lab – **↓ platelets, ↓ fibrinogen (< 100), ↑ PT/ aPTT, ↑ D-dimer/ FDP**; **bedside clot observation test** (clot not formed in 7–10 min / clot lyses); management – **treat cause (deliver)**, FFP, cryoprecipitate, platelets, blood.
- **Pulmonary embolism / DVT** – leading direct cause of maternal death in developed countries; prophylaxis LMWH (high risk), early ambulation.
- **Maternal collapse – "4 Hs & 4 Ts" + pregnancy causes**; CPR in pregnancy – **manual left uterine displacement**, hands slightly higher on sternum.

PYQ style

- PPH definition – **≥ 500 ml vaginal / ≥ 1000 ml CS**; primary within **24 hrs**.
- Commonest cause of PPH – **uterine atony**; commonest cause of maternal death – **PPH**.
- Secondary PPH commonest cause – **retained bits/ infection**.
- First step in PPH – **call for help & uterine massage**; empty **bladder**.
- First-line uterotonic – **oxytocin**; **methergine contraindicated in HTN**; **carboprost contraindicated in asthma**.
- Tranexamic acid – **1 g IV within 3 hrs**.
- Bleeding with contracted uterus – **genital tract tears**.
- Retained placenta – **> 30 min** → **manual removal**.
- Uterine inversion – shock **out of proportion**; Rx – **immediate manual replacement (Johnson)**.
- Earliest sign of uterine rupture – **fetal distress (abnormal FHR)**; commonest cause – **previous CS scar**.
- Cord prolapse – **push presenting part up, knee-chest, LSCS**.
- Shoulder dystocia first manoeuvre – **McRoberts**; never **fundal pressure**.
- Turtle sign – **shoulder dystocia**; injury – **Erb's palsy**.
- Commonest obstetric cause of DIC – **abruptio placentae**.
- Lactation failure after PPH – **Sheehan's syndrome**.

15 Complications of Puerperium & Newborn Problems

15.1 Puerperal Pyrexia & Sepsis

- **Puerperal pyrexia** – temperature **≥ 38 °C (100.4 °F)** on **2 separate occasions 24 hrs apart** (excluding first 24 hrs) **within first 10 days** postpartum.
- Causes: **genital tract infection (puerperal sepsis) – most common**, **UTI**, **breast (engorgement/ mastitis/ abscess)**, **wound infection (episiotomy/ CS)**, **thrombophlebitis**, respiratory (atelectasis), malaria, typhoid.
- **Puerperal sepsis** – infection of genital tract after childbirth (**within 42 days**); **major cause of maternal death**; organisms – **anaerobic Streptococci**, **E. coli**, **Group A β-haemolytic Streptococcus** (most virulent), Staph, Clostridium (gas gangrene).
- **Risk factors**: **prolonged labour**, **PROM > 18 hrs**, **frequent PV examinations**, **unclean delivery (home/ untrained)**, **retained products**, **CS**, PPH, anaemia, malnutrition, manual removal, instrumental, diabetes.
- **Types**: localized – **endometritis (most common)**, perineal/ vaginal/ cervical infection; spreading – **parametritis (pelvic cellulitis)**, salpingitis, **pelvic peritonitis**, thrombophlebitis; generalized – **septicaemia, peritonitis, septic shock**.
- **Signs**: fever, chills, tachycardia, **foul-smelling lochia (or scanty lochia – lochiometra)**, **lower abdominal pain, uterine tenderness, subinvolution**, malaise, headache.
- **Management**: **isolation/ barrier nursing**, **high vaginal swab/ blood/ urine culture before antibiotics**, **broad-spectrum IV antibiotics (ampicillin + gentamicin + metronidazole)**, IV fluids, **Fowler's position (drainage of lochia, localize pus in pelvis)**, perineal hygiene, evacuate retained products, drain abscess, nutrition, analgesics/ antipyretics, monitor vitals 4-hrly, I/O; septic shock – resuscitation.
- **Prevention**: **"6 cleans" (WHO/ clean delivery)** – **clean hands, clean surface, clean blade, clean cord tie, clean perineum, clean cord stump** (+ clean towel); **asepsis**, **limit PV examinations (4-hrly)**, prophylactic antibiotics (CS), early ambulation, nutrition, treat anaemia.

15.2 Other Puerperal Complications

Complication	Key features	Management
UTI/ pyelonephritis	Frequency, dysuria, fever, loin pain (catheterization, trauma)	Fluids, antibiotics (culture)
Urinary retention	Can't void within 6–8 hrs (perineal pain, epidural, oedema, trauma)	Privacy, warm water, ambulation, analgesics → **catheterization**; overdistension → **atony & PPH**
Obstetric fistula (VVF/ RVF)	**Prolonged obstructed labour – pressure necrosis → continuous dribbling of urine** (VVF) from 3rd–10th day; faecal (RVF)	**Continuous bladder drainage 10–14 days** (small fresh fistula may heal), **surgical repair after 3 months**; prevention – **avoid prolonged labour, catheter in obstructed labour**
Breast engorgement	Day 3–5, painful full breasts, low fever	Frequent feeding, warm compress & expression

Complication	Key features	Management
Mastitis/ breast abscess	Wedge-shaped red tender area, fever (Staph aureus)	**Continue feeding**, antibiotics; abscess → drainage
Subinvolution	Uterus larger, boggy, persistent red lochia	Treat cause (retained bits – evacuation, infection – antibiotics), methergine
Secondary PPH	Day 1 – 6 weeks; retained bits, infection	Uterotonics, antibiotics, evacuation
Superficial thrombophlebitis	Varicose veins – tender cord, redness	Elevation, warm compress, analgesics, support stockings
DVT (phlegmasia alba dolens – "white leg")	Calf pain, swelling, **Homans' sign** (dorsiflexion → calf pain – not reliable), fever (2nd week)	**Doppler**; **heparin/ LMWH → warfarin** (breastfeeding safe); **bed rest with leg elevated** initially, **no massage**; stockings
Pulmonary embolism	Sudden dyspnoea, chest pain, tachycardia, collapse	O_2, propped up, heparin, thrombolysis
Haematoma (vulval/ vaginal/ broad ligament)	**Severe perineal pain, tense tender swelling, urinary retention**, shock	Small – ice packs; large/ expanding – **incision & evacuation**
Wound infection/ dehiscence (episiotomy, CS)	Redness, discharge, gaping	Sitz bath, antibiotics, resuturing (secondary)
Postpartum eclampsia	Fits within 48 hrs (up to 6 wks)	$MgSO_4$, BP control
Coccydynia, symphysis separation, foot drop	Birth trauma	Analgesics, support

15.3 Postpartum Psychiatric Disorders

Disorder	Onset	Incidence	Features	Management
Postpartum blues (baby blues/ maternity blues)	**Day 3–5 (peaks day 5)**, lasts **≤ 2 weeks**	**50–80 %**	Tearfulness, mood swings, anxiety, irritability, insomnia; **self-limiting**	**Reassurance, support, rest**; no drugs
Postpartum depression	Within **4 weeks–1 year** (common 2–3 months)	**10–15 %**	Persistent low mood > 2 wks, loss of interest, guilt, poor bonding, suicidal/ **infanticidal thoughts**; **Edinburgh Postnatal Depression Scale (EPDS)** screening	Counselling (CBT), **SSRIs (sertraline – compatible with breastfeeding)**, family support
Postpartum (puerperal) psychosis	**Within first 2 weeks** (abrupt, day 3–14)	**1–2 per 1000**	**Hallucinations, delusions** (often about baby), confusion, severe mood swings, **risk of suicide & infanticide** – **psychiatric emergency**; risk – **bipolar disorder**, previous episode	**Hospitalization (mother-baby unit), antipsychotics, mood stabilizers, ECT**; **don't leave mother alone with baby**; supervised breastfeeding

15.4 Common Neonatal Problems

Birth asphyxia & HIE

- Failure to initiate/ sustain breathing at birth; **APGAR ≤ 3 at 5 min**, metabolic acidosis; **Hypoxic-ischaemic encephalopathy (Sarnat staging I–III)** – lethargy, seizures, hypotonia. Rx – resuscitation, supportive, **therapeutic hypothermia (cooling 33.5 °C × 72 hrs)** in selected, anticonvulsants (**phenobarbitone – drug of choice for neonatal seizures**).

Respiratory distress

- **RDS (Hyaline membrane disease)** – **surfactant deficiency** – **preterm (< 34 wks)**, IDM; onset **within 6 hrs**, **grunting, tachypnoea > 60, chest retractions, nasal flaring, cyanosis**; X-ray – **ground-glass, air bronchogram**; Rx – **CPAP**, **surfactant (intratracheal)**, O_2, ventilation; prevention – **antenatal steroids**. **Silverman-Anderson score** (0–10).
- **Transient tachypnoea of newborn (TTN)** – **delayed clearance of lung fluid** – **term, elective CS**; resolves in 24–72 hrs.
- **Meconium aspiration syndrome (MAS)** – **post-term, fetal distress**; chemical pneumonitis, PPHN.

Neonatal hypoglycaemia

- **Blood glucose < 45 mg/dL (< 40)**; risk – **IDM (infant of diabetic mother – most common)**, preterm, SGA/ IUGR, LBW, hypothermia, sepsis, asphyxia.
- Signs – **jitteriness/ tremors**, lethargy, poor feeding, apnoea, hypothermia, high-pitched cry, **seizures**; may be asymptomatic → screen at-risk babies.
- Rx – **early & frequent breastfeeding**; symptomatic/ < 25 → **IV 10 % dextrose 2 ml/kg bolus** → infusion **6–8 mg/kg/min**.

Neonatal sepsis

- **Early onset (< 72 hrs)** – maternal (PROM, chorioamnionitis, fever) – **E. coli, Group B Strep, Klebsiella**; **late onset (> 72 hrs)** – nosocomial/ community – **Klebsiella, Staph aureus**.
- Signs – **refusal to feed (earliest)**, lethargy, temperature instability (**hypothermia more common**), apnoea, jaundice, vomiting, distension, poor cry, **sclerema**, seizures, **bulging fontanelle (meningitis)**.
- Rx – **blood culture** → **ampicillin + gentamicin** (first-line), supportive; **hand hygiene** = best prevention; **omphalitis** – umbilical redness/ pus.
- **Ophthalmia neonatorum** – conjunctivitis **within 28 days**; **gonococcal (day 2–5, purulent – most serious, corneal ulcer)**, chlamydial (day 5–14); Rx – ceftriaxone (gono), erythromycin (chlamydia); prophylaxis – clean eyes.

Birth injuries

Injury	Features
Caput succedaneum	**Scalp oedema, crosses sutures**, at birth, resolves 2–3 days
Cephalhaematoma	**Subperiosteal blood**, **doesn't cross sutures**, appears after hours, resolves weeks–months; **jaundice risk**; **don't aspirate**
Subgaleal haemorrhage	Below aponeurosis – **vacuum** – can be massive (shock)

Injury	Features
Erb's palsy (Erb-Duchenne)	**C5–C6** – **shoulder dystocia/ breech** – arm adducted, internally rotated, forearm pronated – **"waiter's tip"**; **asymmetric Moro**, grasp present; Rx – **positioning (abduct & externally rotate), physiotherapy** – most recover
Klumpke's palsy	**C8–T1** – claw hand, absent grasp, Horner's
Clavicle fracture	**Most common birth fracture** – crepitus, asymmetric Moro → heals spontaneously
Facial palsy	**Forceps** – CN VII; eye care
Intracranial haemorrhage	Preterm (IVH), breech, precipitate, excessive moulding

Common congenital anomalies (screen at birth)

- **Neural tube defects** – **anencephaly** (incompatible with life), **spina bifida/ meningomyelocele** (protect sac – **prone position, sterile moist saline dressing, no pressure**), encephalocele; prevention – **periconceptional folic acid**.
- **Hydrocephalus** – ↑ head circumference, **bulging fontanelle, sunset eyes**; VP shunt.
- **Cleft lip/ palate** – feeding (special nipple, upright, burp often); **cleft lip repair ~3 months (rule of 10s)**, palate 9–18 months; **elbow restraints** post-op.
- **Oesophageal atresia with TEF** – **polyhydramnios**, **excessive frothy drooling, choking/ cyanosis with feeds**; **can't pass NG tube (coils) → NPO, prop up, continuous suction of pouch**, surgery.
- **Imperforate anus**, **Hirschsprung's** (no meconium in 48 hrs), **congenital hypertrophic pyloric stenosis** (projectile non-bilious vomiting **2–6 wks**, olive mass), **CTEV (club foot – Ponseti casting)**, **DDH (Ortolani, Barlow; Pavlik harness)**, **Down syndrome (trisomy 21 – maternal age, simian crease, flat face, hypotonia)**, **congenital hypothyroidism (TSH screening)**, **congenital heart disease (PDA, VSD – commonest)**.

✎ PYQ style

- Puerperal pyrexia – **≥ 38 °C** on 2 occasions 24 hrs apart within 10 days.
- Commonest puerperal infection – **endometritis**; position – **Fowler's**.
- Prevention of sepsis – **6 cleans**; limit PV examinations.
- VVF cause – **prolonged obstructed labour**; Rx – continuous catheter drainage, repair after **3 months**.
- Postpartum blues – **day 3–5**, self-limiting; psychosis – **emergency, within 2 wks**.
- RDS cause – **surfactant deficiency**; prevention – **antenatal steroids**.
- Neonatal hypoglycaemia – **< 45 mg/dL**; most common in – **IDM**.
- Earliest sign of neonatal sepsis – **refusal to feed**; drug – **ampicillin + gentamicin**.
- Waiter's tip – **Erb's palsy (C5–C6)**; commonest birth fracture – **clavicle**.
- Neonatal seizure drug of choice – **phenobarbitone**.
- Meningomyelocele – **prone, sterile moist dressing**.
- TEF – **frothy drooling, NG tube coils**.

16 Obstetric Operations & Procedures

16.1 Caesarean Section (CS)

D Caesarean section

Delivery of fetus **through incisions on abdominal wall (laparotomy) and uterus (hysterotomy)** after viability. **Lower segment CS (LSCS)** – transverse incision on **lower uterine segment** – **most common (~99 %)**; **Classical CS** – vertical incision on **upper segment**.

	LSCS	Classical
Incision	**Transverse, lower segment** (Kerr's)	**Vertical, upper segment**
Blood loss	Less	More
Healing	Better (passive segment)	Poor
Scar rupture risk later	**Low (0.5–1 %)** – VBAC possible	**High (4–9 %, may rupture before labour)** – **VBAC contraindicated**
Adhesions/ peritonitis	Less	More
Indications	Most cases	**Lower segment inaccessible** – dense adhesions, **fibroid in LUS**, **transverse lie with back down**, very preterm (no LUS), **perimortem CS**, Ca cervix, placenta praevia with large vessels

- **Indications**: **Absolute** – **CPD/ contracted pelvis**, **central placenta praevia**, **obstructed labour (live baby)**, transverse lie (term), brow, **mento-posterior face**, Ca cervix, previous classical CS, previous VVF repair. **Relative** – **previous LSCS (commonest indication currently)**, **fetal distress**, **failed induction/ non-progress of labour**, breech (primi), malpresentation, severe pre-eclampsia/ eclampsia (unfavourable cervix), APH, cord prolapse, twins (1st non-vertex), **elderly primi**, bad obstetric history, macrosomia, HIV (high viral load), active herpes.
- **Categories (urgency – RCOG)**: **1** – immediate threat to life (**decision-to-delivery ≤ 30 min**), 2 – compromise, 3 – needs early delivery, 4 – elective.
- **Robson's 10-group classification** – CS audit (WHO).
- **Anaesthesia** – **spinal (most common)**, epidural, GA (emergency – aspiration risk: **Mendelson's syndrome** – acid aspiration → **antacid/ ranitidine/ PPI, metoclopramide pre-op, cricoid pressure**).

Pre-operative care

- **Informed consent**, **NPO 6–8 hrs** (elective), **pre-op investigations** (Hb, grouping & **cross-match 2 units**, BT/CT, HIV, HBsAg, urine), **antacid prophylaxis**, **IV line**, **Foley catheter** (empty bladder – injury prevention), **skin prep (clipping)**, **prophylactic antibiotic (cefazolin 2 g IV within 60 min before incision)**, remove jewellery/ dentures, **left lateral tilt (15°) on table** (aortocaval compression), FHS check before shifting, neonatal team & resuscitation ready.

Post-operative care

- **Vitals every 15 min (1st hr)**, then 30 min, hourly; **uterine tone & vaginal bleeding (PPH)**; dressing (soakage); **IV fluids**; **oxytocin infusion**; **urine output (catheter removed after 12–24 hrs)**; **pain relief** (opioids, NSAIDs, epidural); **position – lateral/ supine (spinal – flat few hrs not mandatory)**, then semi-Fowler's.
- **Early breastfeeding (within 1 hr – skin-to-skin in OT/ recovery)** – side-lying or **football hold**.
- **Oral fluids** after **6–8 hrs** (if bowel sounds; early feeding encouraged – ERAS), soft diet next day; **early ambulation – within 12–24 hrs** (DVT prevention), deep breathing & coughing (**splint wound**), leg exercises.
- **Wound** – dressing removed on day 2–3; **sutures/ staples removed day 7–8** (subcuticular absorbable – no removal).
- **Complications**: **PPH**, **infection (wound, endometritis, UTI)**, **paralytic ileus**, **DVT/ PE**, bladder/ bowel injury, anaesthesia complications, **spinal headache**, wound dehiscence; long-term – adhesions, **scar rupture**, **placenta accreta/ praevia in next pregnancy**, scar ectopic.
- **Discharge advice**: avoid heavy lifting **6 wks**, **contraception – space next pregnancy ≥ 18–24 months**, **institutional delivery next time (previous CS – high risk)**, wound care, warning signs.

16.2 Instrumental (Operative Vaginal) Delivery

Prerequisites ("FORCEPS" / "ABCDEFGHIJ")

- **Cervix fully dilated**, **membranes ruptured**, **head engaged (≤ 1/5 palpable, station ≥ 0/ +2)**, **position known exactly**, **vertex (or face mento-anterior for forceps; no face for vacuum)**, **no CPD**, **empty bladder**, adequate analgesia (pudendal block/ epidural), **episiotomy**, informed consent, adequate contractions (vacuum), experienced operator, **back-up for CS ("trial in OT")**, neonatologist present.

	Obstetric Forceps	Vacuum (Ventouse)
Inventor	**Chamberlen** (17th century)	**Malmström** (1954 – metal cup); silastic/ Kiwi
Principle	**Traction + rotation** (blades on fetal head – biparietal-bimalar)	**Suction cup on scalp – traction**; **flexion point** (3 cm anterior to posterior fontanelle on sagittal suture)
Maternal injury	**More (tears)**	**Less**
Fetal injury	Facial palsy, facial bruises, skull fracture	**Cephalhaematoma, subgaleal haemorrhage**, retinal haemorrhage, **chignon (artificial caput – disappears in hours)**, jaundice
Can't be used	–	**Preterm < 34 wks**, **face/ brow/ breech**, fetal bleeding disorder, recent scalp blood sampling
Anaesthesia	Needs good analgesia	Less needed
Special uses	**After-coming head of breech (Piper's)**, face (mento-anterior), **rotation (Kielland's)**, CS head delivery (Wrigley's)	Doesn't occupy space – rotation automatic

- Types of forceps – **Wrigley's (outlet – short)**, **Simpson's/ Das's/ Neville-Barnes (mid/ low)**, **Kielland's (rotational – no pelvic curve, sliding lock)**, **Piper's (after-coming head of breech)**.

- Classification (ACOG): **outlet** (scalp visible without separating labia, rotation ≤ 45°), **low (station ≥ +2)**, **mid (station 0 to +2)**; **high forceps – obsolete**.
- **Vacuum** – pressure **0.6–0.8 kg/cm² (≈ 500–600 mmHg)**; **max 3 pulls with descent** / **cup detachment ≤ 2–3 times** / total ≤ 15–20 min → abandon → CS.
- **Indications**: **prolonged 2nd stage**, **maternal exhaustion**, **fetal distress in 2nd stage**, **maternal conditions where bearing down is harmful – heart disease, severe pre-eclampsia/ HTN, severe anaemia, previous CS (cut short 2nd stage)**, ↑ICP, retinal detachment.

16.3 Other Procedures

Procedure	Key points
ARM (Artificial Rupture of Membranes/ Amniotomy)	Induction/ augmentation; **head engaged/ well applied (cord prolapse risk if high)**; **check FHS before & after**, note liquor colour & amount; contraindicated – praevia, HIV (relative), malpresentation
Membrane sweeping (stripping)	Finger separates membranes from lower segment → PG release → labour within 48 hrs
External cephalic version (ECV)	**Breech at 36–37 wks** – convert to cephalic; tocolysis; CTG; **anti-D if Rh –ve**
Internal podalic version	**Only for 2nd twin (transverse)** – hand in uterus, grasp foot, convert to breech → extract
Manual removal of placenta (MRP)	Retained placenta > 30 min/ bleeding – GA/ sedation, aseptic, **antibiotics**, uterotonics
MVA (Manual Vacuum Aspiration)	Incomplete abortion/ MTP **up to 12 wks** – **Karman cannula + 60 ml syringe (−60 to −70 cmHg)** – no electricity, **paracervical block**; **safer than D&C**
D&C (Dilatation & Curettage)	Hegar's dilators + curette – diagnostic (endometrial biopsy)/ evacuation; risk – **perforation**, **Asherman's syndrome (intrauterine adhesions)**
Cervical cerclage	**McDonald (most common)**/ Shirodkar – at **12–14 wks**; **removed at 37 wks** or in labour/ PROM
Destructive operations	**Dead fetus** in obstructed labour (rare now): **craniotomy** (perforate & extract head – hydrocephalus/ obstructed), **decapitation** (neglected shoulder), **cleidotomy** (cut clavicle – shoulder dystocia dead fetus), evisceration
Symphysiotomy	Divide symphysis pubis (rare)
Amnioinfusion	NS into uterine cavity – **variable decelerations from oligohydramnios/ cord compression**, thick meconium
Amniocentesis, CVS, cordocentesis	See fetal wellbeing chapter

16.4 Analgesia & Anaesthesia in Labour

- **Non-pharmacological**: **continuous labour support (birth companion – most effective non-drug)**, breathing techniques, **ambulation/ positioning**, **back massage & counter-pressure**, warm compress, **hydrotherapy**, TENS, music, hypnosis, aromatherapy; **Lamaze (psychoprophylaxis)**, **Grantly Dick-Read (fear-tension-pain cycle – natural childbirth)**, **Bradley (husband-coached)**.

- **Inhalational** – **Entonox (50 % N_2O + 50 % O_2)** – self-administered, start inhaling **30 sec before contraction**.
- **Systemic opioids** – **Pethidine (meperidine) 50–100 mg IM** – avoid **within 2–4 hrs of expected delivery** (**neonatal respiratory depression** – antidote **naloxone 0.1 mg/kg**); fentanyl, tramadol.
- **Regional – Epidural analgesia (gold standard/ most effective)** – **L3–L4 / L2–L3**, catheter, bupivacaine + fentanyl; side effects – **hypotension (most common – preload IV fluids, left tilt)**, prolonged 2nd stage, ↑ instrumental, urinary retention, fever, **post-dural puncture headache**, high block; contraindications – coagulopathy, infection at site, hypovolaemia, patient refusal, ↑ICP.
- **Spinal** (single shot – CS); **CSE** (combined spinal-epidural).
- **Pudendal nerve block** – **S2–S4**, transvaginal near **ischial spine** – **outlet forceps/ vacuum, episiotomy repair**.
- **Paracervical block** – MVA/ D&C (fetal bradycardia if in labour).
- **Local perineal infiltration** – **1 % lignocaine** (max 3 mg/kg; with adrenaline 7 mg/kg) – episiotomy.
- **General anaesthesia** – emergency CS; risks – **aspiration (Mendelson's)**, failed intubation, awareness, neonatal depression, **atony (halothane)**.

PYQ style

- Commonest type of CS – **LSCS (transverse lower segment)**.
- Commonest indication of CS today – **previous CS**.
- VBAC contraindicated after – **classical CS**.
- Mendelson's syndrome – **acid aspiration under GA**; prevention – **antacids, H2 blockers, cricoid pressure**.
- Prerequisites for forceps – **full dilatation, ROM, engaged head, empty bladder**.
- Vacuum contraindicated – **preterm < 34 wks, face, breech**.
- Chignon – **vacuum** (artificial caput).
- Forceps for after-coming head – **Piper's**; rotational – **Kielland's**.
- MVA used up to – **12 wks**; complication of curettage – **Asherman's syndrome**.
- Cerclage removed at – **37 wks**.
- Pethidine avoided near delivery due to – **neonatal respiratory depression**.
- Most effective labour analgesia – **epidural**; common side effect – **hypotension**.
- Pudendal block nerve roots – **S2–S4**.

17 Drugs in Obstetrics

17.1 Uterotonics (Oxytocics)

Drug	Dose / Route	Uses	Side effects / Contraindications / Nursing
Oxytocin (Pitocin/ Syntocinon) – posterior pituitary hormone (synthetic)	**AMTSL: 10 IU IM**; **induction/ augmentation: low dose IV infusion (2.5–5 IU in 500 ml)** start 1–2 mU/min, ↑ every 30 min; **PPH: 20–40 IU in 500–1000 ml**	**Drug of choice – PPH prevention & treatment**, **induction & augmentation**, after LSCS, incomplete abortion	**Uterine hyperstimulation → fetal distress, rupture uterus**; **water intoxication** (ADH-like – headache, vomiting, convulsions); hypotension (IV bolus – **never IV bolus**); **contraindicated – CPD, obstructed labour, malpresentation, praevia, previous classical CS, fetal distress**; **store 2–8 °C**; **infusion pump, monitor contractions & FHS every 15–30 min**
Carbetocin	**100 µg IM/ IV** (heat-stable version available)	PPH prevention (CS)	Long-acting oxytocin analogue
Methylergometrine (Methergine) – ergot alkaloid	**0.2 mg IM/ IV (slow)**; repeat every 15 min (max 5 doses/ 1 mg); oral 0.125 mg TDS (subinvolution)	**PPH (atony)**, subinvolution	**Sustained tonic contraction** – **never before delivery of baby (and 2nd twin)**; **HTN/ vasoconstriction, nausea, vomiting, headache**; **contraindicated – hypertension, pre-eclampsia/ eclampsia, heart disease, peripheral vascular disease, HIV on protease inhibitors**; **light sensitive, refrigerate**
Ergometrine	0.2–0.5 mg	As above	Same
Syntometrine	Oxytocin 5 IU + ergometrine 0.5 mg IM	AMTSL (UK)	
Carboprost (15-methyl PGF2α – Prostodin/ Hemabate)	**250 µg IM (deep)/ intramyometrial**, every 15–90 min, **max 8 doses (2 mg)**	**Refractory PPH (atony)**, 2nd trimester abortion	**Bronchospasm** – **contraindicated in asthma**; **diarrhoea (most common)**, vomiting, fever, shivering
Misoprostol (PGE1 – Cytotec)	**PPH prevention: 600 µg oral**; **PPH Rx: 800 µg sublingual**; **induction: 25 µg vaginal 4-hrly / 25 µg oral 2-hrly**; **MMA: 800 µg** after mifepristone; incomplete abortion 400–600 µg	**PPH (no refrigeration – community)**, **cervical ripening/ induction**, **medical abortion**, missed/ incomplete abortion, IUD	**Shivering, fever**, diarrhoea, vomiting; **hyperstimulation/ rupture (scarred uterus – avoid in previous CS for induction)**; cheap, heat-stable

Drug	Dose / Route	Uses	Side effects / Contraindications / Nursing
Dinoprostone (PGE2 – Cerviprime gel/ Prostin E2)	**0.5 mg gel intracervical**, repeat 6-hrly (max 3); vaginal tablet 3 mg; insert 10 mg	**Cervical ripening** (unfavourable Bishop)	Hyperstimulation; **refrigerate**; patient lies down 30 min after
Mifepristone (RU-486) – antiprogesterone	**200 mg oral** (MMA/ IUD)	Medical abortion with misoprostol, IUD induction, cervical priming	Bleeding, cramps

17.2 Tocolytics (uterine relaxants)

Drug	Dose	Side effects
Nifedipine (Ca channel blocker) – **first-line**	**20 mg oral loading** → 10–20 mg 6–8 hrly	Headache, flushing, hypotension, tachycardia
Ritodrine/ Isoxsuprine/ Terbutaline (β_2-agonists)	Terbutaline **0.25 mg SC** (acute – hyperstimulation, inversion)	**Maternal tachycardia (hold if > 120)**, palpitation, tremor, **hyperglycaemia, hypokalaemia, pulmonary oedema** – avoid in heart disease, diabetes
Atosiban (oxytocin antagonist)	IV infusion	Few side effects; costly
Indomethacin (PG synthetase inhibitor)	50–100 mg	**< 32 wks, ≤ 48 hrs** – **premature closure of ductus arteriosus, oligohydramnios**
$MgSO_4$	IV	Neuroprotection < 32 wks (weak tocolytic)

- Purpose – **delay delivery 48 hrs** for **antenatal steroids & in-utero transfer**.

17.3 $MgSO_4$ & Antihypertensives

- **Magnesium sulphate** – **drug of choice for eclampsia & prevention in severe pre-eclampsia**; Pritchard (IM) / Zuspan (IV) – see Chapter 11; **monitor knee jerk, RR ≥ 16, urine ≥ 30 ml/hr**; **antidote – 10 % calcium gluconate 10 ml IV over 10 min**; therapeutic level **4–7 mEq/L**.

Antihypertensive	Use
Labetalol	**First-line oral & IV** (20 mg IV, then 40, 80 every 10 min; max 300 mg) – avoid asthma
Nifedipine (immediate/ extended release, oral)	Acute severe HTN (10–20 mg oral); **not sublingual**
Methyldopa	**Safe in pregnancy (longest record)**, chronic/ gestational HTN; s/e – sedation, depression, haemolytic anaemia
Hydralazine	IV for hypertensive emergency (5 mg)
Contraindicated	**ACE inhibitors (-pril), ARBs (-sartan)** – fetal renal failure, oligohydramnios, skull defects; **atenolol** (IUGR); **diuretics** (except pulmonary oedema)

17.4 Other Important Drugs

Drug	Use / Dose
Corticosteroids – Dexamethasone 6 mg IM 12-hrly × 4 / **Betamethasone 12 mg IM 24-hrly × 2**	**Fetal lung maturity – preterm 24–34 wks** (↓ RDS, IVH, NEC)
Anti-D immunoglobulin	**300 µg IM** within **72 hrs** postpartum (baby Rh +ve); **28 wks** antenatal; **50 µg** < 12 wks abortion; IM deltoid; **not IV**; store 2–8 °C
Iron (ferrous sulphate/ fumarate)	**60 mg elemental + 500 µg FA** prophylaxis; IV **iron sucrose (100–200 mg)**, **ferric carboxymaltose (up to 1000 mg)**
Folic acid	**0.4 mg (400 µg)** preconception & 1st trimester; **4–5 mg** – previous NTD, epilepsy (valproate), diabetes
Calcium	**1 g elemental/ day** from 14 wks to 6 months PNC (pre-eclampsia prevention, bone)
Low-dose aspirin	**75–150 mg at night from 12–16 wks to 36 wks** – pre-eclampsia prevention (high risk), APLA
Heparin/ LMWH (enoxaparin)	Doesn't cross placenta – **safe**; DVT, mechanical valves, APLA
Warfarin	**Teratogenic (6–12 wks – embryopathy)**; avoid 1st trimester & near term
Insulin	**Drug of choice for diabetes in pregnancy** (doesn't cross placenta)
Metformin	GDM (accepted)
Doxylamine + pyridoxine	Nausea & vomiting of pregnancy
Antibiotics safe	**Penicillins, cephalosporins, erythromycin/ azithromycin, nitrofurantoin (avoid near term), metronidazole (after 1st trim)**
Antibiotics avoid	**Tetracyclines (teeth/ bone)**, **aminoglycosides – streptomycin (ototoxic)**, **fluoroquinolones**, **chloramphenicol (gray baby)**, **co-trimoxazole (near term – kernicterus; 1st trim – folate antagonist)**
Thyroxine	Hypothyroidism – dose ↑ in pregnancy
PTU (1st trim) → carbimazole	Hyperthyroidism
Antiepileptics	**Valproate avoid (NTD)**; lamotrigine/ levetiracetam preferred + folic acid 5 mg
Vitamin K (newborn)	**1 mg IM** (0.5 mg < 1500 g)
Naloxone (newborn)	**0.1 mg/kg** – opioid-induced respiratory depression

- **Drugs in breastfeeding** – most drugs safe; **contraindicated**: **anticancer (cytotoxics)**, **radioactive iodine/ isotopes**, **lithium**, amiodarone, **ergotamine/ bromocriptine (suppress lactation)**, high-dose oestrogen (↓ milk – combined OCP avoid < 6 months), tetracycline (prolonged), chloramphenicol, drugs of abuse. **Safe**: paracetamol, ibuprofen, penicillins, cephalosporins, insulin, heparin, warfarin, thyroxine, most antihypertensives (labetalol, nifedipine, methyldopa, enalapril postpartum), SSRIs (sertraline), ATT, ART.

PYQ style

- Drug of choice for PPH – **oxytocin**; dose in AMTSL – **10 IU IM**.
- Oxytocin side effect (high dose) – **water intoxication**.
- Methergine contraindicated – **hypertension/ pre-eclampsia, heart disease**.
- Carboprost contraindicated – **bronchial asthma**; max dose – **2 mg (8 doses)**.
- Heat-stable uterotonic for community – **misoprostol**.
- Cervical ripening gel – **dinoprostone (PGE2)**.
- Tocolytic first-line – **nifedipine**; β-agonist s/e – **tachycardia, pulmonary oedema**.
- Indomethacin side effect – **premature closure of ductus arteriosus**.
- Antihypertensive contraindicated – **ACE inhibitors**.
- Steroid for lung maturity – **dexamethasone 6 mg IM × 4 doses**.
- Anticoagulant safe in pregnancy – **heparin**; teratogenic – **warfarin**.
- Folic acid dose (high risk NTD) – **4–5 mg**.
- Antibiotic causing gray baby – **chloramphenicol**; teeth staining – **tetracycline**.

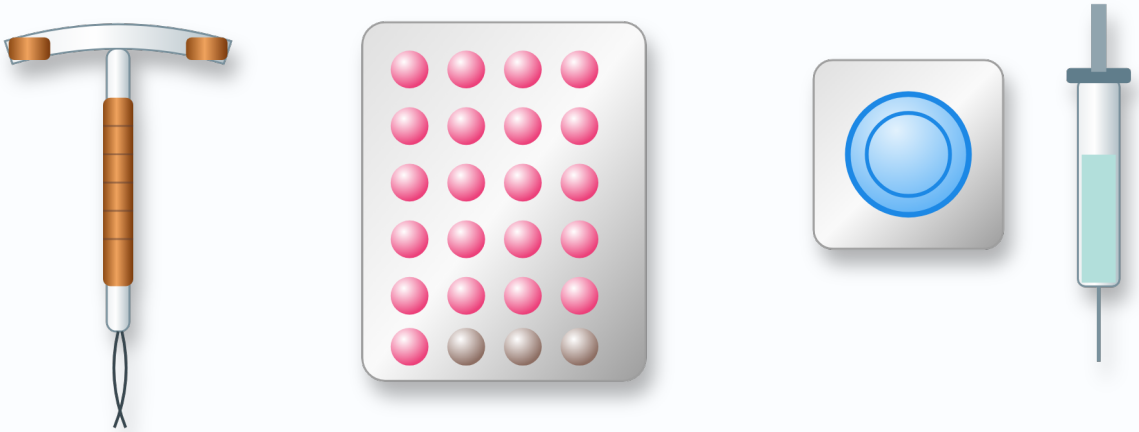
18 Family Planning & Contraception

18.1 Basics

- **Family planning (WHO)** – "A way of thinking and living adopted **voluntarily** upon the basis of knowledge, attitudes & responsible decisions by individuals and couples, to promote health & welfare of family...".
- India – **first country to launch a National Family Planning Programme (1952)**; now part of **RMNCH+A (Family Planning 2020 → FP 2030 commitments)**.
- **Eligible couple** – currently married couple, **wife aged 15–45 yrs** (reproductive age). **Couple protection rate (CPR)** – % of eligible couples protected by any method; **NFHS-5: CPR ~67 % (any method)**, modern methods ~56 %; **female sterilization most used (~38 %)**.
- **TFR India 2.0 (NFHS-5)**; Gujarat ~1.9; replacement level **2.1**.
- **Birth spacing** – **at least 2–3 years (≥ 24 months birth-to-pregnancy interval – WHO)**; after abortion ≥ 6 months.
- **Pearl Index** – **failures per 100 woman-years (HWY) of exposure** = (Number of accidental pregnancies × 1200) ÷ total months of exposure (**× 1300 if counted in 13 cycles**). Lower = more effective.
- **Ideal contraceptive**: safe, effective, reversible, cheap, acceptable, independent of coitus, long-acting, easy to use.

18.2 Classification

Contraceptive Methods – Basket of Choice



Cu-T 380A (IUCD)
10 years · copper 380 mm²
Cu-T 375 – 5 years
Best time: within 10 days of LMP, PPIUCD within 48 hrs/ at LSCS
Check thread after each period

Combined OCP (Mala-N)
21 hormonal + 7 iron (brown) = 28
Start day 1–5 of cycle, same time daily
Miss 1 pill → take ASAP
Chhaya (Centchroman) – weekly, non-steroidal (twice weekly × 3 months)

Condom (Nirodh)
Barrier + STI/ HIV protection

Injectable MPA (Antara)
DMPA 150 mg IM every 3 months

Permanent: Tubectomy (female – Pomeroy, laparoscopic) · Vasectomy (male – NSV)
Vasectomy ke baad ~3 months/ 20 ejaculations tak condom · **Emergency pill:** LNG 1.5 mg within 72 hrs

Contraceptive methods – "basket of choice"

Spacing (temporary/ reversible)	Terminal (permanent)
Barrier – physical (condom, diaphragm, cervical cap, vaginal sponge), chemical (spermicides – nonoxynol-9), combined	**Female sterilization (tubectomy)** – minilap, laparoscopic
IUCDs – copper, hormonal (LNG-IUS)	**Male sterilization (vasectomy)** – conventional, **NSV**
Hormonal – oral pills (COC, POP, Chhaya), injectables (DMPA – Antara), implants, vaginal ring, patch	
Post-conceptional – menstrual regulation, MTP	
Natural/ traditional – abstinence, **coitus interruptus**, **safe period (rhythm/ calendar)**, BBT, cervical mucus (Billings), symptothermal, **LAM**	

Basket of choice (Govt. of India – free)

Condom (Nirodh), **OCP (Mala-N)**, **Centchroman (Chhaya)**, **ECP (E-pill)**, **IUCD (Cu-T 380A, Cu 375)**, **PPIUCD/ PAIUCD**, **Injectable MPA (Antara)**, **Tubectomy**, **Vasectomy/ NSV**; + **Pregnancy testing kits (Nischay)**; ASHAs deliver contraceptives at doorstep (**HDC – Home Delivery of Contraceptives**); **Mission Parivar Vikas (2016)**; **Saas-Bahu Sammelan**, **Nayi Pehel kit**.

18.3 Barrier Methods

- **Condom (male)** – latex, **Nirodh (1963)**; **only method protecting against STI/ HIV** (+ female condom); Pearl index **2–14 (typical ~13)**; use new each act, check expiry, leave space at tip, hold base while withdrawing; **no oil-based lubricants** (latex damage).
- **Female condom (Femidom)** – polyurethane; **diaphragm** (with spermicide, keep **6 hrs after intercourse**, max 24 hrs; toxic shock), **cervical cap**, **vaginal sponge (Today)**.
- **Spermicides** – nonoxynol-9 (foam, jelly, tablet) – may ↑ HIV risk (mucosal irritation) if used frequently.

18.4 Intrauterine Contraceptive Devices (IUCD)

- **Generations**: **1st – inert (Lippes loop – double S, polyethylene)**; **2nd – copper (Cu-T 200, Cu-7, Cu-T 380A, Multiload 250/ 375, Nova-T)**; **3rd – hormone-releasing (Progestasert – progesterone, LNG-IUS/ Mirena – levonorgestrel 20 µg/day, 5 yrs)**.
- **Cu-T 380A – 10 years**; **Cu 375 (Multiload) – 5 years** (GoI). Copper → **sterile inflammatory reaction, toxic to sperm & ovum, prevents fertilization/ implantation**.
- **Timing of insertion**: **within 12 days (10–12) of start of menstruation (interval IUCD)** (os open, less pain, not pregnant), **PPIUCD – within 10 min of placenta (post-placental) / within 48 hrs of delivery / during CS (intra-caesarean)**, or **after 6 weeks**; **PAIUCD – immediately after abortion (within 12 days)**; **emergency contraception – within 5 days (120 hrs) of unprotected intercourse** (most effective EC).
- **Contraindications** (absolute): **suspected pregnancy**, **PID (current/ recent)**, **unexplained vaginal bleeding**, **cancer of genital tract**, **uterine malformations/ fibroids distorting cavity**, **STI**, **Wilson's disease (copper)**, **postpartum sepsis**, **trophoblastic disease** (persistent hCG).
- **Side effects/ complications**: **heavy/ prolonged menstrual bleeding (most common reason for removal) & pain/ dysmenorrhoea**, **spontaneous expulsion (most common in 1st year; first 3**

months/ during menses), **PID**, **uterine perforation** (rare; at insertion), **ectopic pregnancy (if failure)**, pregnancy with IUCD in situ (remove if threads visible – abortion risk), missing threads.

- **Instructions**: **check thread after each menstruation** (for expulsion), **follow-up – after first period (3–6 wks)**, then **3 months**, then yearly; report **"PAINS"** – **P**eriod late/ abnormal spotting, **A**bdominal pain/ dyspareunia, **I**nfection exposure/ abnormal discharge, **N**ot feeling well/ fever/ chills, **S**tring missing/ shorter/ longer.
- **Ideal IUCD candidate** – parous woman, monogamous, normal menses, wants long-term reversible spacing, no PID history.

18.5 Hormonal Contraceptives

Combined oral contraceptive pills (COC)

- **Mala-N (Govt. – free)** / Mala-D (social marketing): **Levonorgestrel 0.15 mg + Ethinyl oestradiol 0.03 mg** – **21 hormonal tablets + 7 iron (brown) tablets (ferrous fumarate 60 mg)** = 28.
- **Mechanism** – **inhibit ovulation (main – suppress FSH & LH)**, thick cervical mucus, endometrium unsuitable.
- **Start on day 1–5 of menstrual cycle**, **1 tablet daily at same time**; new pack immediately after 28.
- **Missed pill**: **1 pill missed** → take **as soon as remembered** (even 2 in one day) & continue; **≥ 2 pills missed** → take one ASAP, continue + **backup (condom) for 7 days**; if missed in last week of hormonal pills → skip iron tablets, start new pack; consider **ECP** if intercourse in previous 5 days.
- **Pearl index ~0.1–0.3 (perfect)**, typical ~7.
- **Benefits (non-contraceptive)**: regular, lighter, less painful periods, **↓ iron deficiency anaemia**, **↓ ovarian & endometrial cancer**, ↓ ectopic, ↓ PID, ↓ benign breast disease, ↓ functional ovarian cysts, acne improvement.
- **Side effects**: nausea, breast tenderness, **breakthrough bleeding/ spotting**, headache, weight gain, mood changes, **hypertension**, **thromboembolism (DVT/ stroke/ MI)**, **↓ lactation**, cholestatic jaundice.
- **Contraindications (absolute)**: **pregnancy**, **breastfeeding < 6 months** (WHO: < 6 wks absolute; 6 wks–6 months generally not), **thromboembolism/ history of DVT, stroke, IHD**, **hypertension (≥ 160/100)**, **smoker ≥ 35 yrs (≥ 15 cig/day)**, **migraine with aura**, **breast cancer**, **liver disease (active hepatitis, tumour)**, **undiagnosed vaginal bleeding**, diabetes with vascular complications, major surgery with immobilization; drug interaction – **rifampicin, anticonvulsants** (↓ efficacy).
- **Warning signs – "ACHES"**: **A**bdominal pain (severe), **C**hest pain/ breathlessness, **H**eadache (severe), **E**ye problems (blurred vision/ loss), **S**evere leg pain (calf) → stop & report.

Other hormonal methods

- **Progestin-only pill (POP/ minipill)** – **lactating women** (from 6 wks), **taken daily without break, same time (3-hr window)**; s/e – irregular bleeding.
- **Centchroman (Ormeloxifene – "Chhaya", earlier Saheli)** – **non-steroidal, non-hormonal SERM** (developed by **CDRI Lucknow, India**); **30 mg – twice a week for first 3 months, then once a week**; first pill on **first day of period**; **safe in lactation**; s/e – **delayed periods (oligomenorrhoea)**; no nausea, no weight gain.
- **Injectable – DMPA (Depot medroxyprogesterone acetate – "Antara" programme)** – **150 mg deep IM every 3 months** (grace period 4 wks early/ late; 2 wks early to 4 wks late); safe in lactation (from 6 wks); s/e – **menstrual irregularity/ amenorrhoea**, **delayed return of fertility (~7–10 months)**,

weight gain, **↓ bone density (long-term)**. NET-EN (2-monthly), monthly combined injectables (Cyclofem).

- **Implants** – **Jadelle (2 rods LNG – 5 yrs)**, **Nexplanon/ Implanon (single rod etonogestrel – 3 yrs)** – upper arm subdermal; most effective reversible.
- **Vaginal ring (NuvaRing – 3 wks)**, **transdermal patch (Ortho Evra – weekly)**.

Emergency contraception (EC)

- **Levonorgestrel 1.5 mg single dose (E-pill/ "i-pill")** (or 0.75 mg × 2, 12 hrs apart) **within 72 hrs** (best within 24 hrs; effective up to 120 hrs with lower efficacy) – prevents/ delays ovulation; **not an abortifacient**; **Cu-IUCD within 5 days (most effective EC, 99 %)**; **Ulipristal acetate 30 mg** (within 120 hrs); **Yuzpe method** (combined pills).
- Indications – unprotected intercourse, condom burst, missed pills, **rape**; **not for regular use**.

18.6 Natural Methods & LAM

- **Calendar/ rhythm (Ogino-Knaus)**: **first fertile day = shortest cycle − 18**; **last fertile day = longest cycle − 11**; **Standard Days Method (CycleBeads)** – cycles 26–32 days – **avoid days 8–19**.
- **BBT** – rise **0.3–0.5 °C** after ovulation → safe after 3 days of elevated temperature. **Cervical mucus/ Billings method** – fertile when mucus clear, slippery, stretchy (spinnbarkeit). **Symptothermal** – BBT + mucus + calendar.
- **Coitus interruptus (withdrawal)** – oldest method; high failure (pre-ejaculatory fluid).
- **LAM (Lactational Amenorrhoea Method)** – effective (**98 %**) only if **all 3 criteria**: (1) **exclusive/ nearly exclusive breastfeeding (day & night)**, (2) **amenorrhoea (no return of periods)**, (3) **baby < 6 months**. If any one fails → other method.

18.7 Permanent Methods (Sterilization)

	Tubectomy (female)	Vasectomy (male)
Procedure	**Minilap** (2–3 cm suprapubic incision – **Pomeroy's technique** – loop ligated with absorbable suture & excised – most common), **Laparoscopic (Falope ring/ Yoon's ring, clips)**	**Conventional (incision)** / **No-Scalpel Vasectomy (NSV – 1974, China; Dr Li Shunqiang)** – vas deferens divided & ligated
Site	**Isthmus** of fallopian tube	Vas deferens (scrotum)
Timing	**Interval** (within 7 days of menses – after ruling out pregnancy), **postpartum (within 7 days; best 24–48 hrs – minilap)**, **with LSCS**, **post-abortion**; **laparoscopic – interval or after 6 wks postpartum/ with 1st trimester MTP** (not postpartum – enlarged uterus)	Any time
Anaesthesia	Local + sedation / spinal	Local
Effectiveness	Immediate; failure ~0.5 %	**Not immediately effective** – sperm stored in seminal vesicles → **use condom for 3 months (or ~20 ejaculations) till semen analysis shows azoospermia (2 consecutive)**

	Tubectomy (female)	Vasectomy (male)
Complications	Anaesthesia, bleeding, infection, bowel/ bladder injury, **failure → ectopic**, regret	Haematoma, infection, sperm granuloma, **no effect on potency/ libido**
Reversal	**Recanalization (tuboplasty)** – ~50–70 % success (Pomeroy/ ring)	Vasovasostomy
Eligibility (GoI)	**Married, female 22–49 yrs, male < 60 yrs**, **at least 1 child aged > 1 yr** (sound health), voluntary informed consent (**spouse consent not mandatory per SC**, but encouraged); one of the couple only	Same

- **Compensation (GoI)** – acceptors ko wage-loss compensation (vasectomy ke liye zyada); **Family Planning Indemnity Scheme** – compensation for death/ failure/ complications.
- **Simpler, safer, cheaper & faster method – vasectomy (NSV)**, yet female sterilization dominates in India.
- **Post-tubectomy care**: rest 24–48 hrs, wound dry, sutures removed day 5–7, **resume intercourse after 1 week (when comfortable)**, light work after 2–3 days, no heavy lifting 2 wks; **report – fever, wound pus, severe pain, missed period (failure/ ectopic)**.
- **Post-vasectomy**: scrotal support, ice pack, avoid heavy work 48 hrs, **condom for 3 months**, semen analysis.

PYQ style

- Eligible couple – wife age **15–45 yrs**.
- Pearl index – failures per **100 woman-years**.
- Only method protecting from STI/ HIV – **condom**.
- Cu-T 380A effective for – **10 years**; Cu 375 – **5 years**.
- Best time for IUCD – **within 12 days of menstruation/ PPIUCD within 48 hrs**.
- Commonest reason for IUCD removal – **bleeding & pain**; commonest complication in first year – **expulsion**.
- Mala-N – **21 hormonal + 7 iron tablets**; start **day 1–5**.
- COC contraindicated – **thromboembolism, breastfeeding < 6 months, smoker > 35, breast cancer**.
- Non-steroidal weekly pill – **Centchroman (Chhaya)**.
- DMPA (Antara) – **150 mg IM every 3 months**.
- Emergency pill – **LNG 1.5 mg within 72 hrs**; most effective EC – **Cu-IUCD within 5 days**.
- LAM conditions – **exclusive BF + amenorrhoea + baby < 6 months**.
- Tubectomy site – **isthmus**; technique – **Pomeroy's**.
- After vasectomy, condom for – **3 months (20 ejaculations)**.
- First country with national FP programme – **India (1952)**.

19 Gynaecological Disorders & Women's Health

19.1 Menstrual Disorders

Term	Meaning
Amenorrhoea – primary	**No menarche by 15 yrs** (with secondary sex characters) / 13 yrs (without) – causes: **Turner's (45 XO) – most common chromosomal**, Müllerian agenesis (MRKH), imperforate hymen (**cryptomenorrhoea** – haematocolpos), Kallmann
Amenorrhoea – secondary	**No periods for ≥ 3 months** (regular cycles) / 6 months (irregular) in a woman who menstruated – **pregnancy (most common – rule out first)**, lactation, menopause, **PCOS**, hypothyroidism, hyperprolactinaemia, **Sheehan's**, **Asherman's**, stress, anorexia, excessive exercise
Menorrhagia (heavy menstrual bleeding)	**> 80 ml** or > 7 days, **regular cycles**
Metrorrhagia	**Irregular, intermenstrual bleeding**
Menometrorrhagia	Heavy + irregular
Polymenorrhoea	Cycle **< 21 days** (frequent)
Oligomenorrhoea	Cycle **> 35 days** (infrequent)
Hypomenorrhoea	Scanty flow
Dysmenorrhoea	**Painful menstruation** – **primary** (no pathology, adolescents, **prostaglandins**; starts with period, lasts 48–72 hrs → **NSAIDs (mefenamic acid)**, OCP, heat, exercise) / **secondary** (pathology – **endometriosis**, adenomyosis, PID, fibroids, IUCD)
PMS (premenstrual syndrome)	Luteal phase symptoms (bloating, irritability, mastalgia), relieved with menses; severe – PMDD
Post-menopausal bleeding (PMB)	Bleeding **≥ 12 months after menopause** → **rule out endometrial carcinoma** (most important), cervical ca, atrophic vaginitis, HRT
AUB (PALM-COEIN – FIGO)	Structural: **P**olyp, **A**denomyosis, **L**eiomyoma, **M**alignancy; Non-structural: **C**oagulopathy, **O**vulatory dysfunction, **E**ndometrial, **I**atrogenic, **N**ot classified

- **PCOS (Polycystic ovary syndrome)** – **Rotterdam criteria (2 of 3)**: (1) **oligo/ anovulation (irregular periods)**, (2) **hyperandrogenism (hirsutism, acne, alopecia)**, (3) **polycystic ovaries on USG (≥ 12/ 20 follicles 2–9 mm or volume > 10 ml – "string of pearls")**; **insulin resistance, obesity, acanthosis nigricans**, infertility, ↑ LH/FSH ratio; long-term – **T2DM, endometrial cancer**, CVD. Rx – **weight loss (first-line)**, **OCP**, **metformin**, **letrozole/ clomiphene (ovulation induction)**, spironolactone (hirsutism).
- **Endometriosis** – **endometrial tissue outside uterus** (**ovary – "chocolate cyst" (endometrioma) – most common site**, POD, uterosacral ligaments); **triad – dysmenorrhoea (secondary, progressive), dyspareunia, infertility**; diagnosis – **laparoscopy (gold standard – powder-burn spots)**; Rx – NSAIDs, OCP, progestins, GnRH analogues, danazol, surgery.
- **Adenomyosis** – endometrium **within myometrium** – multiparous, 40s; **bulky, tender, uniformly enlarged uterus**, menorrhagia, dysmenorrhoea.

19.2 Infertility

- **Inability to conceive after 1 year of regular unprotected intercourse** (**6 months if woman ≥ 35**). **Primary** – never conceived; **secondary** – previous conception.
- Causes: **female (~40–50 %)** – **ovulatory (PCOS – most common)**, **tubal (PID, TB – India)**, uterine (fibroids, Asherman's, anomalies), cervical, endometriosis; **male (~30–40 %)** – oligospermia, azoospermia, varicocele, infections, erectile dysfunction; **unexplained (~10–15 %)**.
- **Investigations**: **Semen analysis (first test – easiest)** (**WHO 2021**: volume ≥ 1.4 ml, count ≥ 16 million/ml, total motility ≥ 42 %, progressive ≥ 30 %, normal forms ≥ 4 %; abstinence 2–7 days); **ovulation** – **mid-luteal (day 21) serum progesterone > 3 ng/ml**, BBT, follicular monitoring USG, LH kits; **tubal patency** – **HSG (hysterosalpingography – day 7–10 of cycle, post-menses pre-ovulation)**, **SSG**, **laparoscopy with chromopertubation (gold standard)**; hormones – FSH, LH, AMH (ovarian reserve), TSH, prolactin.
- Treatment – **ovulation induction (clomiphene citrate, letrozole – PCOS first-line)**, gonadotropins, **IUI** (intrauterine insemination – mild male factor, cervical), **IVF-ET** (tubal block – Louise Brown, 1978 – first IVF baby; India – **Durga (Kanupriya Agarwal) 1978, Dr Subhash Mukhopadhyay**), **ICSI** (severe male factor), surgery; **OHSS** – ovarian hyperstimulation syndrome risk.
- **ART (Regulation) Act 2021 & Surrogacy (Regulation) Act 2021** – commercial surrogacy banned; altruistic only.

19.3 Fibroids, Prolapse & Displacement

- **Uterine fibroid (leiomyoma/ myoma)** – **most common benign tumour of uterus/ female pelvis**; **oestrogen-dependent**; 30–45 yrs, nulliparous, obesity, African; types – **intramural (most common)**, **subserous**, **submucous (most symptomatic – menorrhagia, infertility, abortion)**, cervical, pedunculated; **S/S – menorrhagia (most common)**, pelvic mass, pressure symptoms (frequency, retention, constipation), infertility, pain (**red degeneration – pregnancy (2nd trimester) – most common degeneration in pregnancy**); **hyaline degeneration – most common overall**; **sarcomatous change – rare (< 0.5 %)**. Rx – observation, tranexamic acid, GnRH analogues, **myomectomy (young, fertility desired)**, **hysterectomy**, UAE; shrink after menopause.
- **Uterovaginal prolapse** – descent of uterus/ vaginal walls; causes – **multiparity, prolonged/ difficult labour, premature bearing down, ↑ intra-abdominal pressure (chronic cough, constipation, obesity), menopause (oestrogen ↓)**, congenital weakness; **supports damaged – cardinal & uterosacral ligaments, levator ani**.
  - **Degrees**: **1st** – cervix descends but **within vagina**; **2nd** – cervix **at/ outside introitus**, body inside; **3rd** – cervix & body outside; **Procidentia** – **whole uterus outside** with vaginal walls (complete inversion of vagina) → **decubitus ulcer**. (**POP-Q** system now.)
  - **Cystocele** (bladder – anterior wall; urinary symptoms), **rectocele** (posterior wall; constipation), **enterocele** (POD – small bowel), **urethrocele**.
  - Rx – **pelvic floor (Kegel) exercises** (prevention & mild), **ring pessary** (unfit for surgery, pregnancy, temporary – change 3–6 monthly), **surgery – Fothergill's (Manchester) operation** (young, wants uterus), **vaginal hysterectomy with pelvic floor repair (anterior colporrhaphy + posterior colpoperineorrhaphy) – common in elderly**, Le Fort's colpocleisis (elderly unfit), sling operations (Shirodkar, Purandare – young nulliparous).
- **Retroverted uterus** – 15–20 % normal; mostly asymptomatic; **knee-chest position exercises**; Hodge pessary.

19.4 Reproductive Tract Infections (RTI/ STI) & PID

Infection	Discharge / Features	Treatment
Bacterial vaginosis (Gardnerella – most common cause of vaginal discharge)	**Thin, grey-white, fishy odour (↑ after intercourse)**, **clue cells**, **whiff test +ve (KOH)**, pH > 4.5; no inflammation	**Metronidazole** (oral/ gel)
Candidiasis (moniliasis)	**Thick, white, curdy ("cottage cheese")**, **intense itching**, redness; pH normal (< 4.5); **pregnancy, diabetes, antibiotics, OCP, HIV**	**Clotrimazole vaginal**, fluconazole oral (not in pregnancy)
Trichomoniasis (**T. vaginalis** – protozoa)	**Frothy, greenish-yellow, foul**, itching, **"strawberry cervix"** (punctate haemorrhages), pH > 4.5; **STI – treat partner**	**Metronidazole 2 g single dose / 400–500 mg BD × 7 days (both partners)**; avoid alcohol
Gonorrhoea (**N. gonorrhoeae**)	Purulent discharge, cervicitis, dysuria; neonatal ophthalmia	**Ceftriaxone IM** + azithromycin
Chlamydia (**C. trachomatis**)	Often asymptomatic; **most common bacterial STI**; **PID, tubal infertility, ectopic**	**Azithromycin 1 g single/ doxycycline**
Syphilis (**Treponema pallidum**)	**Painless chancre** (primary), rash (secondary)	**Benzathine penicillin 2.4 MU IM**
Chancroid (**H. ducreyi**)	**Painful** soft ulcer, bubo	Azithromycin/ ceftriaxone
Genital herpes (HSV-2)	Painful vesicles/ ulcers; neonatal herpes → CS if active lesions at term	Acyclovir
Genital warts (HPV 6, 11)	Condyloma acuminata	Podophyllin (not in pregnancy), cryo
LGV / Donovanosis		Doxycycline/ azithromycin

- **Syndromic management (NACO/ NACP – colour-coded STI kits)**: **Kit 1 – Grey** (urethral/ cervical discharge – azithromycin + cefixime), **Kit 2 – Green** (vaginal discharge – secnidazole + fluconazole), **Kit 3 – White** (genital ulcer non-herpetic – benzathine penicillin + azithromycin), **Kit 4 – Blue** (genital ulcer – penicillin allergy – doxycycline + azithromycin), **Kit 5 – Red** (genital ulcer herpetic – acyclovir), **Kit 6 – Yellow** (lower abdominal pain/ PID – cefixime + metronidazole + doxycycline), **Kit 7 – Black** (inguinal bubo – doxycycline + azithromycin). **Suraksha clinics** (DSRC).
- **PID (Pelvic Inflammatory Disease)** – ascending infection (**Chlamydia, gonorrhoea**, anaerobes) → endometritis, **salpingitis**, tubo-ovarian abscess, pelvic peritonitis; S/S – **lower abdominal pain, fever, abnormal discharge, cervical motion tenderness, adnexal tenderness**, dyspareunia; complications – **infertility, ectopic, chronic pelvic pain**; **Fitz-Hugh-Curtis syndrome** (perihepatitis); Rx – antibiotics (both partners), **Fowler's**, rest, remove IUCD if severe.
- **Genital TB** – **India – common cause of tubal infertility**; **fallopian tube most affected (~90–100 %)**, endometrium; amenorrhoea/ infertility; HSG – beaded tubes, **"tobacco pouch"**; ATT 6 months.
- **Vaginal pH** – normal **3.8–4.5**; ↑ in BV & trichomonas.

19.5 Gynaecological Cancers & Screening

- **Cervical cancer** – **2nd most common cancer in Indian women** (after breast); **HPV 16 & 18 (~70 %)**; **squamous cell (most common, ~80–90 %)** at **transformation zone (squamo-columnar junction)**;

risk – early coitus, multiple partners, **multiparity**, smoking, HIV, low SES, OCP long-term; **CIN (precancer) → invasive**; S/S – **post-coital bleeding (earliest/ commonest)**, intermenstrual bleeding, foul watery discharge, late – pain, VVF, uraemia (**commonest cause of death – uraemia/ renal failure**).

  - **Screening**: **VIA (visual inspection with 3–5 % acetic acid – acetowhite area = +ve)** – India NP-NCD, **30–65 yrs every 5 yrs**; **VILI (Lugol's iodine – unstained area)**; **Pap smear** (Papanicolaou – **Ayre's spatula + cytobrush** from SCJ; **avoid during menses, intercourse/ douching 48 hrs before**; fix in **95 % ethyl alcohol** immediately; from 21/ 25 yrs every 3 yrs); **HPV DNA test (every 5 yrs – 30+)**; **colposcopy** + biopsy (abnormal).
  - **Screen & treat** – **cryotherapy / thermal ablation**, **LEEP/ LLETZ**, cone biopsy.
  - **HPV vaccine** – **girls 9–14 yrs (2 doses 6 months apart)**; **Cervavac (indigenous quadrivalent – Serum Institute, 2022)**; Gardasil (4/ 9-valent), Cervarix (2).
  - Staging **FIGO**; Rx – surgery (**Wertheim's radical hysterectomy** – early), **radiotherapy (brachytherapy + external)** – most stages, chemoradiation (cisplatin).

- **Breast cancer** – **most common cancer in Indian women** (and worldwide); **upper outer quadrant**; risk – **BRCA1/2**, early menarche/ late menopause, nulliparity/ late first pregnancy, no breastfeeding, HRT, obesity, alcohol; **painless hard irregular lump**, peau d'orange, nipple retraction; **screening – BSE (monthly, 5–7 days after menses)**, **CBE**, **mammography (40/ 50+ every 1–2 yrs)**; Rx – surgery (BCS/ MRM), chemo, radiation, hormonal (**tamoxifen**).
- **Endometrial cancer** – **post-menopausal bleeding (cardinal symptom)**; risk – **unopposed oestrogen, obesity, nulliparity, PCOS, diabetes, HTN ("corpus cancer syndrome")**, tamoxifen, late menopause; **endometrial biopsy/ fractional curettage**; Rx – hysterectomy + BSO.
- **Ovarian cancer** – "**silent killer**" – vague abdominal symptoms (bloating, distension), late diagnosis; **epithelial (serous – most common)**; marker **CA-125**; **germ cell tumours (young)** – AFP, hCG, LDH; **Krukenberg tumour** (metastatic from stomach – signet ring cells); risk – BRCA, nulliparity, family history; protective – **OCP, pregnancy, breastfeeding, tubal ligation**. **Dermoid cyst (mature cystic teratoma) – most common benign ovarian tumour in young** (torsion – common complication).
- **Choriocarcinoma** – see Chapter 12.

19.6 Menopause

- **Permanent cessation of menstruation** due to loss of ovarian follicular activity – **diagnosed retrospectively after 12 months of amenorrhoea**; **average age 45–50 yrs (India ~46–48)**; **premature menopause < 40 yrs (POI)**; **perimenopause/ climacteric** – transition phase.
- Hormones – **↓ oestrogen, ↓ inhibin → ↑ FSH (> 30–40 IU/L – most sensitive marker)**, ↑ LH.
- **Symptoms**: **vasomotor – hot flushes (most common), night sweats**, palpitations; insomnia, mood swings, irritability, depression, poor memory; **genitourinary syndrome of menopause** – vaginal dryness, **dyspareunia, atrophic vaginitis**, urinary frequency/ urgency/ UTI, **prolapse**; skin thinning, breast atrophy; long-term – **osteoporosis (fractures – vertebra, hip, wrist)**, **cardiovascular disease ↑**, weight gain, dementia.
- **Management**: **lifestyle** (exercise – weight-bearing, **calcium 1200 mg + Vit D**, diet, stop smoking), **HRT (oestrogen ± progesterone – progesterone added if uterus present to prevent endometrial cancer)** – short-term, lowest dose for vasomotor symptoms; **contraindications – breast/ endometrial cancer, undiagnosed vaginal bleeding, thromboembolism, active liver disease, CVD/ stroke**; non-hormonal – SSRIs/ SNRIs, gabapentin, clonidine; vaginal oestrogen/ lubricants; **bisphosphonates** for osteoporosis; **DEXA screening**.

- **18 October = World Menopause Day**.

19.7 Common Gynaecological Procedures – Nursing Points

Procedure	Nursing points
Pap smear	Not during menses; no douche/ intercourse/ vaginal meds 48 hrs; **lithotomy/ dorsal**; **Cusco's (bivalve) speculum – no lubricant (water only)**; fix immediately
Per speculum (P/S)/ per vaginal (P/V) exam	**Empty bladder**, lithotomy/ dorsal, privacy, **female chaperone**, warm speculum, explain
D&C / Endometrial biopsy	NPO, consent, lithotomy, post-op – watch bleeding, perforation, pain
HSG	**Day 7–10 of cycle (after menses, before ovulation)**; iodine allergy check; antibiotics; mild cramps
Colposcopy	Magnified cervix view with acetic acid/ iodine
Laparoscopy	**Trendelenburg** intra-op; **shoulder pain post-op (CO_2 gas – referred)** → reassure, ambulate, analgesics
Hysterectomy (abdominal – TAH / vaginal – VH / laparoscopic – TLH)	Pre-op – bowel prep, catheter, consent; post-op – **vaginal bleeding (pad check), urine output (ureter injury), catheter 24 hrs**, early ambulation (DVT), **no heavy lifting/ intercourse 6 wks**; vaginal vault care; surgical menopause if ovaries removed (BSO)
Vaginal douche	Not routinely (disturbs flora); lukewarm, can height 30–60 cm
Sitz bath	40–43 °C, 15–20 min – perineal wounds, haemorrhoids

PYQ style

- Primary amenorrhoea commonest chromosomal cause – **Turner's syndrome**.
- Secondary amenorrhoea – first rule out **pregnancy**.
- Menorrhagia – **> 80 ml, regular cycles**; polymenorrhoea – **cycle < 21 days**.
- PCOS criteria – **Rotterdam (2 of 3)**; first-line Rx – **weight loss**.
- Chocolate cyst – **endometriosis of ovary**; diagnosis – **laparoscopy**.
- Commonest female infertility cause – **ovulatory (PCOS)**; first test – **semen analysis**.
- HSG done on – **day 7–10 of cycle**.
- Commonest benign uterine tumour – **fibroid**; degeneration in pregnancy – **red degeneration**.
- Procidentia – **whole uterus outside vulva**; ring pessary – **unfit for surgery**.
- Fishy odour, clue cells – **bacterial vaginosis**; curdy white – **candida**; frothy greenish, strawberry cervix – **trichomonas**.
- STI syndromic kit for vaginal discharge – **Kit 2 (green)**; PID – **Kit 6 (yellow)**.
- Commonest cancer in Indian women – **breast**; cervical cancer cause – **HPV 16, 18**.
- VIA uses – **3–5 % acetic acid**; Pap smear fixative – **95 % ethyl alcohol**.
- HPV vaccine age – **9–14 yrs girls**; Indian vaccine – **Cervavac**.
- Post-menopausal bleeding – rule out **endometrial cancer**.
- Menopause marker – **↑ FSH**; commonest symptom – **hot flushes**.

20 Quick Revision – Master Tables & 150 One-Liners

20.1 Master Table – Numbers

Topic	Value
Uterus non-pregnant / term weight	**50–60 g / ~1000 g**
Uterus size	**7.5 × 5 × 2.5 cm**
Fallopian tube / cervix / vagina (post.)	**10 cm / 2.5 cm / 9 cm**
Ovary	**3 × 2 × 1 cm**
Inlet: AP / Oblique / Transverse	**11 / 12 / 13 cm**
Outlet: AP / Transverse	**13 / 11 cm**
Diagonal / obstetric conjugate	**12–12.5 / 10–10.5 cm**
Suboccipito-bregmatic / biparietal	**9.5 / 9.5 cm**
Mento-vertical (brow)	**13.5–14 cm**
Anterior / posterior fontanelle closure	**18 months / 6–8 weeks**
Head circumference at birth	**33–35 cm**
Placenta weight / diameter	**500 g / 15–20 cm**
Cord length	**50 cm** (2 arteries, 1 vein)
Amniotic fluid max / term	**1000 ml (36–38 wks) / 800 ml**
AFI normal	**5–25 cm** (poly > 25, oligo < 5)
FHR normal	**110–160 bpm**
Blood volume increase	**40–50 %**
Weight gain in pregnancy	**10–12 kg (11 kg)**
Hb – anaemia in pregnancy	**< 11 g/dL**; severe < 7
Pregnancy duration	**280 days (40 wks)**
Naegele's rule	**LMP + 9 months + 7 days**
Quickening primi / multi	**18–20 / 16–18 wks**
FHS – Doppler / stethoscope	**10–12 / 18–20 wks**
Gestational sac TVS	**4.5–5 wks**
Fundus at umbilicus / xiphisternum	**22–24 / 36 wks**
ANC visits (India) / WHO	**4 / 8 contacts**
IFA	**60 mg iron + 500 µg FA × 180 days**
Td	**2 doses, 4 wks apart**
Calcium	**1 g/day**

Topic	Value
1st stage primi / multi	**12 / 6 hrs**
2nd stage primi / multi	**1–2 hrs / 30 min–1 hr**
3rd stage	**15–30 min**
4th stage	**1 hr after placenta**
Precipitate labour	**< 3 hrs**
Prolonged labour	**> 18–24 hrs**
Partograph alert line	**1 cm/hr; action 4 hrs right**
FHS 1st / 2nd stage	**every 30 min / every 5 min**
PV in labour	**every 4 hrs**
AMTSL oxytocin	**10 IU IM within 1 min**
PPH	**≥ 500 ml vaginal / ≥ 1000 ml CS**
Retained placenta	**> 30 min**
Methergine	**0.2 mg IM, max 1 mg**
Carboprost	**250 µg IM, max 2 mg (8 doses)**
Misoprostol PPH Rx / prevention	**800 µg SL / 600 µg oral**
Tranexamic acid	**1 g IV within 3 hrs**
$MgSO_4$ Pritchard loading	**4 g IV + 10 g IM**
$MgSO_4$ maintenance	**5 g IM 4-hrly × 24 hrs**
$MgSO_4$ therapeutic level	**4–7 mEq/L**
Dexamethasone (lung maturity)	**6 mg IM 12-hrly × 4**
Anti-D	**300 µg within 72 hrs** (50 µg < 12 wks)
Puerperium	**6 weeks**
Uterus not palpable	**10–14 days**
Lochia rubra / serosa / alba	**1–4 / 5–9 / 10 days – 6 wks**
Puerperal pyrexia	**≥ 38 °C**
Breastfeeding initiation / exclusive	**1 hr / 6 months**
Newborn weight / length	**~2.8–3 kg / 50 cm**
LBW / VLBW / ELBW	**< 2500 / < 1500 / < 1000 g**
Newborn HR / RR	**120–160 / 30–60**
Vitamin K	**1 mg IM**
Neonatal CPR ratio	**3 : 1**
PPV rate	**40–60/min**
Physiological jaundice	**after 24 hrs, peak day 3–5**
Cu-T 380A / Cu 375	**10 / 5 years**
DMPA	**150 mg IM 3-monthly**

Topic	Value
ECP (LNG)	**1.5 mg within 72 hrs**
Medical abortion	**up to 9 weeks (63 days)**
MTP (1 RMP / 2 RMPs)	**≤ 20 wks / 20–24 wks**
Menopause FSH	**> 30–40 IU/L**

20.2 Master Table – Signs & Eponyms

Sign / Name	Meaning
Hegar's	Softening of isthmus (6–10 wks)
Goodell's	Softening of cervix (6 wks)
Chadwick's/ Jacquemier's	Bluish vagina (8 wks)
Osiander's	Pulsation in lateral fornix
Piskacek's	Asymmetric uterine enlargement
Palmer's	Rhythmic uterine contractions (early)
Spalding's	Overlapping skull bones – IUD
Robert's	Gas in fetal vessels – IUD
Bandl's ring	Pathological retraction ring – obstructed labour
Turtle sign	Shoulder dystocia
Kehr's sign	Shoulder tip pain – ruptured ectopic
Cullen's sign	Periumbilical bluish – ruptured ectopic
Couvelaire uterus	Abruption – blood in myometrium
Schultze / Matthews Duncan	Placental separation – central (shiny) / marginal (dirty)
Brandt-Andrews	CCT for placenta delivery
McRoberts	Shoulder dystocia – thigh hyperflexion
Mauriceau-Smellie-Veit	After-coming head of breech
Lovset's	Extended arms in breech
Ritgen's	Perineal support & head control
Johnson's manoeuvre	Replacement of inverted uterus
Pritchard / Zuspan	$MgSO_4$ regimens
Macafee & Johnson	Expectant management of placenta praevia
McDonald / Shirodkar	Cervical cerclage
Pomeroy	Tubectomy technique
Naegele	EDD rule
Leopold	Abdominal grips
Pawlik's	3rd grip – presenting part

Sign / Name	Meaning
Bishop score	Cervical ripeness
Apgar	Newborn score (Virginia Apgar)
Kramer	Jaundice spread
Silverman-Anderson	RDS score
Ballard	Gestational age of newborn
Haase's rule	Fetal length
Rotterdam	PCOS criteria
Caldwell-Moloy	Pelvis types
Sheehan's	Postpartum pituitary necrosis
Asherman's	Intrauterine adhesions
Mendelson's	Acid aspiration under GA
Erb's palsy	C5–C6 – waiter's tip

20.3 150 One-Liners

One-liners 1–50

1. Father of obstetrics – **William Smellie**.
2. Semmelweis – **puerperal fever prevention (hand washing)**.
3. MMR expressed per – **1,00,000 live births**.
4. Most common cause of maternal death – **PPH**.
5. JSY launched – **2005**; JSSK – **2011**.
6. PMSMA – **9th of every month**.
7. Safe Motherhood Day – **11 April**.
8. Grand multipara – **≥ 5 deliveries**.
9. Puerperium – **6 weeks**.
10. Viability (India) – **28 wks**.
11. Site of fertilization – **ampulla**.
12. Tubectomy site – **isthmus**.
13. Main uterine support – **cardinal ligament**.
14. Gynaecoid pelvis – **round inlet, ~50 %**.
15. Largest inlet diameter – **transverse 13 cm**.
16. Clinically measured – **diagonal conjugate**.
17. Anterior fontanelle shape – **diamond**.
18. Caput – **crosses suture lines**.
19. Ovulation (28-day) – **day 14**; LH surge before.
20. BBT rise – **progesterone**.
21. Implantation – **day 6–7 after fertilisation**.
22. Organogenesis – **3–8 weeks**.
23. Fetal heart beats from – **day 22**.
24. Placenta hormone for pregnancy test – **hCG**.
25. hPL – **anti-insulin**.
26. Only Ig crossing placenta – **IgG**.
27. Polyhydramnios – **> 2000 ml / AFI > 24–25**.
28. Oligohydramnios – **renal agenesis (Potter)**.
29. Nitrazine with liquor – **blue**.
30. Highest O_2 in fetus – **umbilical vein**.
31. Ductus arteriosus – **pulmonary artery → aorta**.
32. Foramen ovale → **fossa ovalis**.
33. Blood volume ↑ – **40–50 %**.
34. Cardiac output max – **immediately after delivery**.
35. Supine hypotension – **left lateral**.
36. Chloasma – **mask of pregnancy**.
37. First symptom – **amenorrhoea**.
38. Positive sign – **FHS / USG fetus**.
39. Hegar's sign – **isthmus softening**.
40. EDD – **LMP + 9 m + 7 d**.

41. First ANC – **within 12 wks**.
42. IFA – **180 days**.
43. GDM test (DIPSI) – **75 g, 2-hr ≥ 140**.
44. Leg cramps – **dorsiflex foot**.
45. Kick count – **≥ 10 in 12 hrs**.
46. Most common lie – **longitudinal**; position – **LOA**.
47. Engagement – **BPD through inlet; station 0**.
48. Pawlik's grip – **presenting part & engagement**.
49. Reactive NST – **2 accelerations 15 × 15 in 20 min**.
50. Late deceleration – **uteroplacental insufficiency**.

One-liners 51–100

51. True labour – **cervical dilatation, pain back → front**.
52. Internal rotation – **at pelvic floor**.
53. Smallest diameter – **suboccipito-bregmatic 9.5**.
54. 1st stage ends – **full dilatation**.
55. Alert line – **1 cm/hr**.
56. AMTSL – **oxytocin 10 IU IM, CCT, massage**.
57. Placental separation sign – **cord lengthening**.
58. Common episiotomy – **mediolateral**.
59. 3rd degree tear – **anal sphincter**.
60. Uterus not palpable – **10–14 days**.
61. Lochia alba – **10 days – 6 wks**.
62. Postpartum blues – **day 3–5**.
63. Milk production – **prolactin**; ejection – **oxytocin**.
64. Colostrum – **IgA, protein, Vit A**.
65. Good attachment – **chin touching breast**.
66. Sore nipple cause – **poor attachment**.
67. Mastitis – **continue feeding**.
68. LBW – **< 2500 g**.
69. APGAR – **1 & 5 min, 5 signs**.
70. Golden minute – **PPV within 1 min**.
71. Vitamin K – **1 mg IM**.
72. Physiological jaundice – **after 24 hrs**.
73. Phototherapy – **cover eyes & genitals**.
74. Moro asymmetric – **Erb's/ clavicle fracture**.
75. KMC – **skin-to-skin + EBF**.
76. Hyperemesis – **thiamine before dextrose**.
77. Anaemia in pregnancy – **Hb < 11**.
78. Pre-eclampsia – **HTN + proteinuria after 20 wks**.
79. Eclampsia drug – **$MgSO_4$**.
80. $MgSO_4$ antidote – **calcium gluconate**.
81. Methergine CI – **HTN, heart disease**.
82. Avoid in pregnancy HTN – **ACE inhibitors**.
83. GDM drug of choice – **insulin**.
84. IDM neonate – **hypoglycaemia**.
85. Heart disease critical time – **immediately postpartum**.
86. Anti-D – **300 µg within 72 hrs**.
87. Maternal antibody test – **indirect Coombs**.
88. Twins commonest complication – **preterm labour**.
89. Spalding's sign – **IUD**.
90. Commonest abortion cause – **chromosomal**.
91. Threatened abortion – **os closed**.
92. Cervical incompetence – **McDonald's cerclage**.

93. MTP up to 20 wks – **1 RMP**.
94. Commonest ectopic site – **ampulla**.
95. Ectopic medical Rx – **methotrexate**.
96. Snowstorm – **H. mole**.
97. Mole follow-up – **β-hCG**.
98. APH – **no PV examination**.
99. Painless bleeding – **placenta praevia**.
100. Abruption association – **pre-eclampsia**.

One-liners 101–150

101. Commonest malpresentation – **breech**.
102. Breech commonest type – **frank**.
103. ECV – **36–37 wks**.
104. After-coming head – **MSV manoeuvre**.
105. Face denominator – **mentum**.
106. Brow – **LSCS**.
107. Transverse lie – **LSCS**.
108. OP – **FHS in flank, backache**.
109. Bandl's ring – **obstructed labour**.
110. Obstructed labour – **no oxytocin**.
111. VVF prevention – **continuous catheter**.
112. Tocolytic – **nifedipine**.
113. Steroid – **dexamethasone 6 mg × 4**.
114. PROM – **avoid PV**.
115. Bishop favourable – **≥ 8**.
116. Post-term – **≥ 42 wks**.
117. Oxytocin hyperstimulation – **stop drip**.
118. PPH commonest cause – **atony**.
119. Secondary PPH cause – **retained bits**.
120. PPH first – **call help, massage, empty bladder**.
121. Carboprost CI – **asthma**.
122. Inversion – **shock out of proportion**.
123. Rupture uterus earliest sign – **fetal distress**.
124. Cord prolapse – **push presenting part up, knee-chest**.
125. Shoulder dystocia first – **McRoberts**.
126. DIC commonest cause – **abruption**.
127. Sheehan's – **lactation failure**.
128. Puerperal sepsis prevention – **6 cleans**.
129. Puerperal psychosis – **emergency**.
130. RDS – **surfactant deficiency**.
131. Neonatal sepsis earliest sign – **refusal to feed**.
132. Commonest CS type – **LSCS**.
133. Commonest CS indication – **previous CS**.
134. Classical CS – **VBAC contraindicated**.
135. Vacuum CI – **preterm, face, breech**.
136. MVA – **up to 12 wks**.
137. Epidural s/e – **hypotension**.
138. Pethidine – **neonatal respiratory depression**.
139. Cu-T 380A – **10 yrs**.
140. IUCD commonest removal reason – **bleeding/ pain**.
141. Mala-N – **21 + 7 iron**.
142. Chhaya – **non-steroidal weekly pill**.

143. Antara – **DMPA 3-monthly**.
144. LAM – **EBF + amenorrhoea + < 6 months**.
145. Vasectomy – **condom 3 months**.
146. BV – **fishy odour, clue cells**.
147. Trichomonas – **strawberry cervix**.
148. Cervical cancer – **HPV 16, 18; VIA screening**.
149. Commonest cancer in Indian women – **breast**.
150. Menopause marker – **↑ FSH**; symptom – **hot flushes**.



Exam strategy – Asha Didi ki last advice

- GSSSB paper me **OBG se ~15–20 questions** aate hain – ANC, labour, PPH, newborn care, family planning aur national programmes sabse important hain.
- **Numbers** (diameters, doses, timings) ki tables roz padho; **drugs ke contraindications** (methergine – HTN, carboprost – asthma) zaroor yaad karo.
- Priority questions: **mother's ABC → bleeding control → fetal wellbeing**.
- MCQ Bank PDF ke saare questions **2 baar** solve karo, galtiyan notebook me likho.

— Midwifery & Obstetrical Nursing · GSSSB Staff Nurse Preparation · Padhte raho, revise karte raho! —